

IEEE Standard for Terminology and Test Methods of Digital-to- Analog Converter Devices

IEEE Instrumentation and Measurement Society

Developed by the
Waveform Generation, Measurement, and Analysis
Technical Committee (TC-10)

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Abstract: Terminology and test methods to clearly document prevalent world-wide terms used to describe and test digital-to-analog converters (DACs) are provided. It is restricted to monolithic, hybrid, and module DACs and does not cover systems encompassing DACs.

Keywords: DAC, digital-to-analog converter, IEEE 1658™, SFDR, SNR, spectrum, test methods, test terminology

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Introduction

This introduction is not part of IEEE Std 1658-2023, IEEE Standard for Terminology and Test Methods of Digital-to-Analog Converter Devices.

This standard is divided into 16 clauses plus annexes. [Clause 1](#) is a basic orientation. For further investigation, users of this standard can consult [Clause 2](#), which contains references to other IEEE standards on waveform measurement and relevant International Standardization Organization (ISO) documents. The definitions of technical terms and symbols used in this standard are presented in [Clause 3](#). [Clause 4](#) through [Clause 16](#) present a wide range of tests that measure the performance of a digital-to-analog converter. Annexes, containing the bibliography and informative comments on the tests presented in [Clause 4](#) through [Clause 16](#), augment the standard.

This standard defines the terms, definitions, and test methods used to specify, characterize, and test digital-to-analog converters (DACs). It is intended for the following:

- Individuals and organizations who specify DACs to be purchased
- Individuals and organizations who purchase DACs to be applied in their products
- Individuals and organizations whose responsibility is to characterize and write reports on DACs available for use in specific applications
- Suppliers interested in providing high-quality and high-performance DACs to acquirers

This standard is designed to help organizations and individuals:

- Incorporate quality considerations during the definition, evaluation, selection, and acceptance of supplier DACs for operational use in their equipment
- Determine how supplier DACs should be evaluated, tested, and accepted for delivery to end users

This standard is intended to satisfy the following objectives:

- Promote consistency within organizations in acquiring third-party DACs from component suppliers
- Provide useful practices on including quality considerations during acquisition planning
- Provide useful practices on evaluating and qualifying supplier capabilities to meet user requirements
- Provide useful practices on evaluating and qualifying supplier DACs
- Assist individuals and organizations judging the quality and suitability of supplier DACs for referral to end users

Several standards have previously been written that address the testing of DACs either directly or indirectly. These include the following:

- IEEE Std 1057™, which describes the testing of waveform recorders.
- IEEE Std 1241™, which addresses the testing of Analog-to-Digital Converters.
- IEEE Std 746™, which addresses the testing of Analog-to-Digital and Digital-to-Analog Converters, used for PCM television video signal processing.
- JEDEC Standard 99, addendum number 1, which deals with the terms and definitions used to describe Analog-to-Digital and Digital-to-Analog Converters. This standard does not include test methods.

- IEC 60748-4-1:1993, Semiconductor devices—Integrated circuits—Part 4: Interface integrated circuits—[Section 1](#): Blank detail specification for linear digital-to-analog converters (DAC).
- IEC 60748-4-2:1993, Semiconductor devices—Integrated circuits—Part 4: Interface integrated circuits—[Section 2](#): Blank detail specification for linear analog-to-digital converters (ADC).
- IEC 60748-4-3:2006, Semiconductor devices—Integrated circuits—Part 4-3: Interface integrated circuits—Dynamic criteria for analog-digital converters (ADC).

IEEE Std 1658 for digital-to-analog converters is intended to focus specifically on terms and definitions as well as test methods for DACs for a wide range of applications.

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IEEE Standard for Terminology and Test Methods of Digital-to-Analog Converter Devices

1. Overview

1.1 Scope

This standard defines terminology and test methods to clearly document prevalent world-wide terms used to describe and test digital-to-analog converters (DACs). Although principally applicable to monolithic, hybrid, module DACs, most of the definitions and test procedures can be applied to systems containing DACs.

1.2 Purpose

This standard is written to clear up confusion relating to the terms and definitions concerning DACs. Test methods are defined with expected output and analysis of the data generated from these test methods. Product specifications are defined in clear terms that will be understandable by all international users and manufacturers.

1.3 Word usage

The word *shall* indicates mandatory requirements strictly to be followed in order to conform to the standard and from which no deviation is permitted (*shall* equals *is required to*).^{6,7}

The word *should* indicates that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required (*should* equals *is recommended that*).

The word *may* is used to indicate a course of action permissible within the limits of the standard (*may* equals *is permitted to*).

The word *can* is used for statements of possibility and capability, whether material, physical, or causal (*can* equals *is able to*).

⁶The use of the word *must* is deprecated and cannot be used when stating mandatory requirements, *must* is used only to describe unavoidable situations.

⁷The use of *will* is deprecated and cannot be used when stating mandatory requirements, *will* is only used in statements of fact.

1.4 Discussion of scope and purpose

A digital-to-analog converter (DAC) receives digital input data and converts it to an analog output. The material presented in this standard is intended to provide common terminology and test methods for the testing and evaluation of DACs. The standard graphic symbol for a DAC is given in Figure 1, where N represents the width of the data word (in bits).



Figure 1—DAC graphic symbol

The information in this standard is useful both to manufacturers and users of DACs in that it provides a basis for evaluating and comparing existing devices, as well as providing a template for writing specifications for the procurement of new ones. This standard identifies DAC error sources and provides test methods to measure them. In some applications, the information provided by the tests described in this standard could be used to correct DAC errors, e.g., correction for gain and offset errors.

Note that this standard has many similarities to IEEE Std 1057 [B12]⁸ and IEEE Std 1241 [B13].

1.5 Digital-to-analog converter and analog-to-digital converter differences and similarities

DACs and analog-to-digital converters (ADCs) perform complementary functions. DACs receive a digital signal and generate an analog equivalent, while ADCs receive an analog signal and generate a digital equivalent. Some of the subtleties are summarized in the following list:

- a) Transfer curve differences (see Figure 2)
 - 1) For a DAC, each digital input corresponds to a single analog output value.
 - 2) For an ADC, each digital output corresponds to an interval of analog input values. The analog quantity defined in the transfer curve of an ADC, the code transition level $T[k]$, is usually defined to occur at the point where 50% of the output codes are greater than a specified digital output.

⁸The numbers in brackets correspond to those of the bibliography in Annex D.

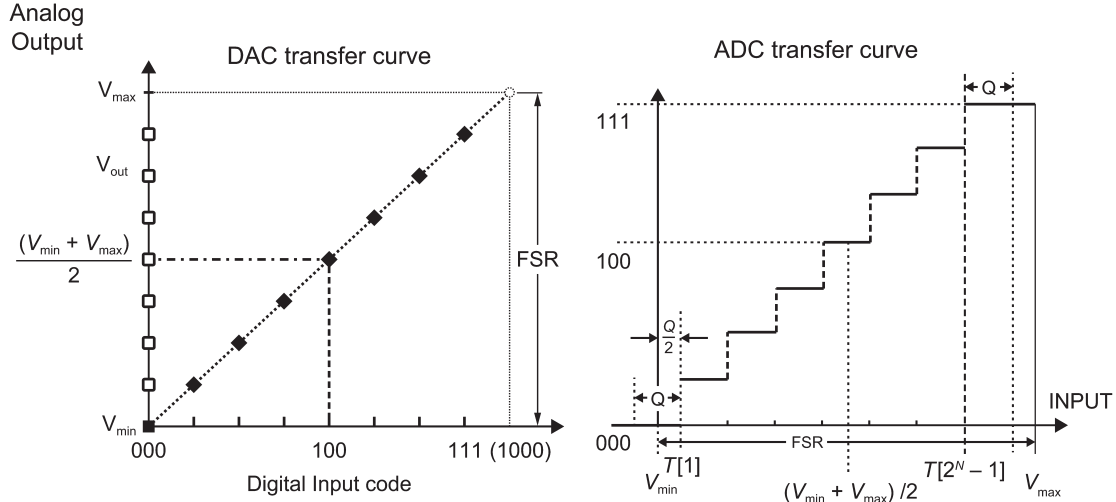


Figure 2—Transfer curves of ideal 3-bit unipolar DAC and ADC

- b) Quantization differences
 - 1) In a DAC, the input is a digital input code and the output is an analog representation of the input.
 - 2) In an ADC, the input is analog and the output is the quantized representation of the input.
- c) Time-domain differences
 - 1) A DAC with a constant input code produces a continuous nominally fixed analog output that may contain additive noise.
 - 2) An ADC with a constant input may generate a distribution of quantized outputs because of the presence of noise.
- d) Frequency-domain differences
 - 1) A DAC driven by a single-frequency digital sine wave at its input will produce at its output a single frequency between dc and 1/2 the output update rate, i.e., the rate at which codes are provided to the internal D/A section. Additionally, other frequencies will be produced that are at the multiples of the output update rate plus/minus the input frequency.
 - 2) An ADC with any fixed frequency analog input will produce a data record that will appear to contain frequencies only up to one half the sampling rate.
- e) Both a DAC and an ADC require an internal or external reference current or voltage that will cause them to have an output that can be in some way traced to standard units of voltage or current.

There are also some subtle differences between ADCs and DACs in the area of aliasing and image frequency generation. Aliasing is a phenomenon usually associated with ADCs while the generation of image frequencies is usually associated with DACs. The two phenomena are very closely related. In both cases there is a reference frequency, f_s . For an ADC this is the sampling frequency, the rate at which samples are taken. For a DAC this is the output update rate, the rate at which data words are supplied (after possible digital pre-processing) to the internal D/A section. Two positive frequencies, f_1 and f_2 , are said to be aliases of one another if they satisfy Equation (1):

$$f_2 - f_1 = lf_s, \text{ or } f_2 + f_1 = lf_s \quad (1)$$

where

l is any integer

Note that the integer l can be positive, negative, or zero. Recall that the n^{th} Nyquist band consists of all frequencies, f , satisfying $(n-1)f_s/2 \leq f < nf_s/2$. It is fairly easy to show that any frequency, f , has exactly one alias in each Nyquist band.

When an input signal of frequency, f , is supplied to an ADC the resulting samples will be a signal whose frequency is the alias of f in the first Nyquist band. Thus, many different frequencies get mapped into a single frequency. This phenomenon is called aliasing. Conversely, when an input sequence with a frequency in the first Nyquist band is supplied to a DAC, the output signal will generally contain components at all of the aliases of the input signal frequency. These spurious output frequencies are called images of the input frequency. Note that a sine-wave input sequence to a DAC can always be considered to be in the first Nyquist band because the sampled values of a sine wave in a higher Nyquist band are exactly equal to the sampled values of the signals alias in the first Nyquist band.

The aliasing phenomenon has complementary implications for the measurement of frequency response using sine-wave inputs for ADCs and DACs. For a DAC, inputs only need to be generated in the first Nyquist band, but the outputs need to be measured at the images of the input frequency in all bands of interest. For an ADC, the input signals shall be in all Nyquist bands of interest, but the outputs always appear at the alias in the first Nyquist band of the input signal.

1.6 Digital-to-analog converter background

This standard considers only those DACs whose output values have discrete values for discrete input words. Although different methods exist for representing a discrete sequence of binary words as a continuous analog signal, an underlying model implicit in many of the tests in this standard assumes that the relationship between the input code and the output value (transfer curve) approximates a straight line as depicted in [Figure 3](#) and [Figure 4](#). Applying this model to a code-input DAC, the full-scale output range (FSR) at the DAC is divided into uniform intervals, with nominal separation of Q . The number of code levels in the discrete transfer curve is equal to 2^N where N is the number of bits of the DAC. The number of intervals is $2^N - 1$. The full-scale outputs are generally defined as physical quantities (voltage or current) and are usually given as the values with code 0 applied, and code 2^N applied. The code for 2^N is generally a virtual code that does not exist but serves as a good definition of the DAC. Thus, a 0 V to 10 V 10-bit DAC will theoretically never actually generate 10.0 V. If the code $2^N - 1$ is applied (the last physically realizable code), the ideal output will be approximately 9.9902 V. The 3-bit DAC modeled in [Figure 3](#) could be described as an 8 V FS unipolar DAC, where each binary least significant bit would correspond to a 1 V increment in the output. Note that the DAC could not actually generate a +8 V output.

In addition to unsigned binary shown in [Figure 3](#), DACs may use two's complement, one's complement, sign-magnitude, Gray, Binary-Coded-Decimal (BCD), or other digital coding schemes. The 3-bit DAC modeled in [Figure 4](#) could be described as a ± 4 V bipolar two's complement DAC, where each binary least significant bit would correspond to a 1 V increment in the output. Note that the DAC could not actually generate a +4 V output.

DAC errors can be divided into the categories of static and dynamic. The static errors are usually caused by mismatch of the components and limit the accuracy at low speed. The static performance is often described by measures of differential and integral nonlinearities (DNL[k] and INL[k], where the DNL and INL are defined for each input word k). The dynamic errors, such as settling errors, glitches, etc., are important and they become more significant with higher update rates and when higher frequency waveforms are being synthesized by the DAC. To analyze the effect of errors, it is usually easier to consider the DAC's behavior in frequency domain using measures, such as the spurious free dynamic range (SFDR) and signal-to-noise and distortion (SINAD) ratio.

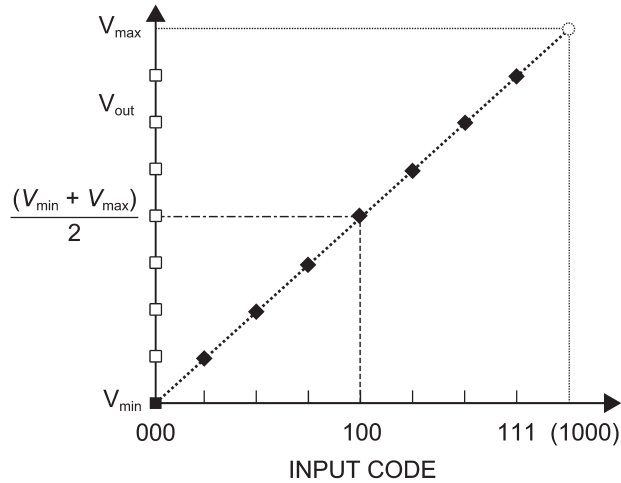


Figure 3—Elements of an ideal linear DAC transfer curve (unipolar)

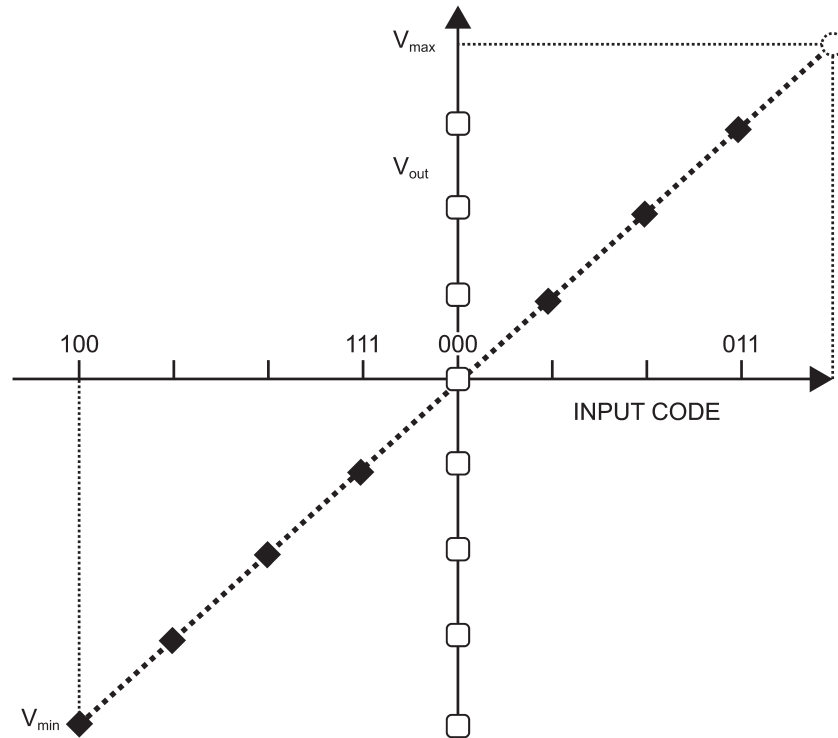


Figure 4—Elements of an ideal two's complement DAC transfer curve (bipolar)

DAC input architectures can be divided into parallel and serial. Parallel input converters receive parallel input codes that simultaneously control internal DAC switches that select a number of weights that are summed. Serial converters receive a serial input that is internally converted to a parallel word after a defined number of clocking cycles. The internal parallel word then functions in the same manner as a parallel input converter.

The general single-channel DAC architecture is shown in [Figure 5](#), which illustrates the main functional blocks that include the following:

- The digital pre-processing of the input data codes provided to the DAC at the input data rate f_{DATA} , which typically consists of one or more of the following procedures: filtering, pre-emphasis, interpolation, and re-sampling with higher resolution. Its output codes, which can have a different width N_{out} , are provided at the output update rate f_s . If no upsampling is applied, then $f_s = f_{DATA}$.
- The internal clock generation, which makes available a reference clock at f_s to be used for operating the internal D/A section.
- The internal D/A section, by which the stream of data codes at the output of the digital pre-processing is converted into an analog signal. This block marks the boundary between digital and analog domains.
- The analog post-processing, typically a low-pass filter. In some cases, a variable gain is also available. Its output is an analog signal that represents the output of the DAC.

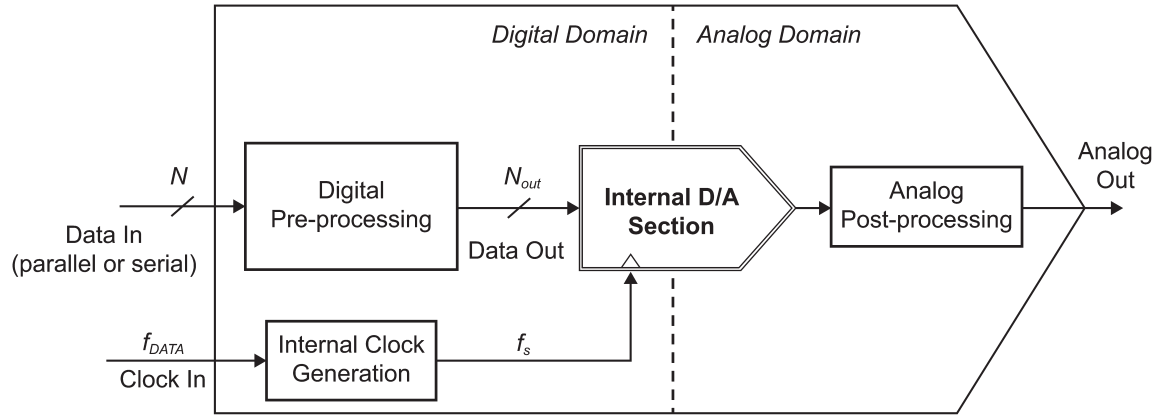


Figure 5—General architecture of a DAC (single channel, single D/A core)

From a circuital standpoint, the typical DAC consists of the following four major elements:

- The logic circuitry
- Some type of resistor or capacitor network
- A means of switching either a reference voltage or current to the proper input terminals of the network as a function of the digital value of each digital input bit
- A reference voltage or current

The following mathematical model is useful for discussing and analyzing the dynamic performance of a DAC. However, it should be kept in mind that this is an equivalent approximate mathematical model and does not necessarily reflect the physical operation of the DAC. This model constructs the output signal of the DAC by [Equation \(2\)](#):

$$V_{out}(t) = \left[\sum_{k=-\infty}^{\infty} w_k \delta(t - k\Delta t) \right] * f_R(t) = \sum_{k=-\infty}^{\infty} w_k f_R(t - k\Delta t) \quad (2)$$

where

- w_k is the value of the word of width N_{out} at the input of the internal D/A section at time $k\Delta t$
- $\Delta t = 1/f_s$ is the time interval between words at the input of the internal D/A section
- $\delta(t)$ is the Dirac delta distribution

$f_R(t)$ is the impulse response of the reconstruction network implemented in the internal D/A section
* is the convolution operator

For the sake of simplicity, no analog post-processing is considered in the model of [Equation \(2\)](#).

For the special case of the zero-order-hold DAC, the impulse response is represented by [Equation \(3\)](#).

$$f_R(t) = f_{ZH}(t) = \begin{cases} 1 & \text{for } 0 \leq t < \Delta t \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

A return-to-zero (RZ) DAC has an impulse response represented by [Equation \(4\)](#).

$$f_R(t) = f_{RZ}(t) = \begin{cases} a & \text{for } 0 \leq t < \tau \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

where

$a > 1$ is a constant introduced to recover from gain losses
 τ is the duration of the output pulse from the internal D/A section, which is assumed to be less than Δt

The purpose for a return-to-zero DAC is to generate images of the applied sequence in higher Nyquist bands.

The zero-order-hold DAC is illustrated in [Figure 8](#), where a complex analog waveform is loaded into a DAC as a sequence of digital words. The DAC output is retained at its initial value until the next word is loaded.

An internal D/A section with a zero-order hold may be followed by an analog low-pass filter in order to make the reconstruction process more accurate by filtering out aliases of the applied sequence still arising above the first Nyquist band. In this case, the impulse response of the overall filtering procedure is the convolution of that of the analog filter and that of the zero-order hold in [Equation \(3\)](#).

In many cases it is desirable to consider the input word values, w_k , to be the sampled values of a function, $V_{in}(t)$, of continuous time. In that case, [Equation \(2\)](#) can be written as shown in [Equation \(5\)](#).

$$V_{out}(t) = \left[V_{in}(t) \sum_{k=-\infty}^{\infty} \delta(t - k\Delta t) \right] * f_R(t) \quad (5)$$

From this it follows that the Fourier transforms of the input and output signals are related by [Equation \(6\)](#).

$$\hat{V}_{out}(\omega) = \left[\hat{V}_{in}(\omega) \sum_{l=-\infty}^{\infty} \delta(\omega - l\omega_s) \right] \hat{f}_R(\omega) \quad (6)$$

where

ω_s is $2\pi f_s$
 $\hat{f}_R(\omega)$ is the Fourier transform of $f_R(t)$, i.e., the frequency response of the reconstruction network

The case study of $V_{in}(t) = \cos(\omega_{in} t)$, or $\hat{V}_{in}(\omega) = \frac{1}{2}(\delta(\omega - \omega_{in}) + \delta(\omega + \omega_{in}))$, is of particular interest. In this case, [Equation \(6\)](#) yields [Equation \(7\)](#).

$$\hat{V}_{out}(\omega) = \left[\frac{1}{2} \sum_{l=-\infty}^{\infty} (\delta(\omega - l\omega_s - \omega_{in}) + \delta(\omega - l\omega_s + \omega_{in})) \right] \hat{f}_R(\omega) \quad (7)$$

The frequencies $lf_s \pm f_{in}$, for $|l| \geq 1$, are called the image frequencies of f_{in} . Note that both positive and negative values of l and f_{in} are required to generate all of the image frequencies. In most cases it is desired for the reconstruction network to remove the image frequencies while retaining the input frequency. As the input frequency approaches one-half of f_s , the input frequency and the lowest image frequency become very close, placing stringent requirements on the reconstruction network.

The requirements on the reconstruction network and the analog filter are reduced with an interpolating DAC (also called an over-sampling DAC), whose internal D/A section operates at a rate f_s that is a multiple of the input data rate f_{DATA} . A block diagram of an interpolating DAC is shown in Figure 6, as a particular case of the general scheme of Figure 5 (only interpolation is addressed here; nonetheless, additional functionalities from Figure 5 can be found in an interpolating DAC, as well):

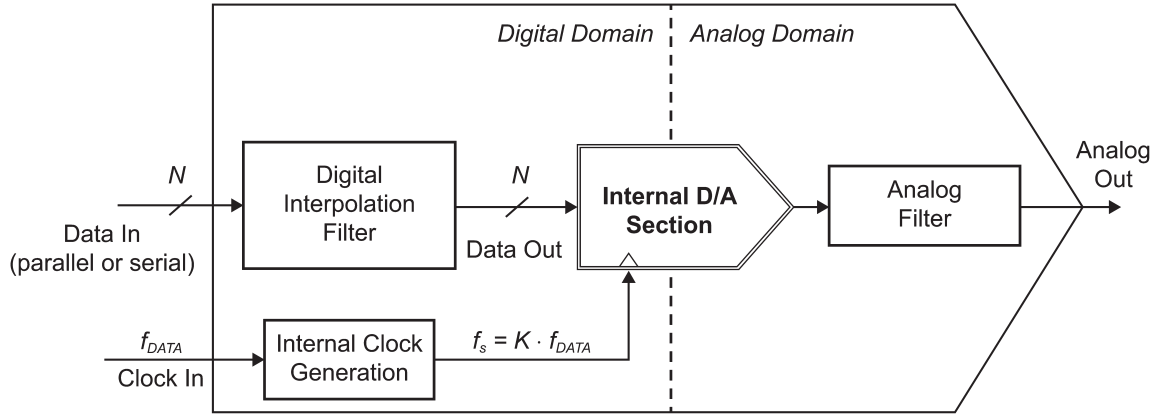


Figure 6—Block diagram for an interpolating DAC

The diagram in Figure 6 is a representation of a typical interpolating, or oversampling, DAC. If $K = 1$, then the DAC is a non-interpolating DAC, where the internal D/A section inputs are provided at a rate that coincides with the input data rate, i.e., $f_s \equiv f_{DATA}$. As in the case of Figure 5, when the term DAC is used without additional specification, it refers to the complete diagram, not to the internal D/A section. An interpolating DAC will usually have additional inputs to select parameters such as the oversampling factor K and the interpolation filter characteristics.

The interpolation filter is a digital block that converts the input word stream into a higher-rate digital time series that relaxes the design requirements of the reconstruction network and the analog filter.

The interpolation filter has an impulse response of the form shown in Equation (8).

$$f_i[n] = \sum_{k=-\infty}^{\infty} a_k \delta_{n, \frac{k}{K}} \quad (8)$$

where

- δ_{ij} is the Kronecker delta operator
- a_k is the parameters used to design the filter, which provide a frequency response that is periodic in frequency with period $f_s = K f_{DATA}$

In some sources it is stated that the image frequencies for an interpolating DAC are separated by f_s rather than by f_{DATA} , but this is incorrect. The interpolation filter block preserves digital image frequencies arising

from discrete-time operation at the same locations as those for a non-interpolating DAC with the same input data rate. However, the interpolation digital filter can be designed to reduce more effectively these image frequencies. Instead, it is true that the interpolation procedure gives rise to *additional* image frequencies around $\pm lf_s$, but these, too, can be even more effectively mitigated by the reconstruction network and the analog filter that follows in Figure 6.

This process can be better understood with frequency domain and time domain plots. Consider a PCM audio DAC with an input data rate of 44.1 kHz. The “Data in” data stream is set to generate 10 tones starting at 2 kHz and ending at 20 kHz. The desired spectrum of the waveform encoded by the “Data in” sequence is shown in Figure 7.

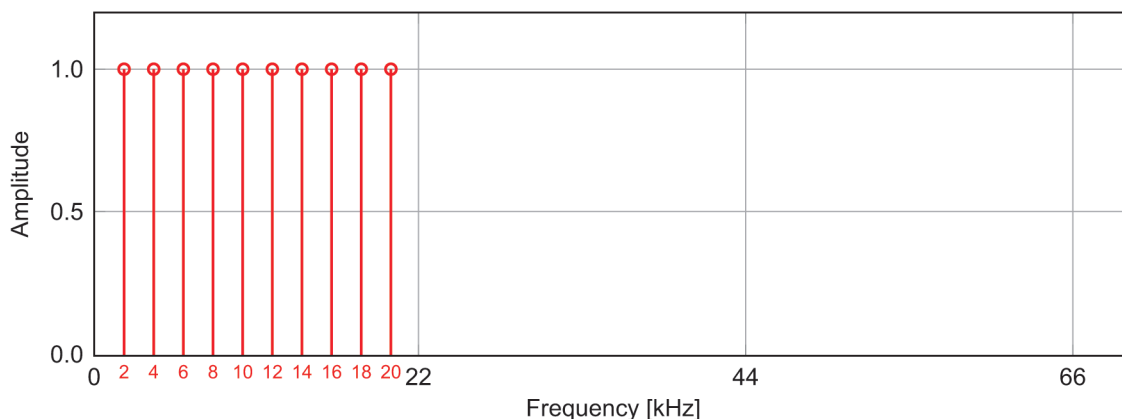


Figure 7—Spectrum of 10 tones of an audio signal that is applied to the DAC with an input data rate of 44.1 kHz shown in Figure 5 (normalized units)

If the factor K is set to 1, the zero-order-hold internal D/A section output will resemble the waveform shown in Figure 8 that is identified as “Unfiltered D/A section Output.” The desired output is also shown (without the time delay that is associated with the filtering process).

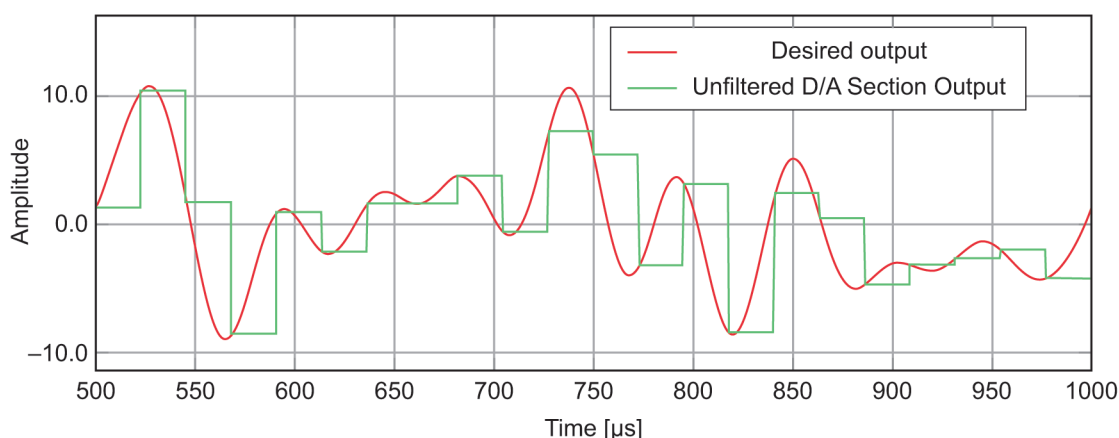


Figure 8—Plots of desired continuous signal and unfiltered D/A section output when $K = 1$ (exclusive of time delays) for the circuit shown in Figure 5

The spectrum of the analog waveform observed at the input of the analog filter will resemble that shown in Figure 9.

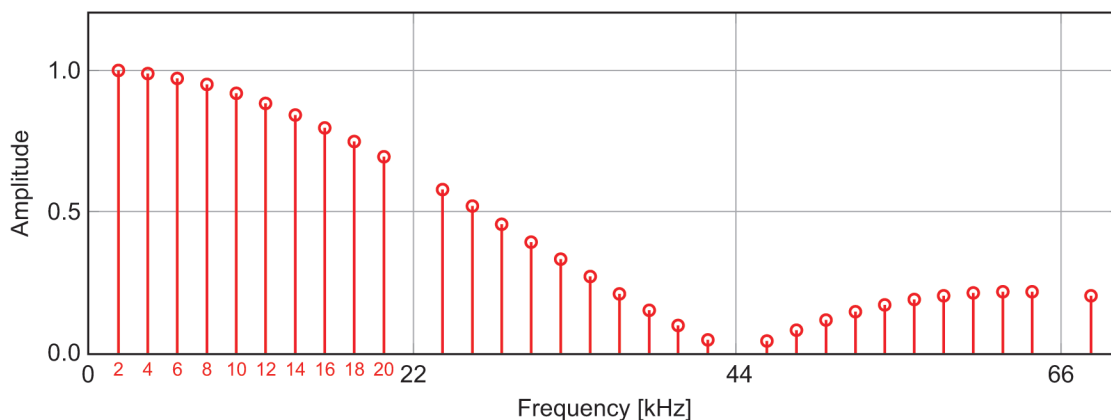


Figure 9—Spectrum of signal at the input to the analog filter for the circuit shown in Figure 5 when $K = 1$

The ideal analog output should have the same spectrum as that shown in Figure 7. In order to accomplish this, the analog filter should attenuate the unwanted image frequencies appropriately. It is very difficult to design an analog filter which will perform to the standards demanded by PCM audio users since the filter should implement (i) a precise post-emphasis to make the spectrum flat again in band and (ii) a sharp roll-off around 22 kHz.

For $K > 1$, a digital interpolation filter is used. It consists of an upsampling and a digital lowpass filter. If $K = 16$, the upsampling consists of adding $K-1 = 15$ zeros between each couple of adjacent samples. The digital lowpass filter should preserve the signal bandwidth and attenuate the image frequencies. The magnitude of the frequency response of a lowpass filter that can be used for this purpose is shown in Figure 10.

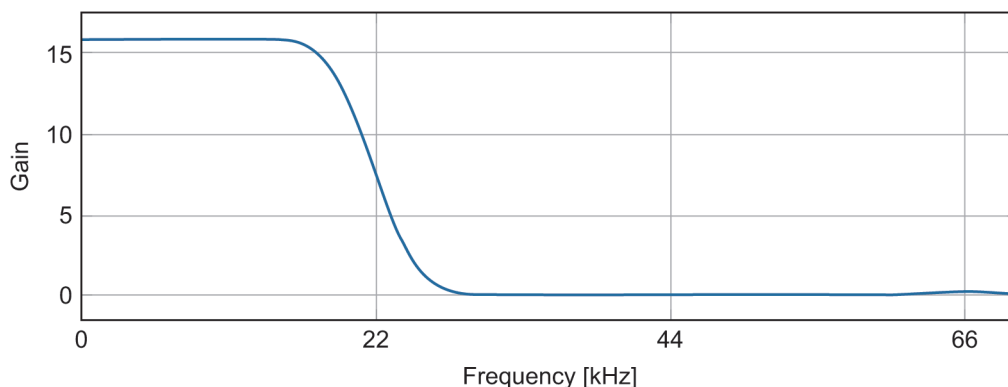


Figure 10—Magnitude of the frequency response of the lowpass digital filter for an interpolation of $K = 16$

The signal resulting from the interpolation is represented by the green line shown in Figure 11.

The time delays that are inherent in the reconstruction process are removed to clarify the description. It can be seen that the interpolated D/A section output is much closer to the desired output than that without interpolation (Figure 8).

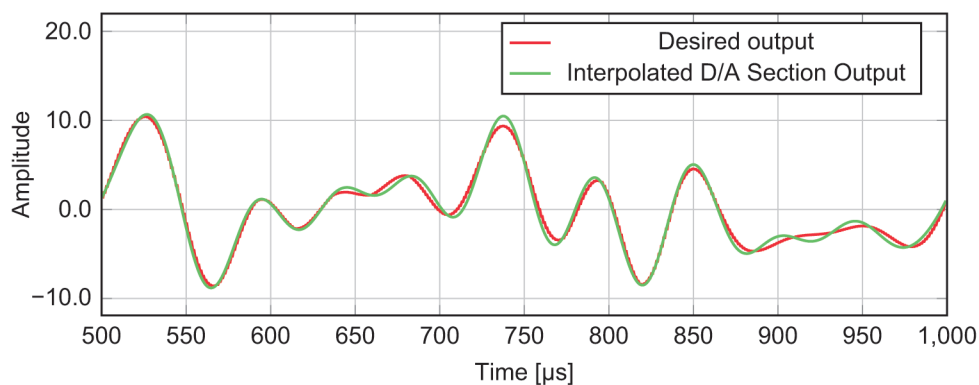


Figure 11—Plot of time domain signal for $K = 16$, and the desired output signal for the circuit shown in Figure 5 (time delays removed for clarity)

The spectrum of the signal generated downstream of the interpolation filter is shown in Figure 12. Note that $K = 16$ implies that the DAC update rate, f_s , is 16 times the input data rate of ~ 44.1 kHz or ~ 705.6 kHz. In the practice of DACs for high-performance audio applications K is not an integer, but it is slightly higher than 16, so that the update rate is 768 kHz.

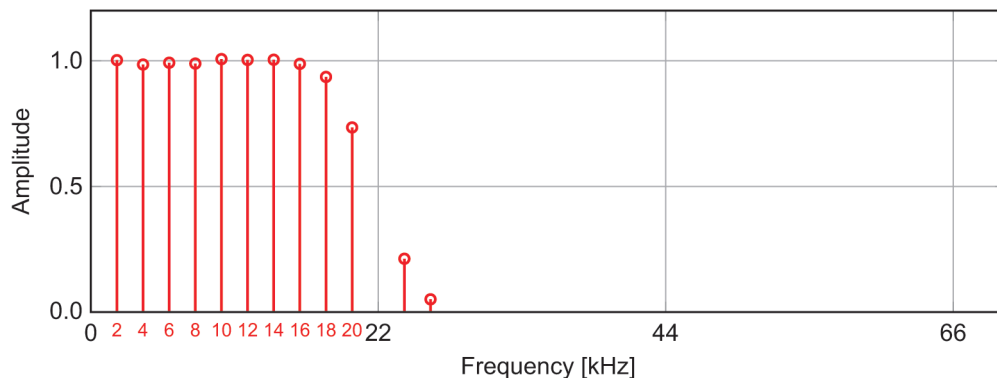


Figure 12—Spectrum of signal after the interpolation filter for the circuit shown in Figure 5

It can be observed that with the interpolation, the spectrum is flatter than without interpolation and that the image components are attenuated. However, due to the distortion of the filter, the spectrum may not be reasonably flat and some images can be still present. The analog filter can then be used to further remove the image components.

1.7 Guidance to the user

1.7.1 Interfacing

DACs present unique interfacing challenges, and without careful attention users can experience substandard results. As with all mixed-signal devices, DACs perform as expected only when the analog and digital domains

are brought together in a well-controlled fashion. The user should fully understand the manufacturer's recommendations with regard to proper signal buffering and loading, input signal connections, transmission line matching, circuit layout patterns, power supply decoupling, reference decoupling, and operating conditions. Edge characteristics for load-data pulse(s) and clock(s) should be carefully chosen to ensure that input signal purity is maintained with sufficient margin up to the digital input pin(s). Most manufacturers now provide excellent DAC evaluation boards, which demonstrate recommended layout techniques, signal conditioning, and interfacing for their DACs. If the characteristics of a new DAC are not well understood, then these boards should be analyzed or used before starting a new layout.

1.7.2 Coding

Some commonly used input data formats for digital-to-analog conversion are described in the paragraphs that follow.

Binary codes are most familiar in representing integers. In a natural binary integer code having N bits, the least significant bit (LSB) has a weight of 2^0 , the next bit has a weight of 2^1 , and so on up to the most significant bit (MSB), which has a weight of 2^{N-1} . The value of a binary number is obtained by adding up the weights of all nonzero bits. When the weighted bits are added up, they form a unique number having any value from zero to $2^N - 1$. Each additional trailing zero bit, if present, essentially doubles the size of the number.

In converter technology, because full scale is independent of the number of bits resolution, a more useful coding is fractional binary, which is always normalized to full scale. Integer binary can be interpreted as fractional binary if all integer values are divided by 2^N . The MSB has a weight of $1/2$, the next bit has a weight of $1/4$, and so forth down to the LSB, which has a weight of $1/2^N$. When the weighted bits are added up, they form a number with any of 2^N values, from zero to $(1 - 2^{-N})$, of the full scale.

With BCD, each decimal digit is represented by a group of four binary-coded digits (or “quad”).

In fractional BCD, the LSB of the most significant quad has a weight of 0.1, the LSB of the next has a weight of 0.01, the LSB of the next has a weight of 0.001, and so on. Each quad has 10 permissible levels with weights zero to nine. Group values in excess of nine are not permitted.

In Gray code, each bit represents a binary-weighted segment of the range and each code corresponds to a unique location in the range; but the bit weights do not readily combine to form a binary magnitude. In Gray code, as the number value changes, the transitions from one code to the adjacent code involve transitions of only one bit.

In complementary codes, all bits are represented by their complements. Complementary binary (or inverted binary) is the negative true binary system. It is identical to the binary code except that all binary bits are inverted. Thus, zero scale is all ones while full scale is all zeros. Complementary BCD, for each quad of a BCD coded converter, is the code obtained by representing all bits by their complements. In complementary BCD, zero is represented by 1111, and nine is represented by 0110.

For bipolar DACs, the most often used binary codes are as follows:

- Two's complement
- Sign-magnitude
- Offset binary
- One's complement

Two's complement code consists of a binary code for positive magnitudes (zero sign bit) and the two's complement of each positive number to represent its negative. The two's complement is formed arithmetically by complementing the code for the positive number and adding one.

One's complement code consists of a binary code for positive magnitudes (zero sign bit) and the one's complement of each positive number to represent its negative. The one's complement code for negative numbers is formed arithmetically by complementing the code for the positive number.

A sign-magnitude code determines the code appropriate for the magnitude of signed analog quantities and adds a polarity bit. The polarity bit can be either zero or one for positive values. Table 1 assumes a one for positive values.

In offset binary code, the weights are given by Equation (9).

$$w_m = 2^{m-1} \quad (9)$$

where

$$\begin{aligned} m & \text{ is } 1 \\ N & \text{ is } -1 \\ w_N & \text{ is } 2^{N-1} \end{aligned}$$

The offset binary code is widely used.

Note that the +FS output is represented by a virtual code that is 1 LSB greater than the maximum number that can be applied to the DAC. A similar phenomenon occurs at the −FS values for one's complement and sign-magnitude coding.

The parameter k represents the index of the input code. The total number of actual physical codes is 2^N for all the tabulated coding methods.

Table 1—Bipolar DAC input codes

Scale	k	Offset binary (and binary)	Two's complement	One's complement	Sign magnitude
+Full scale	2^N	—....—	—....—	—....—	—....—
+Full scale-1 LSB	2^N-1	1111....1111	0111....1111	0111....1111	1111....1111
+0.75 Full scale		1110....0000	0110....0000	0110....0000	1110....0000
+0.5 Full scale		1100....0000	0100....0000	0100....0000	1100....0000
+0.25 Full scale		1010....0000	0010....0000	0010....0000	1010....0000
+1 LSB		1000....0001	0000....0001	0000....0001	1000....0001
+0		1000....0000	0000....0000	0000....0000	1000....0000
−0		—....—	—....—	1111....1111	0000....0000
−1 LSB		0111....1111	1111....1111	1111....1110	0000....0001
−0.25 Full scale		0110....0000	1110....0000	1101....1111	0010....0000
−0.5 Full scale	2^{N-2}	0100....0000	1100....0000	1011....1111	0100....0000
−0.75 Full scale		0010....0000	1010....0000	1001....1111	0110....0000
−Full scale+ 1 LSB	1	0000....0001	1000....0001	1000....0000	0111....1111
−Full scale	0	0000....0000	1000....0000	—....—	—....—

1.7.2.1 Adjustment range

Many converter types require no adjustment as purchased to meet specified accuracy. However, there are cases where (i) adjustments are called for in the specifications, (ii) cost-savings can be affected by trimming inexpensive devices, with modest specs, for use over narrow ranges for better than specified accuracy, or (iii) long-term corrections are necessary after long periods of operation.

The adjustment range is defined by many DAC manufacturers, as the range that a recommended adjustment mechanism can be used to change the full scale range and the offset of a DAC. The adjustment mechanism can be used to correct for the gain and offset deviations from the nominal, or to compensate for other gain and offset errors in the system, or it can be used to change slightly the range of a DAC. A typical case of such a change would involve converting a ± 5 -V DAC to a ± 5.12 -V DAC. Adjustment of a DAC beyond the specified adjustment range may compromise the linearity and stability obtainable from the DAC.

The adjustment can be implemented through many mechanisms, including the following:

- a) Internal registers and DACs that can be programmed to adjust the gain and the offset.
- b) Nodes that can be connected to potentiometers to adjust the gain and the offset. The use of potentiometers is generally not recommended since the reliability of a potentiometer setting is reduced when many adjustments of parameters are made near the same region of a potentiometer setting.
- c) A node that can be connected to an external DAC to perform the functions described in mechanism a).

Proper adjustment of zero and full scale output in DACs is a procedure that requires great care and high-quality instruments. The meter used to read the output of a DAC shall have enough resolution so that the test instrument does not limit the adjustment value at both ends of the range of the converter, (at zero and full scale or minus full scale and plus full scale).

Bipolar converters require knowledge of the DAC architecture to be properly adjusted. Some adjustment mechanisms will adjust the offset at the negative full scale of the converter without affecting the range. Other adjustment mechanisms will have interaction between the gain and the offset adjustments.

Many bipolar binary converters that use offset binary or two's complement coding employ analog offsetting to convert a unipolar design into bipolar. For example, a zero to +10 V DAC may have its output amplifier offset by -5 V, resulting in an output of -5 V corresponding to 00...0 input and +5 V (minus 1 LSB) corresponding to a 11...1 input. Such a converter should have its "zero" adjusted at -5 V (100...0 in two's complement).

The specific instructions published in product datasheets should be followed.

It is not simple to calibrate a DAC based on the best-fit-straight-line algorithm. The best-fit approach is usually used to determine the worst-case INL error, where gains and offsets are not a factor. It is recommended that the adjustment of gain and offset be done at two specific codes in the range of the DAC. Typically, these codes correspond to minus Full Scale ($-FS$) and plus Full Scale-1 LSB ($+FS-1$ LSB). In some applications, the two points may be 20% of the FSR and 80% of the FSR.

1.7.3 Test conditions

DAC test specifications can be split into two groups: test conditions and test results. Typical examples of the former are temperature, power supply voltages, clock frequency, and reference voltages. Examples of the latter are power dissipation, effective number of bits, spurious free dynamic range (SFDR), and integral nonlinearity (INL). The test methods defined in this standard describe the measurement of test results for given test conditions.

DAC specification sheets will often give allowed ranges for some test conditions (e.g., power supply ranges). This implies that the DAC will function properly and that the test results will fall within their specified ranges for all test conditions within their specified ranges.

Since the test condition ranges are generally specified in continuous intervals, they describe an infinite number of possible test conditions, which obviously cannot be exhaustively tested. It is up to the manufacturer or tester of a DAC to determine from design knowledge and/or testing the effect of the test conditions on the test result, and from there to determine the appropriate set of test conditions needed to accurately characterize the range of test results. For example, knowledge of the design may be sufficient to know that the highest power dissipation (test result) will occur at the highest power supply voltage (test condition), so the power dissipation test need be run only at the high end of the supply voltage range to check that the dissipation is within the maximum of its specified range. It is very important that relevant test conditions be stated when presenting test results.

1.7.4 Test equipment

It shall be ensured that the performance of the test equipment used for these tests significantly exceeds the desired performance of the DAC under evaluation. Additional signal conditioning in the form of filters and pulse shapers may need to be included. Accessories such as terminators, attenuators, delay lines, and other such devices are usually needed to match signal levels and to provide signal isolation to avoid corrupting the output signal.

Quality testing requires following established procedures, most notably those specified in standard ISO 17025. In particular, traceability of instrumental calibration to a known standard is important. Commonly used test setups are described in [4.2](#).

1.7.5 Test selection

When choosing which parameters to measure, one should follow the outline and hints in this clause to develop a procedure that logically and efficiently performs all needed tests on each unique setup. The standard has been designed to facilitate the development of these test procedures. For example, most users require that the DAC meets or exceeds a minimum SINAD limit for their application and that the nonlinearity of the DAC be specified. Certainly, the extent to which this standard is applied will depend upon the application; hence, the procedure should be tailored for each unique characterization plan.

1.8 Manufacturer supplied information

1.8.1 General information

Manufacturers shall supply the following general information:

- a) Device part number
- b) Physical characteristics: dimensions, packaging, pin outs
- c) Power requirements
- d) Environmental conditions: safe operating, non-operating, and specified performance temperature range; altitude limitations; humidity limits, operating and storage temperature ranges; vibration tolerance; and compliance with applicable electromagnetic interference specifications
- e) Any special or peculiar characteristics
- f) Compliance with other specifications
- g) Control signal characteristics
- h) Output signal characteristics

- i) Pipeline delay (if any)
- j) Exceptions to the above parameters where applicable

1.8.2 Minimum specifications

The manufacturer shall provide the following specifications. See appropriate clauses for definitions.

- a) Number of bits N_{out}
- b) Input data rates and widths
- c) Analog bandwidth (minimum, maximum, typical)
- d) Output signal full-scale range with nominal reference signal levels
- e) Output impedance
- f) Output mode (current or voltage, differential or single-ended)
- g) Reference signal levels to be applied (minimum, maximum, typical)
- h) Supply voltages (minimum, maximum, typical)
- i) Supply currents (minimum, maximum, typical)
- j) Power dissipation (minimum, maximum, typical)
- k) Logic levels at digital inputs
- l) Number of channels

1.8.3 Additional specifications

- a) Gain error (minimum, maximum, typical)
- b) Offset error (minimum, maximum, typical)
- c) Differential nonlinearity (DNL) (minimum, maximum, typical)
- d) Harmonic distortion and spurious response (minimum, maximum, typical)
- e) Integral nonlinearity (INL) (minimum, maximum, typical)
- f) Unadjusted static error (minimum, maximum, typical)
- g) Signal-to-noise ratio (minimum, typical)
- h) Signal-to-noise and distortion ratio (SINAD) (minimum, typical)
- i) Total harmonic distortion (THD) (maximum, typical)
- j) Total harmonic distortion plus noise (THD+N) (maximum, typical)
- k) Spurious free dynamic range (SFDR) (minimum, typical)
- l) Effective number of bits (minimum, typical)
- m) Wideband effective number of bits (minimum, typical)
- n) Random noise and associated measurement bandwidth (maximum, typical)
- o) Frequency response
- p) Settling time (maximum, typical)

- q) Transition duration of step response (maximum, typical)
- r) Slew rate limit (minimum, maximum, typical)
- s) Timing uncertainty (short-term instability)
- t) Crosstalk (channel-to-channel, input-to-output, etc.)
- u) Monotonicity
- v) Intermodulation distortion (maximum, typical)
- w) Noise power ratio (minimum, typical)

1.8.4 Critical DAC parameters

Table 2 is presented as a guide for a few common DAC applications. The wide range of DAC applications makes a comprehensive listing impossible. This table is intended to be a helpful starting point for users to apply this standard to their particular applications.

Table 2—Critical DAC parameters

Typical applications	Critical DAC parameters	Performance issues
Audio	SNR, THD, SINAD (THD+N)	Audio quality
Video/RAMDAC	GE, DNL, INL, DG, DP, OV, SFDR, TR, CM, ST	Picture quality
Control	DNL, T_s , GE	DNL monotonicity is often critical
Communications	IMD, MTPR, ACLR	Communications quality of service

ACLR—adjacent channel leakage power ratio
 CM—channel matching
 DNL—differential nonlinearity
 ENOB—effective number of bits
 GE—gain error
 IMD—intermodulation distortion
 INL—integral nonlinearity
 OV—output voltage compliance range
 MTPR—Missing Tone Power Ratio
 SINAD—signal-to-noise and distortion ratio
 SFDR—spurious free dynamic range
 SNR—signal-to-noise ratio
 T_s —settling time
 THD—total harmonic distortion
 THD+N—THD plus noise
 TR—transient response

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they shall be understood and used, so each referenced document is cited in text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEEE Std 1057™, IEEE Standard for Digitizing Waveform Recorders.^{9,10}

IEEE Std 1241™, IEEE Standard for Terminology and Test Methods for Analog-to-Digital Converters.

⁹IEEE publications are available from The Institute of Electrical and Electronics Engineers, Inc., 445 Hoes Lane, Piscataway, NJ 08854, USA (<https://standards.ieee.org/>).

¹⁰The IEEE standards or products referred to in this clause are trademarks of The Institute of Electrical and Electronics Engineers, Inc.

IEEE Std 181™, IEEE Standard for Transition, Pulses, and Related Waveforms.

ISO 17025, General requirements for the competence of testing and calibration laboratories.¹¹

MIL Std 810F, Environmental Engineering Considerations and Laboratory Tests.¹²

3. Definitions, symbols, acronyms, and abbreviations

3.1 Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.¹³

NOTE—The linearity specifications to follow (gain, offset, INL, DNL) depend on which transfer curve is used as a reference. One can have either “end-point” linearity or “best-fit” linearity. The first takes as reference curve a straight line between the minimum and maximum measured points. The second uses as reference a best-fit regression line through all measured points. Since “best-fit” linearity seems to be the most widely used, that definition is used as the standard. The main advantage of “best-fit” linearity is that it reduces the linearity error (INL) and thus in general will result in higher parametric yield. End-point linearity is relevant when post-processing does not allow a linear regression.¹⁴

absolute worst-case (total) error: For a device that uses an internal reference, or a specified external reference, the absolute value of the worst-case difference between the actual output value and the nominal output value for any input code, without gain, and offset correction (see Figure 13).

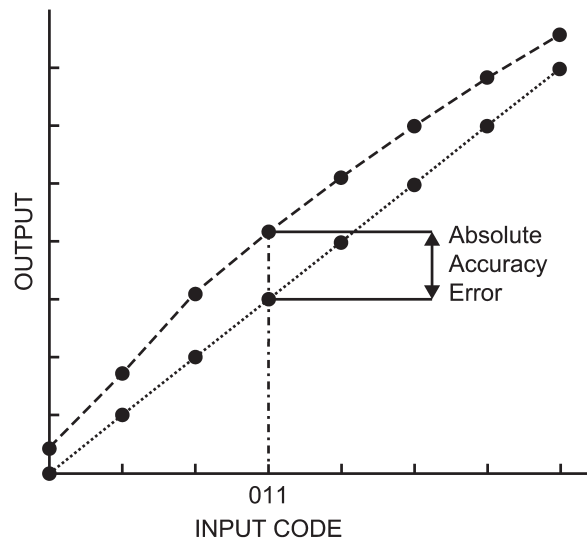


Figure 13—Absolute worst-case error of a DAC

¹¹ISO publications are available from the ISO Central Secretariat, 1, ch. de la Voie-Creuse, Case postale 56, CH-1211 Geneva 20, Switzerland (<http://www.iso.ch/>). ISO publications are also available in the United States from the Sales Department, American National Standards Institute, 25 West 43rd Street, 4th Floor, New York, NY 10036, USA (<https://www.ansi.org/>).

¹²MIL publications are available from Customer Service, Defense Printing Service, 700 Robbins Ave., Bldg. 4D, Philadelphia, PA 19111-5094.

¹³*IEEE Standards Dictionary Online* is available at: <http://dictionary.ieee.org>. An IEEE account is required for access to the dictionary, and one can be created at no charge on the dictionary sign-in page.

¹⁴Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

adjacent channel leakage ratio (ACLR): A ratio in dB between the measured power within a frequency channel relative to adjacent frequency channels.

bandwidth: The frequency at which the magnitude of the Fourier transform of the digital-to-analog converter's (DAC's) impulse response computed from the time derivative of its full-scale step response, is attenuated by $1/\sqrt{2}$ from its dc value. This definition of bandwidth is sample-rate independent and does not account for the $\sin(\pi fT)/(\pi fT)$ response arising from zero-order-hold waveform reconstruction.

channel matching (when applicable): Specifies the degree of matching between two or more digital-to-analog converters (DACs) in a single package.

clock and data feedthrough: Specifies the amount of energy in the digital-to-analog converter (DAC) output originating from clock and/or data signals respectively.

clock jitter: The standard deviation of variation in time of the active edge of a clock signal.

clock pulse width (latch pulse width): The minimum amount of time that the clock pulse should be left in the logic "1" state to achieve rated performance.

clock signal duty cycle: The fraction of the time the clock signal spends in excess of the logic threshold.

clock signal slew rate: The time derivative of the clock signal at the point where it crosses the logic threshold.

code bin (k): A digital input that corresponds to a particular nominal analog output value.

crosstalk: Undesired energy appearing in a digital-to-analog converter (DAC) channel as a result of coupling with another DAC channel in the same package.

cumulative distribution function (cdf): The integral of the probability density function defined in the definition for **output propagation delay**. This function is positive and monotonic.

differential nonlinearity (DNL): The differences between the output at adjacent input codes, divided by the measured average step width of the digital-to-analog converter (DAC) under test minus 1.

differential-output impedance: For a differential-mode output digital-to-analog converter (DAC), the impedance between the positive output and the negative output.

digital feedthrough: This is the glitch transferred to the output of a digital-to-analog converter (DAC) channel in response to a change in an input. It includes data feedthrough and clock feedthrough.

digital-to-analog converter (DAC): A converter that translates a digital input into an analog output.

dynamic range: *See:* **spurious free dynamic range (SFDR)**.

effective number of bits (ENOB): A measure of the signal-to-noise-and-distortion ratio used to compare actual digital-to-analog converter (DAC) performance to the expected performance of an ideal DAC.

full-scale (FS) errors: The difference between the actual digital-to-analog converter (DAC) output and the expected nominal output when a digital code is applied, which should generate the maximum and the minimum outputs. Unlike gain error, offset error is included in the FS error measurement.

full-scale range (FSR): The difference between the most positive and most negative analog outputs of a converter's operating range. Generally, FSR includes the virtual codes at the minimum and the maximum outputs. Thus, a -10 V to $+10\text{ V}$ digital-to-analog converter (DAC) will have FSR equal to 20 V .

gain and offset: Gain and offset are the values by which the ideal output values are multiplied and then to which the ideal output values are added, respectively, to cause the output values to satisfy specified conditions. Two conditions are commonly used: “independently based” where the sum of the squared deviations from the ideal output values is minimized, and “terminal-based” where the ideal outputs are matched at the first and the last code.

gain matching: In a multichannel converter, the difference between the largest gain and the smallest gain, expressed in % FSR or dB.

glitch: A transient that leaves an initial state, enters the boundaries of another state for a duration less than the duration for state occurrence, and then returns to the initial state.

glitch area (GA): The time integral of the analog value of the glitch transient.

glitch energy (GE): The time integral of the electrical power of the glitch transient.

harmonic: A frequency that is an integer multiple of the fundamental.

harmonic distortion: For an assumed sine-wave signal, the amplitudes of the components at frequencies that are multiples of the signal frequency by integers greater than one.

hold time (t_H): The amount of time required for the digital inputs to remain deterministic after a trigger clock edge.

ideal code bin width (Q): The ideal full-scale output range (FSR) divided by the total number of codes.

impulse: A time signal that approximates a Dirac delta distribution.

impulse DAC: An idealized digital-to-analog converter (DAC) for which the output is a sequence of impulses proportional to the input word at the update rate.

impulse response: Response of a linear network to an impulse applied at the input.

input data rate (f_{DATA}): The rate at which input words are supplied to the digital-to-analog converter (DAC).

integral nonlinearity at code bin (k) (INL(k)): The difference between the ideal and actual analog output value corresponding to the code bin (k) after correcting for gain and offset.

integral nonlinearity (INL): The maximum difference between the ideal and actual analog output values after correcting for gain and offset.

intermodulation distortion (IMD): The ratio of amplitude of the intermodulation product with the largest amplitude to the largest amplitude generated test tone.

interpolation filter: The filter in an interpolating digital-to-analog converter (DAC) that converts the digital data at the input data rate to digital data at the update rate.

interpolation ratio: The ratio of the update rate to the input data rate in an interpolating digital-to-analog converter (DAC).

jitter: The deviation of a sequence of reference instants from their ideal values.

least significant bit (LSB): Synonymous with one ideal code bin width (Q), which corresponds to the digital input bit in a data converter code that has the smallest mathematical weight.

major carry transition: The midscale point where the MSB changes from low to high and all other bits change from high to low, or where the MSB changes from high to low and all other bits change from low to high. This point is often where the worst switching noise occurs.

monotonic digital-to-analog converter (DAC): A DAC that has output values that do not decrease (increase) for a uniformly increasing (decreasing) input codes, disregarding random noise. *See:* **monotonicity**.

monotonicity: A property of the transfer curve that helps to ensure the consistent increase or decrease of the analog output in response to a consistent increase or decrease of the digital input. (Figure 14 illustrates non monotonic conversion). There are some data converter architectures that are monotonic by design, e.g., thermometer-coded digital-to-analog converters (DACs).

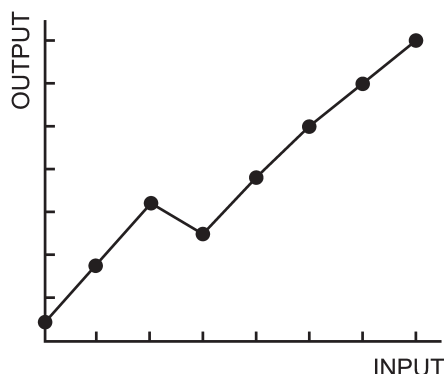


Figure 14—Transfer curve of a non monotonic DAC

most significant bit (MSB): The digital input bit in a data converter code that has the greatest mathematical weight.

multiplying DAC: A digital-to-analog converter (DAC) that allows a signal to be applied to the reference input pin to scale the DAC output.

multitone power ratio (MTPR): The spurious free dynamic range containing multiple carrier tones of equal amplitude. It is measured as the difference between the rms amplitude of the largest carrier tone to the peak spurious signal in the region of a removed tone.

noise: Any deviation between the desired nominal output signal and the actual output signal except deviations caused by linear time invariant system response (gain and phase shift), a dc level shift, total harmonic distortion, or an error in the applied update rate. For example, noise includes the effects of random fluctuations, nonlinearities producing harmonics at frequencies greater than those used in measuring total harmonic distortion, apparent quantization error, and spurious signals.

noise and distortion (NAD): The sum of the effects of noise and total harmonic distortion.

noise power ratio (NPR): It is the measure of the spectral power of all contributed errors, such as intermodulation and harmonic distortion, in a narrow frequency-slot within the baseband of the composite signal being processed.

noise spectral density: The digital-to-analog converter (DAC) output noise normalized to a 1 Hz bandwidth.

offset: *See:* **gain and offset**.

offset binary coding: For bipolar signals; offset binary is a digital coding scheme in which the most negative value is represented by all zeros (00000000) and the most positive value is represented by all ones (11111111).

output compliance range: The range of allowable voltage at the output of a current output digital-to-analog converter (DAC). Operation beyond the maximum compliance limits may cause either output stage saturation or breakdown resulting in nonlinear performance.

output propagation delay (t_{PD}): The delay between the 50% point of the clock transition and the time when the DAC output traverses 50% of the output transition. The propagation delay includes the effect of digital pipelines and analog filtering.

pass band: Frequency band in which a sine wave waveform generated by a digital-to-analog converter (DAC) is not significantly attenuated.

phase noise: Random fluctuations in the phase of a periodic signal.

pipeline delay: A measure of the latency in terms of digital-to-analog converter (DAC) clock cycles between the clock transition that initiates sampling of the digital input and the presentation of the output value of that sample.

power-supply rejection (PSR): The amount of change in the converter's output (often measured at full scale), as the power-supply voltage is changed from its nominal value.

power-supply rejection ratio (PSRR): The ratio of the amount of change in the converter's output (often measured at full scale), relative to the change of the power-supply voltage.

power-supply sensitivity (PSS): Synonymous with power-supply rejection (PSR).

probability density function (pdf): For a continuous random variable, the function $f_X(x)$, which has the interpretation that $f_X(x)dx$ is the probability that the random variable, X , lies in the interval $(x, x + dx)$.

quantization: The mathematical operation of truncating or rounding a value of arbitrary precision to a lower fixed resolution.

quantization error: The error caused by conversion of a variable having a continuous range of values to a quantized form having only discrete values, as in analog-to-digital conversion. The error is the difference between the original value and its quantized representation translated into input units. A digital-to-analog converter (DAC) does not generate quantization error. The error that is caused by the resolution limitation of a DAC is called the truncation error.

quantization noise: Equivalent noise produced during the quantization of a continuous range variable. A quantization process of step size Q produces an rms quantization noise of $Q/\sqrt{12}$ when the continuous variable range is much larger than Q .

random noise: A non-deterministic fluctuation in the output of a digital-to-analog converter.

reconstruction network: The equivalent circuit that relates the input data sequence of an internal D/A section of a DAC to its output signal.

record of data: A sequential collection of samples.

relatively prime: Describes integers whose greatest common divisor is 1.

residuals: In curve fitting, the differences between the recorded data and the fitted function.

resolution: The number of output levels the digital-to-analog converter (DAC) is designed to produce. Usually stated as the number of **bits** used (the base two **logarithm** of the number of levels). A 1-bit DAC is designed to produce 2 (2^1) levels while an 8-bit DAC is designed to produce 256 (2^8) levels.

root-mean-square (rms): For a given set of data, the square root of the arithmetic mean of the squared values of each of the data.

root-sum-square (rss): For a given set of data, the square root of the sum of the squared values of each of the data.

sampling: The process of assigning discrete time values to a continuous time signal.

settling time interval: The time at which the step response enters and subsequently remains within a specified error band around the final value, measured from the 50% reference level instant of the response. The final value is defined to occur 1 s after the beginning of the step unless otherwise specified.

setup time interval (t_{SU}): The amount of time required for the digital inputs to remain deterministic prior to a leading transition of the update signal.

short-term settling time interval: Measured from the 50% point of the output, the time at which the step response enters and subsequently remains within a specified error band around the final value. The final value is defined to occur at a specified time less than or equal to one second after the beginning of the step.

signal-to-noise-and-distortion ratio (SINAD): For a pure sine-wave input of specified amplitude and frequency, the ratio of the root-mean-square (rms) amplitude of the digital-to-analog converter (DAC) filtered reconstructed output sine wave to the rms amplitude of the output noise and distortion.

signal-to-noise ratio (SNR): For a pure sine-wave input of specified amplitude and frequency, the ratio of the root-mean-square (rms) amplitude of the digital-to-analog converter (DAC) filtered reconstructed output sine wave to the rms amplitude of the output noise (excluding the harmonics).

signed binary coding: A coding scheme in which the MSB represents the sign (positive or negative) of a binary number. In this scheme, with an 8-bit word, “–2” is represented by 10000010 and “2” is represented by 00000010.

slew rate: The maximum rate at which the digital-to-analog converter (DAC) output can change without resulting in a deviation from the response expected from a linear system.

spurious components: Persistent sine-wave components at frequencies other than harmonic frequencies of the fundamental.

spurious free dynamic range (SFDR): Specifies the available signal range as the difference in magnitude between the amplitude of the fundamental and the amplitude of the largest spurious component in the frequency band of interest.

NOTE—Some industry spec sheets include harmonics along with spurious components in their specification of SFDR. It is important to specify for any SFDR spec whether or not the harmonics are included.

stop-band rejection: The amount of attenuation of a frequency outside the pass band applied to the digital-to-analog converter (DAC) relative to a full-scale signal applied at the DAC input within the pass band.

total harmonic distortion (THD): The ratio of the square root of the sum of the squares of the rms amplitudes of the harmonic components of the output sine wave (i.e., of sinusoidal components appearing at multiples of

the fundamental output sine wave frequency) to the rms value of the output sine wave. It is typically expressed in decibels or percent.

total harmonic distortion plus noise (THD+N): The ratio of the square root of the sum of the squares of the amplitudes of all components of the output signal (excluding the fundamental) to the amplitude of the fundamental output over a specified frequency band. It is typically expressed in decibels or percent. It is similar to THD, but also includes the noise.

transfer curve: The curve that describes the relationship between the digital input and the analog output.

transient response: The measured output response to an input code change having designated state levels corresponding to a full-scale step signal. Transient response includes the combined behavior of transition duration, aberrations, and settling error.

transition duration (t_d): The difference between the two reference level instants of the same transition. Unless otherwise specified, the two reference levels are the 10% and 90% reference levels [B11].

NOTE 1—See Figure 15 and Figure 16.

NOTE 2—The following are deprecated terms: risetime, falltime, leading edge, rising edge, trailing edge, falling edge, and transition time. The terms risetime, falltime, and transition time, although widely used, are deprecated because they are ambiguous and confusing. First, the use of the word time in this standard refers exclusively to an instant and not an interval. Also, if the first transition of a waveform within a waveform epoch (IEEE Std 181) happens to be a negative transition, some users may refer to its transition duration as its risetime, and some others may refer to its transition duration as its falltime. If the use of these deprecated terms is required, then risetime is synonymous with the transition duration of a positive-going transition, and falltime is synonymous with the transition duration of a negative-going transition. If the upper and lower state boundaries of the two states are not the user-defined reference levels (e.g., the 10% and 90% reference levels), then the duration of a transition is not equal to the transition duration.

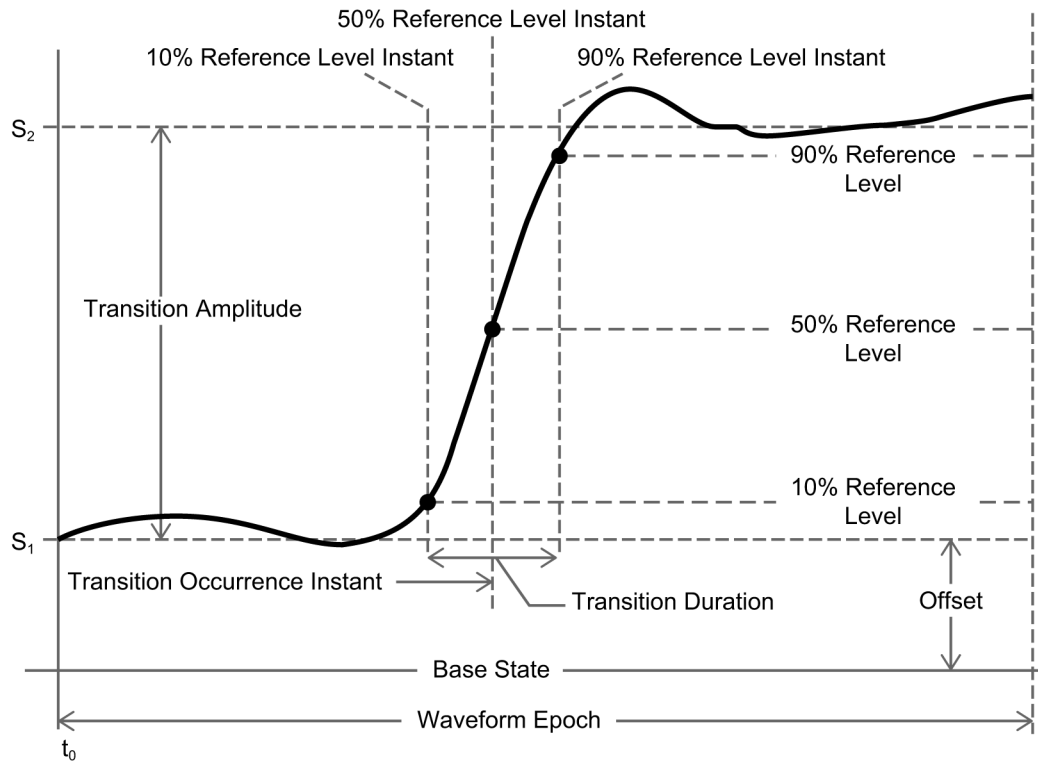


Figure 15—Positive single transition

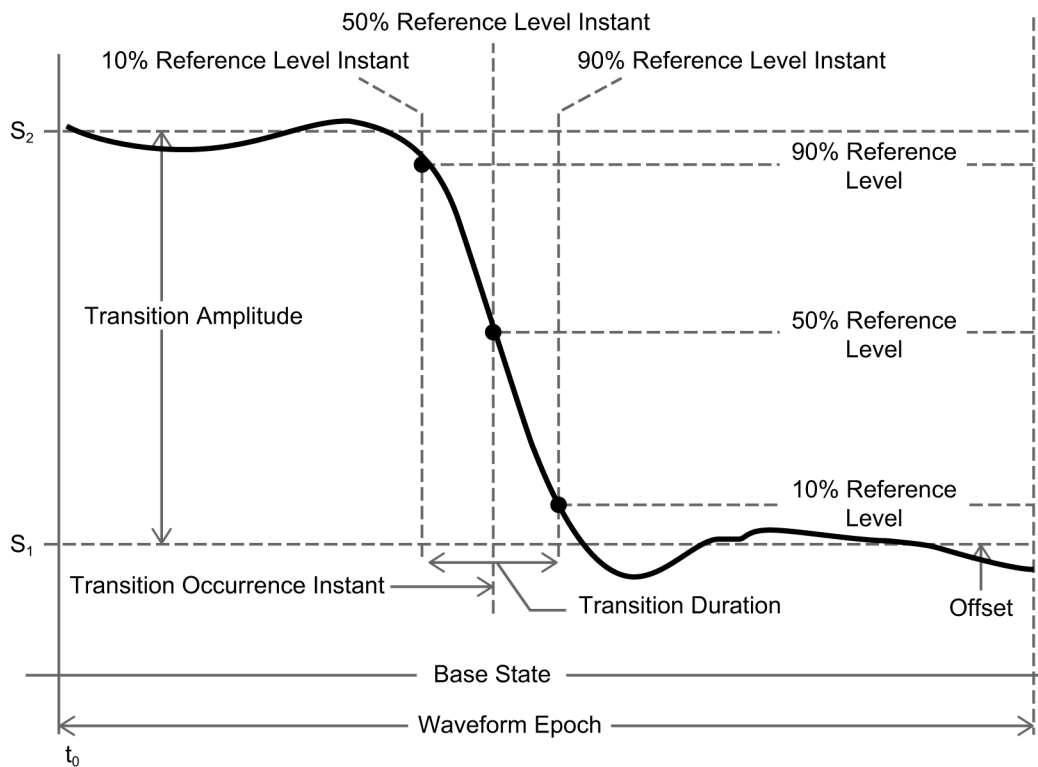


Figure 16—Negative single transition

two tone intermodulation distortion (IMD): The ratio of the rms amplitude of either of two equal-amplitude input tones to the rms amplitude of the worst third order intermodulation product; reported in dBc.

two tone SFDR: The ratio of the rms amplitude of either equal-amplitude input tone to the rms amplitude of the peak spurious component. The peak spurious component may or may not be an intermodulation distortion (IMD) product. It also may be reported in dBc (i.e., degrades as signal level is lowered) or in dBFS (i.e., always relates back to converter full scale).

unipolar: A signal that swings from zero to the positive full-scale, thus having only a positive polarity.

update rate (f_s): The rate at which analog values are updated at the output of the digital-to-analog converter (DAC).

waveform: A representation of a signal (e.g., a graph, plot, oscilloscope presentation, discrete time series, equations, or table of values). This term refers to a measured or otherwise-defined estimate of the physical phenomenon or signal.

waveform epoch: An interval to which consideration of a waveform is restricted for a particular calculation, procedure, or discussion. Except when otherwise specified, it is assumed to be the span over which the waveform is measured or defined.

zero-order-hold DAC: An ideal digital-to-analog converter (DAC) that produces an analog output proportional to its input word immediately after the data clock pulse corresponding to the word and maintains this output value until the next data clock pulse. This is mathematically equivalent to an impulse DAC followed by a filter whose impulse response is a constant proportional to the value of the input impulse for one data clock pulse duration and zero elsewhere.

3.2 Symbols

$\delta(t)$	Dirac delta distribution
δ_{ij}	Kronecker delta operator
ε	error, used for total error and error band
$\varepsilon[k]$	difference between $U[k]$ and ideal value of $U[k]$ computed from G and V_{os}
$\varepsilon_G(f)$	gain flatness error of frequency f ; ϕ = phase, expressed as radians
π	constant, ratio of the circumference of a circle to the diameter
σ	standard deviation; sometimes used as a measure of noise amplitude, which is the standard deviation of the random component of a signal
σ_t	standard deviation of the jitter
$\omega = 2\pi f$	angular frequency, expressed in radians per second
ω_{in}	angular input frequency in radians/second
$\omega_s = 2\pi f_s$	update angular frequency
Δt	update period, the inverse of f_s
ΔV_{out}	change in the rms output level
ΔV_{PS}	change in the power supply voltage
a	real constant
A	sinusoidal amplitude

B	sinusoidal amplitude
$B W_m$	measurement bandwidth
C	offset
$DNL[k]$	differential nonlinearity of code k
DNL_{RMS}	rms value of the DNL
$e_{noise\ rms}$	rms value of the DAC output corresponding to an input code that nominally generates a zero output
f	frequency
$f[n]$	sine-wave component of output sample $y[n]$
f_{DATA}	the input data rate of the DAC
f_{dig}	digitizer sample rate
f_h	frequency of a harmonic of the input frequency
$f_l[n]$	impulse response of the interpolation filter
f_{in}	actual input frequency or approximation of the desired input frequency or frequency of interest
f_m	the m -th frequency component of a magnitude spectrum produced by a DFT
f_{ref}	reference frequency
$f_R(t)$	impulse response of the reconstruction network implemented in the internal D/A section of the DAC
$\hat{f}_R(\omega)$	frequency response of the reconstruction network implemented in the internal D/A section of the DAC
f_s	update rate
f_t	test frequency
FSR	DAC Full Scale Range
G	static gain of the DAC under test
$G(f)$	dynamic gain of the DAC under test, as a function of frequency f
$h[n]$	discrete time impulse response of a system
$INL[k]$	integral nonlinearity at output code k
i	general purpose index
J	number of periods in a waveform
k	code bin
L	general purpose integer
l	general-purpose index
M	number of sequential samples in a record
M_{test}, N_{test}	number of cycles and number of samples in the digitized test signal, respectively
N_{dig}, M_{dig}	number of cycles and number of samples acquired of the analog test output, respectively
N	number of digitized bits (note that for certain DACs, N may not be an integer value)

n	sample index within a record
Q	ideal code bin width, expressed in input units
R	general purpose integer
$s(t)$	DAC step response
$T_{\text{jitter, rms}}$	rms jitter of digitizer clock
T_G	glitch duration
T_S	signal period
t_{DATA}	reciprocal of f_{DATA}
t_d	transition duration
t_n	discrete sample times
$U[k]$	analog output value corresponding to the digital input code k
U_0	ideal value corresponding to $U[0]$
V_G	analog value of the glitch transient
V_{OS}	output offset of DAC
$V_{\text{in}}(t)$	analog input voltage
$\hat{V}_{\text{in}}(\omega)$	Fourier transform of the analog input signal
$V_{\text{out}}(t)$	analog output voltage
$\hat{V}_{\text{out}}(\omega)$	Fourier transform of the analog output signal V_{ripple} = rms ripple voltage added to power supply
V_{PS}	power supply voltage
$W[k]$	corrected analog output difference for code bin k , or $G(U[k+1] - U[k])$
$w[n]$	window function coefficient (for a DFT)
w_k	the value of the word at the input of the internal D/A section $X_{\text{avg}}[m]$ = the averaged spectral magnitude at the m -th frequency component f_m after a DFT
$y[n] = y_n$	the n -th output data sample within a record
$y_{n'}$	best fit points to a data record

3.3 Acronyms and abbreviations

CMR	common mode range
DNL	maximum differential nonlinearity over all k
ENOB	effective number of bits
ENBW	equivalent noise bandwidth
FS	full-scale
FSR	full-scale range
IMD	intermodulation distortion
INL	integral nonlinearity
MTPR	multi-tone power ratio

N	number of digitized bits (note that for certain DACs, N may not be an integer value)
NAD	noise and distortion
NPR	noise power ratio
PSR	power supply rejection
PSRR	power supply rejection ratio
SINAD	signal-to-noise-and-distortion ratio
SNR	signal-to-noise ratio
THD	total harmonic distortion
THD+N	total harmonic distortion plus noise
THE	total harmonic energy
VSWR	voltage standing wave ratio

4. Test methods

4.1 General comments on test methods

It is assumed that the user/manufacturer has defined safe operating limits for the device under test. These limits are categorized as absolute (the limit beyond which the device will be destroyed) and operating (the limit beyond which the device will not operate properly). These limits will vary from device to device depending on the design. It is not the intention of this document to describe the method of setting these limits, only to verify the operation within them.

All test procedures described herein apply only to parameters of a device that is operated within its specified limits.

4.2 Test setup

Figure 17 depicts the generic form of the test setup used for testing DACs. DAC testing can be differentiated into static and dynamic. The first is basically a sequence of dc voltage acquisitions using either a digital voltmeter (DVM) or a digital waveform recorder. The equipment used should have significantly better accuracy and resolution than the specified limits. If repeatability is a significant component of the inaccuracy, the accuracy can be improved by averaging multiple repeated acquisitions. If lack of repeatability is due to independent random errors in each acquisition, the associated uncertainty contribution reduces by a factor $\frac{1}{\sqrt{L}}$ where L is the number of acquisitions taken. Where appropriate, the usual precautions shall be taken in board design such as decoupling close to the DUT, separate analog and digital ground planes, or very low impedance single ground planes, separate analog and digital power supplies and shielding of low-level analog signals from noisy digital signals or from coupling with switching power supplies.

For dynamic testing, coherent sampling test conditions are recommended. This means that all signals in the system are frequency locked with one another. These include the sine-wave array of DAC input data codes generated by the digital pattern generator and its clock, and the digitizer clock (f_{dig}). The record of N_{dig} samples acquired by the digitizer should cover an integer number M_{dig} of periods of the digital sine wave generated by the pattern generator.

The resulting analog output is fed through an interface to a digitizer. The interface and/or digitizer may contain some analog pre-processing to make the signal suitable for measurement. For instance, if the dynamic range of the test signal exceeds that of the digitizer, it may be necessary to compress this range through notch filtering or some other means. Some level shifting may be needed to shift the resulting signal within the measurement

range of the digitizer. Ideally, this may reduce the rms amplitude of the fundamental in the test signal by a given amount and may have to be corrected by a post-processing stage. A notch potentially creates a problem because it may mask close-in sidebands that may be present in the DAC's output, and the technique limits the choice of test frequency because the notch filter can usually only accommodate a small band of frequencies around the notch. Many state-of-the-art testers have sufficient dynamic range in their digitizer to accommodate test signals with a (very) high dynamic range directly.

Filtering is usually a part of the test signal conditioning preceding the digitizer. This filtering step then determines the measurement bandwidth BW_m . If the required measurement bandwidth exceeds the Nyquist bandwidth (i.e., $= f_{\text{dig}}/2$) of the digitizer, the aliasing produced by the digitizer may be used to extract the wanted information from the folded spectrum. Such measurements require the timing of the samples to be carefully placed with respect to the DAC output transitions. This may also mask potential low-frequency components generated by non-repetitive glitch energy. For systems containing DACs, such as arbitrary waveform generators, some of the blocks in Figure 17 are included in the device under test.

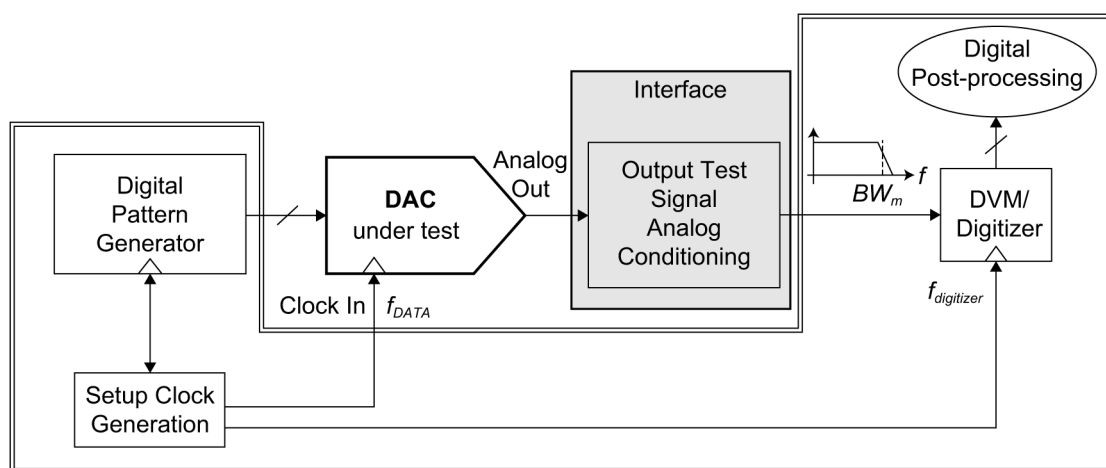


Figure 17—Test set-up for DACs

For the above setup, the following parameters should be considered:

- f_{DATA} : DAC input data rate
- f_i : frequency of the test sine-wave provided by the pattern generator
- f_{dig} : digitizer sample rate
- $M_{\text{dig}}, N_{\text{dig}}$: number of test signal periods and number of samples acquired by the digitizer
- BW_m : measurement bandwidth

4.3 Static testing

Static parameter testing is implemented by way of a system similar to that shown in Figure 17.

For static parameter testing, the following steps should be followed:

- 1) Select the codes that will be tested.

- 2) Apply successive codes. For all-codes testing, the codes should proceed from the code that corresponds to negative full scale up to the code that corresponds to positive full scale and/or vice versa. At each point allow the output to settle, and the digitizer to acquire the DAC output value of voltage or current. The value at code k should be stored in the array U , at location $U[k]$.

In the following it is assumed that the DAC does not suffer from failure modes or hysteresis effect. Therefore, the digital codes can be applied in any order, for instance, as it happens in most cases, in increasing order. However, for some DAC devices, it may be desired to scramble the codes so that repeatability of transfer curve can be assessed. When averaging needs to be done to improve accuracy, the system should either average several acquisitions at each step, or the total set of input codes can be applied multiple times. Once all the acquisitions are done, processing of the data may be performed by fitting a least-squares regression curve through the acquired points and subsequently calculating the linearity errors. Alternatively, an end-point calibration can be performed, and the linearity error can be computed from the deviation of U from the line connecting the end points.

NOTE—The user may decide to not test all codes but may opt to test only codes around major bit transitions. In either case, the worst-case codes should be determined and tested.

4.3.1 Considerations about static testing for high-resolution DACs

In order to make a dc measurement of a DAC output when it changes from one input code to another, it is first necessary to set a limit on the confidence or precision expected in the result. It is usually convenient to assume that the DAC has a first order response to a code change and has a time constant that can be calculated and measured. On this basis, it is possible to define the expected precision in output voltage in terms of the number of time constants required.

It follows that the largest output change, i.e., full scale, will require the longest time to settle to the required voltage. It is necessary for the input voltage applied to the first order network (i.e., the reference voltage) to be sufficiently stable during the DAC settling time so that the output voltage is measured with sufficient precision. If the reference voltage drifts with time, then clearly this will limit the maximum number of time constants that can be employed in the measurement process.

In order to test all the codes of a high-resolution DAC it is first necessary to produce a code test sequence that takes into account dc drift and settling time. A code sequence that produces an output change of one least significant bit amplitude is likely to require the smallest test time. However, if the dc reference drifts too much during this time, then there will be a gain or dc offset error in the result. One way to estimate this error is to periodically repeat a measurement made earlier in the test run and use this to correct results at the end. This may result in a larger voltage change requiring a longer settling time. When maximum precision in estimating drift is required, this code change should result in a full-scale output.

4.4 Dynamic testing

To measure a DAC's dynamic performance, the test stimulus is a digital signal at the input of the DAC, and the analog output is measured. Note that this digital signal is usually a quantized version of an ideal waveform. An ideal DAC will produce the exact analog signal equivalent to the input pattern without any deterioration. But a non-ideal DAC will produce a different version of the analog signal, which may be different in amplitude, and which may contain distortion and noise.

When a quantized sine wave is used as the stimulus of a DAC, the output of the converter observed with a spectrum analyzer contains the fundamental and artifacts, which may include harmonics, images of all of the signals generated by the reconstruction process, spurious frequencies, and noise. If the output is being measured with a waveform recorder, the signal and its harmonics will be measured directly if the frequency being measured is below the Nyquist frequency of the recorder and the update rate is more than twice the frequency being measured. If the location of a harmonic is above the Nyquist frequency of the test digitizer or

the update rate is less than half the frequency being measured, the aliased frequency will be measured instead. In normal DAC applications, the DAC output is low-pass filtered to reduce the unwanted frequency content. For DAC testing, it is occasionally recommended that the low-pass filter not be used and that all the unwanted components be measured directly. The performance of the filtered DAC can be extrapolated from the non-filtered version of the output. Measuring without an anti-aliasing filter can create a host of potential problems since noise outside the band of interest will be folded back into the baseband.

Any instrument used to measure the output of a DAC needs to have better performance than the DAC itself so that the distortion and the noise of the instrument can be ignored in the measurement. In practice, this condition may be difficult to attain. To overcome this condition, a band-reject or notch filter is often used to reduce the signal amplitude of the fundamental, thus reducing the dynamic range of the signal to be measured.

For all the tests, the following three basic steps are followed:

- 1) Acquire a waveform containing M_{dig} periods with N_{dig} samples, which follow the relationship

$$\frac{f_t}{f_{\text{dig}}} = \frac{M_{\text{dig}}}{N_{\text{dig}}} \quad (10)$$

where

- f_t is the test frequency (the fundamental tone generated by the DAC)
 - f_{dig} is the digitizer sample rate
 - M_{dig} is the number of periods of the fundamental tone acquired from the analog test output
 - N_{dig} is the number of samples acquired of the analog test output
- 2) Perform a DFT on the waveform (see specific test section for details).
 - 3) Calculate the relevant parameters.

In many cases, the digitizer parameters (f_{dig} , M_{dig} , and N_{dig}) will be equal to that of the stimulus (f_s , M_{test} , and N_{test}). If not, set the system parameters such that the following relationships are valid:

$$f_t = \frac{M_{\text{test}}}{N_{\text{test}}} f_s = \frac{M_{\text{dig}}}{N_{\text{dig}}} f_{\text{dig}} \quad (11)$$

where

- f_t is the test frequency (the fundamental frequency of the signal being synthesized by the DAC)
- M_{test} is the number of periods of the fundamental of the test signal
- N_{test} is the number of samples in the digitized test signal
- f_s is the internal D/A section update frequency
- f_{dig} is the digitizer sample rate
- M_{dig} is the number of periods acquired of the analog output
- N_{dig} is the number of samples acquired of the analog output

Because of the finite number of samples acquired combined with the quantization error of the digitizer and the presence of random noise in either the DAC or the measurement system, these estimates show a variance that is usually a function of N_{dig} . When measuring the rms amplitude a_i of a harmonic signal, the standard deviation of the measurement is due to the following:

- a) The quantization noise and thermal noise of the digitizer, which are (approximately) equally distributed over all DFT spectrum.
- b) The resolution limitations of the DAC.

- c) System noise from switching power supplies, or radio station pickup, or power line interference.
- d) Noise from reference and resistor or capacitor noise in the DAC circuit.
- e) The digitizer nonlinearity.

The spectrum of the measured signal is usually characterized by a noise floor with harmonic and spurious components. As the N_{dig} is increased, the noise floor is reduced, while the harmonic and spurious components remain fixed. This is illustrated in Figure 18.

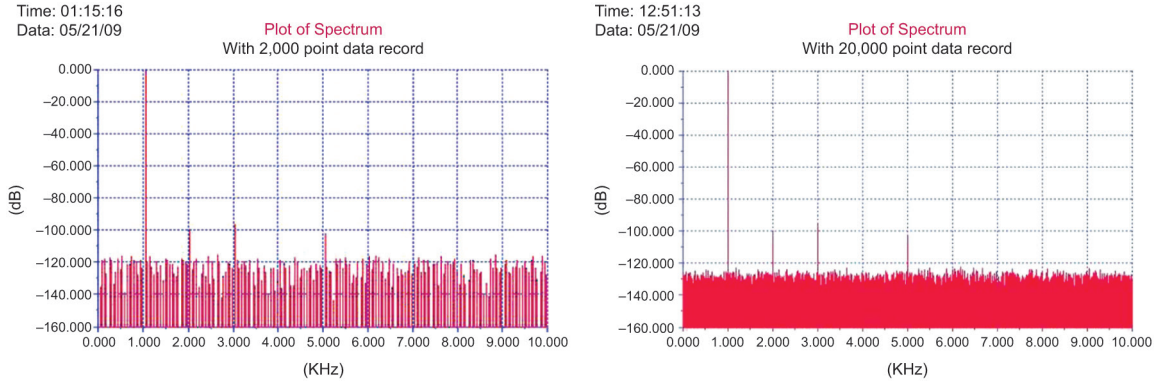


Figure 18—Comparison of computed spectrum with $N_{\text{test}} = 2000$ points and $N_{\text{test}} = 20\,000$ points

Note that the noise floor is 10 dB lower with $N_{\text{test}} = 20\,000$ points in the waveform than it is with $N_{\text{test}} = 2000$ points in the waveform. Note that the fundamental and the harmonics remain fixed regardless the value of N_{test} .

Digitizer and update clock jitter affect the apparent measurement noise floor. If the digitizer and the update clocks have a jitter effect of $T_{\text{jitter,rms}}$ then for a test signal with period $T_s = 1/f_i$ the measurement signal-to-noise ratio, for a full-scale input equals [B4]:

$$SNR = 20 \log_{10} \left(\frac{T_s}{2\pi T_{\text{jitter,rms}}} \right) \approx 20 \log_{10} \left(\frac{T_s}{T_{\text{jitter,rms}}} \right) - 15.96 \quad (12)$$

where

SNR is the signal-to-noise ratio, expressed in units of dB
 T_s is the test signal period (the period of the fundamental frequency), expressed in units of s
 $T_{\text{jitter,rms}}$ is the digitizer and update clock rms jitter, expressed in units of s

Equation (12) can be used effectively when examining the repeatability of the test system. In most test systems, quantization noise, random noise, and jitter-induced noise are the dominant noise sources.

4.5 Taking a waveform

The DAC being tested should be mounted in a test set-up whose block diagram is shown in Figure 17. For static testing, a digital voltmeter can be used. For dynamic testing, a digitizer can be replaced by a distortion analyzer or a spectrum analyzer.

5. Fitting sine waves

5.1 Curve fitting test method

Several DAC tests require that the device-under-test produce a sine-wave output. This is accomplished by applying an appropriate pattern of codes to the DAC input as suggested in 11.1.1. The analog signal observed at the output is then analyzed for noise and distortion to determine several DAC parameters. The analysis can be made by digitizing the DAC output with a waveform digitizer. The data thus obtained can be digitally processed. One of the traditional processing operations carried out is fitting the data to an ideal sine wave given by Equation (13).

$$x[n] = A_0 \cos(2\pi f_0 t_n) + B_0 \sin(2\pi f_0 t_n) + C_0 \quad (13)$$

where

t_n	is the nominal time associated with the n^{th} data value
A_0, B_0, C_0 , and f_0	are determined to minimize the least squares difference between the data and the best-fit sine wave

This is called the four-parameter fit. If the signal frequency is known before testing, it is substituted for the parameter f_0 and the remaining three parameters are estimated with what is called a three-parameter sine fitting algorithm. There are many algorithms for performing these least squares curve fits, which can be found in IEEE Std 1057 [B12].

Subtract the fitted signal from the recorded signal to obtain the residuals.

The sine-wave curve fit is used in several specific test methods described later in this standard. Analyze the residuals and fit parameters using methods described for the specific test being performed.

The four parameters mentioned above can be converted to an amplitude and phase form corresponding to a sine wave described by Equation (14).

$$A_0 \cos(2\pi f_0 t) + B_0 \sin(2\pi f_0 t) + C_0 = A \cos(2\pi f_0 t + \phi) + C_0 \quad (14)$$

where

A	is $\sqrt{A_0^2 + B_0^2}$
ϕ	is $\arctan(-B_0/A_0)$
$\arctan$	is the inverse tangent

Arctan is usually implemented with a function which returns a value in the range of 0 to 2π or in the range from $-\pi$ to π .

5.1.1 Three-parameter versus four-parameter fit

The four-parameter method with a record containing at least five periods of the signal is sometimes used even if the input frequency is accurately known a priori. The four-parameter method usually determines the frequency to be different. This difference may be caused by the fact that the frequency is not really known, or the difference may mask harmonic distortion errors and make the device under test appear to be better than it actually is. The computations are more complicated with the four-parameter method. Software for carrying out these computations is available (Annex B). For waveforms containing less than five periods, the four-parameter method can underestimate harmonic distortion by compensating for it with a change in frequency. In this case, the three-parameter method may be more appropriate. When frequencies are chosen precisely as

recommended here, with an exact integer number of periods in a waveform, a DFT will give the same results as a three-parameter sine-wave fit.

5.2 Choice of frequencies and waveform epoch

There are several factors that enter into the selection of frequencies. These factors affect frequency selection on three different scales: fine, medium, and coarse. Although the frequency selection decisions are made in the order: coarse, medium, then fine, the criteria are described in the reverse order.

5.3 Fine-scale frequency selection

On the fine scale, it is optimal to select frequency for which the input values to the DAC will all be different. For example, if a frequency of 200 kHz was used to test a DAC with an update rate of 1 MHz, the same five different phases of the sine wave would be generated many times. The recommended approach is to use a waveform epoch, N_{test} , and a frequency, f_{in} , such that N_{test} uniformly distributed phases will be produced. This is easily accomplished by choosing f_{in} as follows:

$$f_{in} = \frac{J}{N_{test}} f_s \quad (15)$$

where

J is an integer that is relatively prime to N_{test}
 f_s is the update rate
 N_{test} is the waveform epoch

The condition of being relatively prime means that N_{test} and J have no common factors (i.e., their greatest common divisor is one) and that for the recommended frequency there are exactly J periods in the waveform epoch. If N_{test} is a power of two, then any odd value for J meets the relatively-prime condition.

For a DAC with N bits of resolution and an ideal transfer characteristic, the minimum N_{test} that will help ensure that every code word appears as an input value is $2^N\pi$, when the input frequency is chosen as above. The smallest power-of-two waveform epoch is four times 2^N .

5.4 Medium-scale frequency selection

The rules for fine-scale frequency selection give many frequencies that optimize the spacing between the samples recorded. On the medium scale, frequencies are selected to cause errors from different sources to occur at different frequencies. For example, for a DAC with an update rate $f_s = 2$ GSa/s if a frequency of 400 MHz were selected, third harmonic distortion would be at a frequency of 1200 MHz. Since this is above the Nyquist frequency of 1000 MHz, an additional image might appear at the 800 MHz, alias of this frequency. This is the same frequency as second harmonic distortion, so the two would be indistinguishable. With a frequency of 420 MHz, the second harmonic is at 840 MHz while the third harmonic aliases down to 760 MHz, allowing one to distinguish between the two.

5.5 Coarse-scale frequency selection

In the coarse scale frequency selection, round numbers are normally selected (e.g., 250 MHz, 500 MHz). These round numbers then have to be modified to separate the errors from different harmonics. It is important to take aliasing into account in this step. The resulting frequencies should then be modified a second time to meet the criteria for fine-scale frequency selection.

5.5.1 Comments on fine, medium, and coarse frequency selection

Perform tests at several frequencies spanning the range of major expected frequency components in the final-use output signal. The highest frequency should have a maximum slew rate (derivative with respect to time) at least as large as that of the final-use output signal.

The test frequencies can be categorized as low, medium, and high. Low frequencies are low enough to not cause significant dynamic errors (e.g., frequency dependent distortion and time jitter) in the DAC. Frequencies less than a few percent of the DAC bandwidth are generally safe to consider low. Medium frequencies are those high enough to cause some dynamic effects, but still well below the DAC bandwidth. These will generally be in the range of 10% to 30% of the DAC bandwidth. High frequencies are near enough to the DAC bandwidth that the amplitude roll-off is a significant factor. The test frequencies should include at least one frequency in each category.

5.6 Selecting signal amplitudes

Signal amplitudes of full scale should be used for each frequency unless otherwise specified. This amplitude is the amplitude of the fundamental signal at the output of the DAC. Because of the interaction between DAC clocking rate and the amplitude of the fundamental frequency at the DAC output, the user should expect that the output will be attenuated by a calculable amount.

6. Digital input

6.1 Coding

The DAC maps the digital input codes via the converter's transfer curve into an analog output signal. The DAC input codes represent finite-resolution values that are represented by such coding schemes as those described in [1.7.2](#).

6.2 Clock and data feedthrough

Clock and data feedthrough are unwanted spurious signals at the output of the converter as a direct result of coupling the digital clock and input data signals, respectively. This occurs because a DAC is forced to exist on the frontier between the noisy environment of high-speed digital logic and the sensitive analog domain. The problems of this interfacing are particularly acute when demands of high speed (greater than 10 MHz switching times) and high precision (12 bits or more) are combined. No amount of design effort can perfectly isolate the analog portions of a DAC from the spectral components of a digital input signal. Inevitably, once this digital signal is brought onto the chip, some of its higher frequency components will find their way to the sensitive analog nodes, producing transients on the analog output of the DAC. This phenomenon is represented by a parameter generically called digital feedthrough or clock and data feedthrough (Analog Devices, AD568 [B1]). The unwanted spurious signals are typically caused by the switching speeds and levels of both the digital input data and the sampling clock applied to the DAC. They are then coupled through device packaging parasitics, the DAC reference, board design and layout, etc. See Figure 19 for an example of clock and data feedthrough.

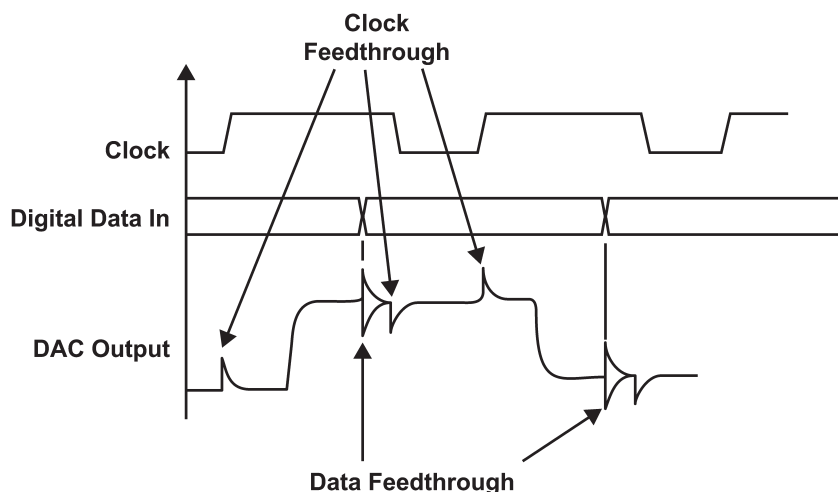


Figure 19—Example of digital feed-through

Feedthrough can introduce significant spurious frequency components at the DAC output. In particular, the clock feedthrough can introduce both harmonic distortion and distortion tones at multiples of the clock signal. Moreover, clock feedthrough typically does not exhibit any code dependency (Hendricks [B10]).

6.2.1 Clock and data feedthrough test method

The setup shown in Figure 17 should be used to evaluate clock and data feedthrough. The measuring instrument shown in the block diagram shall be a high bandwidth oscilloscope for glitch parameter measurements. A 1 LSB repetitive code change is applied on the digital input, which would nominally create an output change of 1 LSB. Unless otherwise specified, the code change should be a transition between the following two binary codes:

Code 1 1111111111... (–1 LSB in a two’s complement coded DAC)

Code 2 0000000000... (0 in a two’s complement coded DAC)

In other coding schemes, the transition should be appropriate for the coding used in the DAC. The coding should exercise as many bits as possible so as to enhance the data dependent feedthrough effects.

The resulting output is acquired by the oscilloscope over multiple clock cycles as shown in Figure 19. The number of clock cycles should be chosen depending on the oscilloscope time resolution, the DAC update rate, and input-output time delay. The clock and data feedthrough is then calculated as the area in each of the transients, in the recorded output waveform, that can be exclusively ascribed to the clock and data feedthrough. The transients are specified in units of nV s. The clock and data feedthrough shall be tested at the maximum DAC update clock frequency unless otherwise specified.

Clock and data feedthrough may cause spurious tones to appear in ac tests. The spurious tones should be detected in the SFDR tests (Roberts and Burns [B27]).

6.3 Static input parameters

Digital logic inputs may be of many different types, depending on the DAC. The logic family name (CMOS, TTL, ECL, LVDS, etc.) should be used if the digital signals adhere to the standards for the family. If there are deviations from the standards, then the differences should be clearly stated.

Users of DACs will not usually need to measure the input parameters of the logic inputs, but manufacturers should do so to help ensure compliance with accepted logic standards.

6.4 Timing parameters

The performance of the DAC is dependent on the timing relationship between the position of the clock edge and the time in which the input changes. SNR is sensitive to the timing relation of this clock edge relative to the changing state of the input data. This timing parameter is typically referred to as setup and hold timing.

Setup timing (t_s) is the period of time that the input data is to be held constant at a valid logic level relative to the DAC clock transition/latching edge.

Hold time (t_h) is the period during which the input data is to be held constant after the latching clock edge.

The time reference is typically measured relative to the valid logic level. Propagation delay of the clock signal and timing jitter specifications need to be understood and accounted for to set the nominal timing requirements and the available timing margin respectively. Figure 20 represents a positive setup and hold timing relationship relative to a positive latching edge of the clock signal.

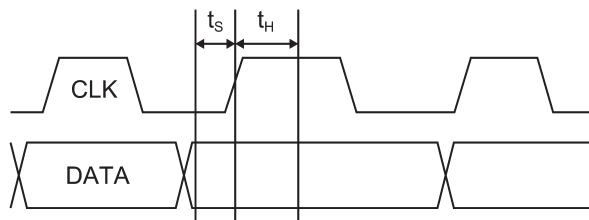


Figure 20—Example of setup and hold timing

7. Analog inputs

Normally, the only analog inputs are one or more power supply voltages and a reference voltage or current if the reference is not internal to the DAC. For simple DACs, there may be only one power supply input that also acts as the reference voltage. More complex DACs may require one or more analog power supplies, a digital power supply, and voltage or current references. In all cases, the nominal (or typical), minimum, and maximum operating voltages shall be specified, and the absolute maximum input levels should also be given. See 1.8 for power supply parameters, definitions, and test methods.

7.1 Comments on analog inputs

The reference inputs to a DAC should be more stable and accurate than the analog output specifications of the DAC. The noise content of the reference should be small enough to not cause a deterioration in DAC performance. The parameters of the reference input signals that are important include the following:

- The noise density in units of $V/\sqrt{Hz}$ (sometimes called spot noise) as a function of frequency over the bandwidth of the reference transfer function to the output
- The long term stability in units of (percent-of-value)/day
- The voltage (or current)

In addition to these characteristics of the reference input signal, the following characteristics of the load at the DAC reference terminal should be known:

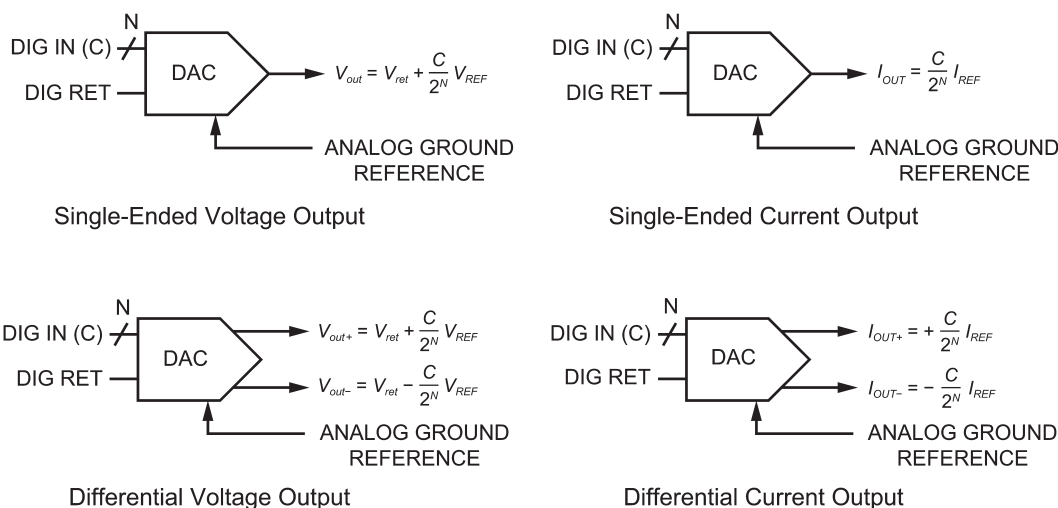
- The nominal input impedance
- The variations in the current drawn from a reference voltage as a function of the DAC codes and, correspondingly, the variations in the compliance voltage seen by a reference current

8. Analog output (single-ended and differential)

DACs generally have either voltage or current outputs.

Both current- and voltage-output DACs are sometimes configured with differential outputs. This allows the DACs to have outputs that do not reference themselves to a local DAC ground. Block diagrams of the topologies described are shown in Figure 21. Note that V_{ref} refers to a reference voltage that can either be external or internal to the DAC. I_{ref} refers to a reference current that can either be external or internal to the DAC. V_{ref} is the voltage on pin designated as “ANALOG GROUND REFERENCE.”

For purposes of this discussion, the DACs are assumed to be unipolar DACs with outputs that are nominally 0 with a code of $C = 0$ applied, and nominally Full Scale – 1 LSB when C is set to $2^N - 1$. Other coding schemes exist.



8.1 Figure 21—Block diagram of DAC topologies

8.2 Output impedance

Voltage-output DACs generally have low-output impedances. In high-speed applications, voltage DACs may have a series resistor in the output with a typical value of 50 Ω to match transmission lines that are driven by the DAC.

Current-output DACs generally have high-output impedances. Ideally, current-output DACs should be connected to zero ohms at ground potential. In practice, it is actually not zero, and the deviation from zero that the DAC is able to tolerate is defined under the heading “compliance” [B18].

In high-speed applications, the DAC may have an internal shunt resistance with a typical value of 50 Ω to match transmission lines that are driven by the DAC.

8.2.1 Output impedance test method

Voltage-output DACs with series resistors and current-output DACs with shunt resistors can have their impedances measured by programming the DAC to a full-scale output and going through the following procedure:

- Measure the output voltage into the variable V_1 .
- Apply a load of known resistance R_0 between the output and ground.
- Measure the new output voltage into variable V_2 .
- Calculate the output resistance, R_{out} , from the equation $R_{out} = \frac{V_1 - V_2}{V_2} R_0$.

Voltage-output DACs with nominally low output impedances (less than 1 Ω) can be difficult to measure by the method described above since the difference between V_1 and V_2 will be small when a reasonable load is applied. To more accurately measure the output impedance of a voltage DAC, program the DAC to a nominally zero output and implement the following sequence of steps:

- Measure the output voltage into the variable V_1 .
- Apply a current source of known value I_0 . (The output impedance of the current source should be much larger than the expected value of R_{out}).
- Measure the new output voltage into the variable V_2 .
- Calculate the output resistance, R_{out} , from the equation $R_{out} = \frac{V_2 - V_1}{I_0}$.

Current-output DACs with nominally high output impedances (greater than 10 k Ω) require another approach. To accurately measure the output impedance, connect the output of the DAC to a voltage source with current measuring capability, and go through the following sequence of steps:

- Program the voltage source to a value equal to the highest specified compliance voltage V_P . Program the DAC for a full scale output current.
- Measure the current flowing in the voltage source into the variable I_P .
- Program the voltage source to a value equal to the lowest specified compliance voltage V_M .
- Measure the current flowing in the voltage source into the variable I_M .
- Calculate the output shunt resistance, R_{shunt} , from the equation $R_{shunt} = \frac{V_P - V_M}{I_P - I_M}$.

The results from this calculation should be considered carefully, since a perfect current DAC will have $I_P = I_M$, and a divide by 0 error can occur, or a negative shunt resistance can be calculated. When this phenomenon occurs, the usual practice is to test the denominator before performing the calculation, and revise the acceptance criterion to allow for the uncertainty in the repeatability of the measurement.

8.3 Short-circuit current

A voltage output DAC can usually supply only a limited amount of current before the DAC output drivers go into a current limiting mode.

8.3.1 Short-circuit current test method

Connecting a very low impedance load and measuring the current flowing through the load will allow for measurement of the short-circuit current. The short-circuit current is specified in the DAC data sheet. The short-circuit current may not be a fixed value—it may vary with the temperature and the power supply voltages. Internal power-limiting circuits may also cut in and reduce the current when internal power dissipation exceeds certain limits.

If a voltage source is connected to the DAC output, internal current limiting may cause the short-circuit current to vary with the value of the applied voltage.

The short-circuit test should not be performed on DACs that do not have internal mechanisms for limiting the output current.

The short-circuit current measured on a current-output DAC should be equal to the programmed DAC output current.

8.4 Compliance voltage for current-output DACs

A current-output DAC has a range of output voltages that can be driven without significantly impacting the transfer curve of the DAC. This range of voltages is called the compliance voltage range of the DAC.

8.4.1 Compliance voltage test method

- a) Connect the output of the DAC to a voltage source that can also measure current with an internal ammeter.
- b) Program the DAC to the positive full-scale current output.
- c) Program the voltage source for the maximum specified compliance voltage.
- d) Measure the current using the voltage source ammeter.
- e) Program the voltage source for the minimum specified compliance voltage.
- f) Measure the current using the voltage source ammeter.
- g) Verify that the measured currents are consistent with the DAC specifications and the specified internal shunt resistance.

8.5 Load current for voltage-output DACs

A voltage-output DAC will operate in a linear fashion within some range of output currents. This property can be verified by performing a load current test.

8.5.1 Load current test method for voltage-source DACs

Connect the DAC to a current source, and implement the following steps:

- a) Program the DAC to a positive full-scale value.
- b) Set the current source to draw the maximum allowed load current as specified on the DAC data sheet.
- c) Measure the output voltage and verify that the output voltage is within the specified limits.
- d) Program the DAC to a negative full-scale value.

- e) Set the current source to draw the maximum allowed load current as specified on the DAC data sheet.
- f) Measure the output voltage and verify that the output voltage is within the specified limits.

8.6 Maximum usable dynamic range

8.6.1 Definition of maximum usable dynamic range

The maximum usable dynamic range of a DAC is the ratio between the largest amplitude single frequency that can be generated and the noise floor in the absence of a signal. The frequency range over which the noise floor is measured shall be defined. This parameter is not identical to Spurious Free Dynamic Range (SFDR), which describes the ratio between a fundamental frequency and any other spurious components that are generated in the presence of a signal.

8.6.2 Test method for maximum usable dynamic range

The rms value of the largest single tone output that can be generated from a DAC with a given FSR can be approximated by [Equation \(16\)](#):

$$\text{Largest Signal} = \frac{FSR}{2\sqrt{2}} \quad (16)$$

The noise floor should be measured while the DAC is being loaded at its maximum clock rate with an input code that nominally generates a zero output. The DAC output is measured by a system whose bandwidth is defined by the defined frequency range. This is generally most efficiently measured by connecting the DAC output to a waveform recorder with a bandwidth that is greater than the specified frequency range. A waveform is collected, and a DFT is performed on the data. The rms value of the signal present in the specified frequency range is defined as $e_{\text{noise rms}}$ and the dynamic range is defined by [Equation \(17\)](#):

$$\text{Maximum Dynamic Range} = 20 \log_{10} \frac{\text{Largest Signal}}{e_{\text{noise rms}}} \quad (17)$$

9. Gain and offset (static and dynamic)

Gain and offset are the values by which the ideal DAC output values are multiplied and to which the ideal output values are added, respectively, to minimize deviation from the actual DAC output values. There are two definitions of static gain and offset, independently based and terminally based, corresponding to two methods of calculating the deviation between the ideal output values and the actual output values.

9.1 Static gain and offset (independently based)

Independently-based gain and offset are the values by which the ideal DAC output values are multiplied and to which the ideal output values are added, respectively, to minimize the mean squared deviation from the actual DAC output values. Unless otherwise specified, static gain and offset will be taken to mean independently-based static gain and offset.

9.1.1 Static gain and offset test method

Measure the code output levels as per [4.3](#). The transfer curve can then be represented by [Equation \(19\)](#).

$$U[k] = G \cdot (Q \cdot k + U_0) + V_{OS} + \varepsilon[k] \quad (18)$$

where

$U[k]$	is the k^{th} output value corresponding to the k^{th} code applied to the inputs. The code applied to the DAC depends on the DAC coding (see 1.7.2)
U_0	is the ideal value corresponding to $U[0]$
V_{OS}	is the output offset in units of the output quantity, nominally equal to zero
G	is the gain, nominally equal to unity
Q	is the ideal width of a code, that is, the full-scale range (FSR) divided by the total number of codes ($\text{FSR}/2^N$)
$\varepsilon[k]$	is the residual error corresponding to the k^{th} input code, that is the deviation of the actual DAC output values from the ideal values multiplied and added by the gain and offset, respectively

Equation (18) can be rewritten as follows:

$$\frac{U[k] - V_{OS} - \varepsilon[k]}{G} = Q \cdot k + U_0 \quad (19)$$

In this way, the expression on the right side gives the ideal code level, in input units, as a function of k , where $0 \leq k \leq 2^N - 1$, and N corresponds to the number of bits in the DAC. The variable, k , is assumed to be the value of the binary coded input as described in 1.7.2.

Using conventional linear least-squares estimation techniques, independently-based static offset and gain are the values of V_{OS} and G that minimize the mean squared value of $\varepsilon[k]$ over all k . The value of G that minimizes the mean squared value of $\varepsilon[k]$ is given by Equation (20).

$$G = \frac{\sum_{k=0}^{2^N-1} k \cdot \sum_{k=0}^{2^N-1} U[k] - 2^N \sum_{k=0}^{2^N-1} k U[k]}{Q \left(\left(\sum_{k=0}^{2^N-1} k \right)^2 - 2^N \sum_{k=0}^{2^N-1} k^2 \right)} \quad (20)$$

The value of V_{OS} that minimizes the mean squared value of $\varepsilon[k]$ is given by Equation (21).

$$V_{OS} = \frac{\left(Q \sum_{k=0}^{2^N-1} k + 2^N U_0 \right) \sum_{k=0}^{2^N-1} k U[k] - \left(Q \sum_{k=0}^{2^N-1} k^2 + U_0 \sum_{k=0}^{2^N-1} k \right) \sum_{k=0}^{2^N-1} U[k]}{Q \left(\left(\sum_{k=0}^{2^N-1} k \right)^2 - 2^N \sum_{k=0}^{2^N-1} k^2 \right)} \quad (21)$$

Given these values for G and V_{OS} , $\varepsilon[k]/G$ is the independently-based integral nonlinearity.

9.2 Static gain and offset (terminally based)

Terminally-based gain and offset are the values by which the ideal output values are multiplied and to which the ideal output values are added, respectively, to cause the deviations from the outputs to be zero at the terminal points, that is, at the first and last codes.

9.2.1 Static gain and offset test method

Measure the code output levels as per 4.3. The transfer curve can then be represented by Equation (18).

Terminally-based gain and offset are the values of G and V_{OS} that cause $\varepsilon[0] = 0$ and $\varepsilon[2^N - 1] = 0$, where N corresponds to the number of bits in the DAC and $2^N - 1$ is code bin of the highest code defined.

The formulas for terminally-based G and V_{OS} become:

$$G = \frac{U[2^N - 1] - U[0]}{Q(2^N - 1)} \quad (22)$$

$$V_{OS} = U_0 - GU[0] \quad (23)$$

Given these values for G and V_{OS} , $\varepsilon[k]/G$ is the terminally-based integral nonlinearity.

9.3 Dynamic gain and offset

The dynamic gain of the DAC under test at a frequency f is the magnitude of the frequency response at that frequency. Specifications of dynamic gain shall include the data rate, update rate, and interpolation filter used.

9.3.1 Dynamic gain error

The dynamic gain error is the difference between the dynamic gain, $G(f)$, of the DAC at a given frequency f and its gain at a specified reference frequency divided by its gain at the reference frequency. The reference frequency is chosen to be a frequency whose gain is at or near the peak gain of the DAC passband; typically, it is the same frequency as the one used in the frequency response test described in [Clause 15](#). For dc-coupled DAC, the reference frequency is typically dc ($f=0$).

To determine the dynamic gain error, first determine the dynamic gain (e.g., use one of the frequency response test methods described in [Clause 15.3](#)). The gain error at frequency f is:

$$\varepsilon_G(f) = \frac{G(f) - G(f_{ref})}{G(f_{ref})} \times 100\% \quad (24)$$

where

ε_G is the gain error
 f_{ref} is the chosen reference frequency

10. $G(f)$ is the dynamic gain of the DAC at frequency f . Linearity (static and dynamic)

10.1 Integral nonlinearity

The integral nonlinearity is the difference between the ideal and measured output values after correcting for static gain and offset. Integral nonlinearity is usually expressed as a percentage of full scale or in units of LSBs. It will be independently-based or terminally-based depending on how static gain and offset are defined. When the integral nonlinearity is given as one number with no code bin specification, it is the absolute maximum integral nonlinearity over the entire range.

10.1.1 Integral nonlinearity test method

Solve [Equation \(19\)](#) for the linearity error, $\varepsilon[k]$. Find the static gain and offset according to the method in [9.1.1](#) or [9.2.1](#), as appropriate for independently-based or terminally-based static gain and offset. The static integral nonlinearity as a function of k is given in percent by:

$$INL[k] = 100\% \cdot \frac{\varepsilon[k]}{2^N \cdot Q \cdot G} = 100\% \cdot \frac{\varepsilon[k]}{FSR \cdot G} \quad (25)$$

where

$INL[k]$	is the integral nonlinearity at input code bin k in units of percent of FSR
$\varepsilon[k]$	from Equation (19) in output units
Q	is the ideal code bin width, expressed in output units
G	is the gain, nominally equal to unity
FSR	is the output Full Scale Range expressed in output units

The maximum INL is the maximum absolute value of $INL[k]$ for all k .

10.2 Differential nonlinearity

Differential nonlinearity (DNL), as defined by Equation (26), is the difference, after correcting for static gain, between the outputs corresponding to two consecutive input codes minus the difference between two consecutive ideal output values (Q), divided by Q . When given as one number without specification, it is the absolute maximum differential nonlinearity over the entire range.

$$DNL[k] = \frac{W[k] - Q}{Q} \quad (26)$$

where

$W[k]$	is the corrected analog output difference for code bin k , or, $G \cdot (U[k+1] - U[k])$
Q	is the ideal width of a code, that is, the Full-Scale Range (FSR) divided by the total number of codes ($FSR/2^N$)
G	is the static gain
k	is the input code bin $0 \leq k \leq 2^N - 2$

An ideal DAC has no differential nonlinearity, that is $DNL[k] = 0$, for any k .

The maximum differential nonlinearity is the maximum absolute value of $DNL[k]$ for all k . In addition, the rms value of the DNL can be given by Equation (27).

$$DNL_{RMS} = \left[\frac{1}{2^N - 1} \sum_{k=0}^{2^N-2} (DNL[k])^2 \right]^{1/2} \quad (27)$$

10.3 Monotonicity

A monotonic non-inverting DAC produces average output values that are consistently increasing with increasing input codes and consistently decreasing with decreasing input codes, changing in the same direction relative to the change in input stimulus. If the input stimulus and output values change consistently in opposite directions (e.g., a higher input code produces a lower output value), the DAC is monotonic and inverting.

10.4 Spurious free dynamic range (SFDR)

Spurious free dynamic range is used as a figure of merit for DACs. A tone is synthesized, and the non-fundamental frequency components are measured. The spread between the amplitude of the fundamental and the highest non-fundamental component is defined as the spurious free dynamic range.

There is a difference of opinion in the industry concerning the harmonics of the fundamental. Some argue that any component, even if it is a harmonic, should be counted in the SFDR measurement. Others argue that

the harmonics are measured as a part of a total harmonic distortion (THD) measurement. The controversy is visible in the following example (Figure 22) of an SFDR result.

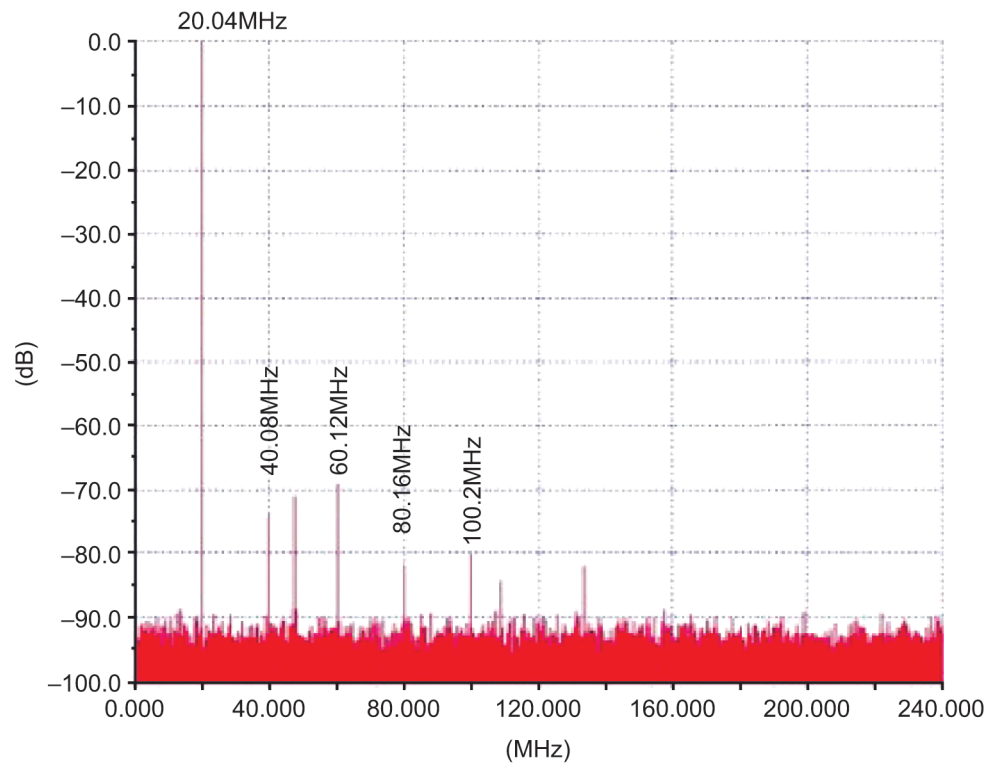


Figure 22—Plot of DAC SFDR data

The frequency components are tabulated as follows:

Frequency	Response	Comment
20.04 MHz	0 dB	Fundamental
40.08 MHz	−75.0 dB	2nd Harmonic
47.53 MHz	−71.0 dB	Spur
60.12 MHz	−69.0 dB	3rd Harmonic
80.16 MHz	−82.0 dB	4th Harmonic
100.2 MHz	−80.0 dB	5th Harmonic
108.471 MHz	−84.0 dB	Spur
133.571 MHz	−81.0 dB	Spur

If the harmonics are included, the SFDR would be evaluated as 69.0 dB. If the harmonics are not included, the SFDR would be 71.0 dB.

It is recommended to estimate SFDR by including harmonics, as it provides a more realistic indicator about the signal content, related to the fundamental.

10.4.1 Spurious free dynamic range test method

The DAC under test should be connected as shown in Figure 17. There are many dimensions to the testing of SFDR. These include the following:

- a) The DAC update rate, f_s
- b) The DAC input pattern length
- c) The frequency of the fundamental
- d) The amplitude of the fundamental
- e) The measurement sampling frequency
- f) The measurement waveform epoch

Unless otherwise specified, the DAC update rate, f_s , should be set to the maximum rate specified for the device.

Unless otherwise specified, the DAC input pattern length, N_{test} , should be such as to include all the DAC codes. For an N -bit DAC this should be greater than $2^N\pi$.

Unless otherwise specified, a number of frequencies should be applied. They should start at the lowest frequency as determined by the input pattern length. The lowest frequency, f_{min} , that can be generated is f_s / N_{test} . Each frequency f should be selected such that an integer number of cycles are generated by the pattern, and ff_{min} is an integer that is relatively prime to N_{test} .

The maximum frequency that should be generated is determined by the DAC application and the reconstruction network. In PCM audio, the maximum appropriate frequency is 20 kHz, and f_s is 44.1 kHz.

Unless otherwise specified, the amplitude of the fundamental should be the maximum allowable.

The measurement sampling frequency should be at least twice the maximum frequency component that is being considered. Any significant energy above the Nyquist frequency will appear to be present in the band below the Nyquist frequency.

The measurement waveform epoch, N_{dig} , should be long enough to include the entire DAC input pattern.

In the example shown in Figure 22, the waveform epoch N_{dig} was 25 000, the measurement sampling frequency was 1 GHz, f_s was 400 MHz, and the lowest frequency that could be generated was 40 kHz. The pattern length is 10 000, which is appropriate for an 11-bit DAC.

A typical SFDR plot is illustrated in Figure 23.

The DAC input pattern length is unspecified but is presumably appropriate for generating each of the frequencies shown.

The frequency of the fundamentals generated were 10 MHz, 20 MHz, 40 MHz, 70 MHz, 100 MHz, and 120 MHz.

The amplitudes of the fundamentals were 0 dB relative to full scale, -3 dB relative to full scale, and -6 dB relative to full scale.

The measurement sampling frequency and waveform epoch are unspecified.

10.4.2 Spurious free dynamic range discussion

SFDR can be also measured with more than one tone test signal. It is important to note that a single-tone measurement result does not reflect device multi-tone performance. While a single-tone test might be performed at full scale, multi-tone tests require each tone to be backed off further from full scale to avoid digital overload of the DAC. The phase of each individual tone needs to be chosen to avoid alignment of peaks of sine waves. A multi-tone test is often used to simulate the DAC behavior under typical conditions, especially for modern communication.

An example of SFDR plots is in Figure 23. The plot illustrates how data from multiple frequencies and amplitudes are displayed on one plot. Approximately 12 fundamental frequencies were tested at three different amplitudes to generate the plot. Additional examples can be found in DAC5675-EP Product Data Sheet [B7].

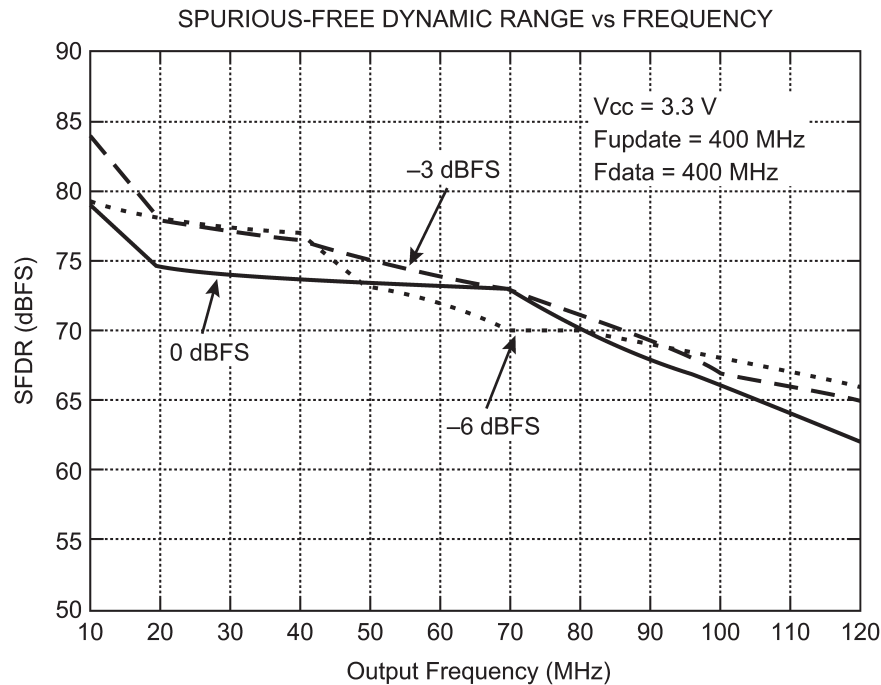


Figure 23—Example SFDR plots

The presented SFDR shall be always accompanied by a listing of the test signal frequency (frequencies), amplitude(s), DAC clock signal frequency, and frequency band of interest (if the last one is not specified, it is assumed the SFDR value is valid in the entire range 0 to Nyquist).

Many manufacturers have used the term SFDR to only describe spurs that are non-harmonically related to the fundamental signal. If the user prefers this definition, the user shall specify that the measurements exclude harmonic components.

11. Noise

Noise has historically been an ambiguous term. In this standard, noise is any deviation between the desired nominal output signal and the actual output signal except deviations caused by linear time invariant system response (gain and phase shift), a dc level shift, total harmonic distortion, or an error in the applied update rate. For example, noise includes the effects of random fluctuations, nonlinearities producing harmonics at

frequencies greater than those used in estimating total harmonic distortion, apparent quantization error, and spurious signals.

11.1 Signal-to-noise and distortion ratio

Signal-to-noise and distortion ratio is the ratio of output sine-wave signal power to the noise and distortion. Unless otherwise specified, it is assumed to be the ratio of root-mean-square (rms) signal to rms noise and distortion for sine-wave output signals. SINAD depends on the amplitude and frequency of the applied digital sine array. The amplitude and frequency at which the measurement is made shall be specified. For testing of a DAC, the signal amplitude typically spans the full output range of the DAC. A bandwidth for the measurement shall also be specified. Audio SINAD is generally measured from 20 Hz to 20 kHz.

11.1.1 Test setup for SINAD

The DAC test setup should be configured as shown in Figure 17. This test requires a digital code generator to produce a signal by which a sinusoidal output waveform from the DAC under test is analyzed in the time or frequency domain. The code generator should be capable of producing valid data codes at a rate that is at least as fast as the maximum update rate supported by the DAC. It is important that waveform samples be generated so that a maximum number of phases are included in the test code pattern. This is to help ensure that all codes of the DAC are exercised. For a test pattern consisting of N_{test} samples and J waveform periods, waveform samples may be computed with Equation (28).

$$y_n = \text{round}\left(\frac{2^N - 1}{2} \cos\left(\frac{2\pi n J}{N_{test}}\right) - 0.5\right)$$

$$n = 0, 1, \dots, N_{test} - 1$$
(28)

If the DAC is unipolar, gray code, one's complement, or offset binary, the bits should be appropriately modified. The test pattern will contain the maximum number of distinct phases possible (N_{test}) if N_{test} is a power of 2 and J is odd, such that N_{test} and J be mutually prime. The precise code pattern used to test the DAC depends on the test frequency and the DAC update rate. Low test frequencies may require only modest DAC update rates and will allow many samples per waveform period. Higher test frequencies may require higher DAC update rates and more waveform periods in the test pattern. The test frequency, f_t , that the DAC generates is determined by the relationship shown in Equation (29).

$$f_t = f_s \frac{J}{N_{test}}$$
(29)

where

- J is the integer that determines the fundamental frequency that is generated by the DAC
- the number of cycles within the number of samples, N_{test}
- N_{test} is the integer that determines the repetition rate of the overall pattern the number of samples
- f_s is DAC update rate

It should be noted that the length of the pattern of words, N_{test} , that is used to generate the sine wave will not excite all codes of the DAC unless the pattern is longer than $\pi 2^N$.

If the instrument measuring the DAC output is a digitizer or waveform recorder, and the output waveform captured is not coherent with the recorder sample frequency there is the potential of leakage artifacts affecting the quality of the measurement result. In that case, sine-fitting can be used to remove most leakage effects. If desired, windowing or other data correction methods can be applied.

11.1.2 Test method for SINAD in the time domain

SINAD can be computed by appropriately analyzing the waveform from a waveform recorder that is measuring the DAC output. Regardless of the measurement method used, SINAD is the ratio of the rms output signal to the rms noise-and-distortion in a specified bandwidth.

The DAC test setup should be configured as shown in [Figure 17](#).

Apply an input code sequence to produce a DAC output sine wave of specified amplitude and frequency. Take a record of data. Compute the SINAD using [Equation \(30\)](#).

$$SINAD = \frac{A/\sqrt{2}}{\sqrt{\frac{1}{N_{test}} \sum_{n=1}^{N_{test}} (y_n - y'_n)^2}} \quad (30)$$

where

- y_n is the waveform sample
- y'_n is the sample of the best-fit sine wave
- A is the measured amplitude of the sine wave
- N_{test} is the number of samples

SINAD is usually expressed in dB using [Equation \(31\)](#):

$$SINAD_{dB} = 20 \log_{10} \left(\frac{A/\sqrt{2}}{\sqrt{\frac{1}{N_{test}} \sum_{n=1}^{N_{test}} (y_n - y'_n)^2}} \right) \quad (31)$$

If a notch filter is used to remove the fundamental, the noise near the fundamental will also be attenuated so that the SINAD could be underestimated. In this situation, there are two different approaches for this problem.

If the insertion loss versus frequency of the notch filter can be measured prior to the SINAD measurement, then the noise level of the DAC at each frequency bin could be compensated with the insertion loss. Be aware that in the region where the insertion loss is very high, the digitizer noise could become the dominant noise.

If the insertion loss is not measured, then the noise energy around the fundamental bin can be excluded and a correction factor introduced with the assumption that the noise is almost evenly distributed.

[Figure 17](#) illustrates the block diagram of a circuit configuration, which can measure SINAD, SNR, and THD. A plot of the resultant spectrum is shown in [Figure 24](#).

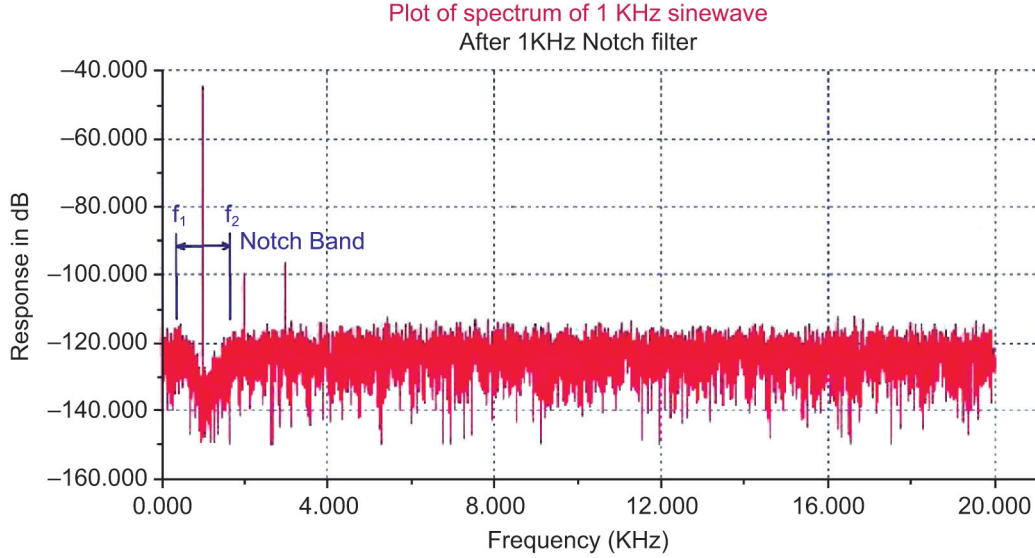


Figure 24—Resultant spectrum plot

The correction is performed in the frequency domain by modifying the data in Equation (32). The values of $X_{\text{avg}}[m]$ in the notch band are replaced by an equal number of values in a region that contains no spurs. In Figure 24, this can correspond to the region between 8 kHz and 9 kHz.

11.1.3 Test method for SINAD in the frequency domain

SINAD can also be determined by measuring the spectrum of the DAC output with a waveform recorder having SINAD at least four times better than that specified for the DAC under test. Care should be taken to synchronize the waveform recorder sampling with DAC updating so that waveform recorder samples are taken in a repeatable manner. The DAC sample frequency, in general, should be at a rate that is at least twice the highest frequency component present at the DAC output. It is best to collect waveforms that are an exact integral number of cycles of the fundamental DAC output sine wave. See Benetazzo, et al. [B5] for details.

SINAD can be determined equivalently from the frequency domain as a consequence of Parseval's Theorem if the DAC output waveform and the digitizer sampling frequency are such that a data record consists of an integral number of cycles of the DAC output's waveform. Compute the DFT of the DAC output waveform. SINAD is the ratio of the rms of the fundamental frequency of the DAC output signal to the rms noise and distortion (NAD). Both quantities can be determined from the DFT of waveforms as done for THD that is described in 12.1.1. The rms signal, A_{rms} , is obtained from Equation (48) in 12.1. NAD is found from the sum of all the remaining Fourier components after the bins at dc and at the test frequencies have been deleted from the spectrum.

$$NAD = \frac{1}{\sqrt{N_{\text{test}}(N_{\text{test}} - 3)}} \left[\sum_{m \in S_0} X_{\text{avg}}[m]^2 \right]^{1/2} \quad (32)$$

where

- S_0 is the set of all integers between 0 and $N_{\text{test}} - 1$, excluding the two values that correspond to the fundamental frequency and the zero-frequency term
- X_{avg} is the averaged spectral magnitude, as defined in Equation (C.6)

SINAD is then given by substitution of NAD and A_{rms} into Equation (33).

$$SINAD = A_{rms}/NAD \quad (33)$$

SINAD is often expressed in dB through Equation (34).

$$SINAD_{dB} = 20 \log \left(\frac{A_{rms}}{NAD} \right) \quad (34)$$

11.1.4 Comments on attainable SINAD

For a zero-order-hold DAC the number of samples per waveform period [N_{test}/J as defined in Equation (28)] affects the measurement result since the DACs zero-order-hold approximation of the waveform introduces its distortion. For a noiseless unfiltered DAC reproducing a sine wave with perfect precision, i.e., with no error due to finite word length, $SINAD_{dB}$ versus number of samples per waveform period, N_{test}/J , can be computed as follows in Equation (35):

$$20 \log_{10} \left(\frac{1 - A_1^2}{A_1^2} \right)^{\frac{1}{2}} \quad (35)$$

where

$$A_1 = \frac{\sin(\pi J / N_{test})}{\pi J / N_{test}} \quad (36)$$

Table 3 lists SINAD for a range of samples per waveform period from 16 to 8192 when the zero-order hold dominates the frequency response of the DAC. SINAD is seen to be inversely proportional to the number of samples per waveform period.

Table 3—SINAD versus number of samples per waveform

Samples per waveform period (N_{test}/J)	SINAD _{db} (all-pass filter)
16	18.9
32	24.9
64	30.9
128	37.0
256	43.0
512	49.0
1024	55.0
2048	61.1
4096	67.1
8192	73.1

If sufficient filtering is used, the SINAD can be reduced to less than the zero-order-hold limit of the DAC.

The best SINAD that can be expected from a perfect N -bit DAC with a reconstruction network that removes all spurious components greater than the Nyquist frequency, can be evaluated with Equation (37).

$$\text{SINAD}_{\text{dB,BEST}} = 20 \log_{10}(2^N \sqrt{1.5}) \approx 6.02N + 1.76 \quad (37)$$

This is based on the assumption that the rounding error introduced in Equation (28) is uniformly distributed between $-1/2$ and $+1/2$. The values of SINAD corresponding to this formula are tabulated in Table 4. Note that it is possible to get performance better than the tabulated results if the DAC is clocked at higher speeds and the input data is tailored to reduce the quantizing error.

Table 4—Best SINAD versus DAC resolution

N	2^N	$\text{SINAD}_{\text{dB,BEST}}$
4	16	25.8
6	64	37.9
8	256	49.9
10	1024	62.0
12	4096	74.0
14	16 384	86.0
16	65 536	98.1
18	262 144	110.1

11.2 Signal-to-noise ratio

This is the ratio of the signal to the noise. Unless otherwise specified, it is assumed to be the ratio of root-mean-square (rms) signal to rms error of the least squares sine-wave fit. SNR generally depends on the amplitude and frequency of the applied sine wave that shall be specified. The bandwidth of the measuring system shall also be specified. The 2nd through 10th harmonics are excluded from SNR calculations unless otherwise specified. The signal-to-noise ratio is the same as SINAD with the harmonic distortion power (THP) excluded.

The general formula for rms noise, η , is given by Equation (38).

$$\eta = \sqrt{NAD^2 - THP} \quad (38)$$

where

NAD is noise and distortion as defined in Equation (32)
 THP is total harmonic power as defined in Equation (46)

11.2.1 Coherent-sampling test method for SNR

Capture an appropriate sine wave from the DAC under test as described in 11.1.1. Compute the DFT of the measured waveform. SNR is obtained from the DFT by excising the signal harmonic frequencies, $f_{h[n]}$, identified in Equation (46) from the test signal spectrum. The input rms signal, A_{rms} is obtained from Equation (48). The rms noise is then found from the sum of all the remaining Fourier components.

$$\eta = \frac{1}{\sqrt{N_{\text{test}}(N_{\text{test}} - N_d - 1)}} \left[\sum_{m=1}^{M-1} X_{\text{avg}}[f_m]^2 \right]^{\frac{1}{2}} \quad (39)$$

where

X_{avgfh} is the averaged DFT spectrum, X_{avg} , with the components (bins) at: dc, the test frequencies f_{in} and $(f_s - f_{in})$, and the specified harmonic frequencies $f_{h[n]}$, all set to the average noise floor
 N_d is the number of DFT spectral components deleted ($N_d = 10$, unless otherwise specified)

SNR is then given by the substitution of these measures into Equation (40).

$$SNR_{dB} = 20 \log_{10} \left(\frac{rms\ signal}{rms\ noise} \right) = 20 \log_{10} \left(\frac{A_{rms}}{\eta} \right) \quad (40)$$

If a notch filter is used, the correction methods (11.1.2) may be used.

11.2.2 Comments on SINAD and SNR

These ratios are both proportional to the test signal, A_{rms} . It is customary to use a near full-scale signal for these measures. As opposed to ADCs, clipping should never occur if the input signal is generated without exceeding the DAC code range. In addition, these measures are generally a function of amplitude, $X_{avm}(f_{in})$, and the frequency, f_{in} , of the input sine wave. Thus, the amplitude and frequency of the input, for which the SINAD and/or SNR measurements are made, shall be specified.

11.3 1/f Noise

11.3.1 General discussion of 1/f noise

Resistor noise is characterized by having a spectral density that is uniform over frequency; thus, if resistor noise is passed through a 1 Hz bandwidth band-pass filter, the rms of the measured noise will be constant, independent of the band-pass filter's center frequency, f .

Active circuits containing amplification generally have a property that below some frequency, the noise density is no longer uniform, and increases at approximately the rate of $1/f^\alpha$ for $1 \leq \alpha < 2$, i.e., the noise becomes very large as f approaches 0. The frequency at which the noise density increases is called the 1/f corner frequency. Figure 25(a) illustrates this effect. In the figure, a corner is exhibited at about 20 Hz. Additional examples are given in DAC8311 Product Data Sheet [B8]. Note that the wide band noise is largest on the midscale plot, presumably because there is more of a resistive component at midscale than there is at zero or full scale, where presumably the output shows only the reference noise, or the residual noise.

11.3.2 Test method for 1/f noise

Configure the DAC as shown in Figure 17. A low-pass filter should be in place that will pass all frequencies below the 1/f corner frequency. Collect a waveform (this will take a long time). A typical waveform is shown in Figure 25 (b). The data should be windowed before a DFT is performed so as to eliminate record edge effects.

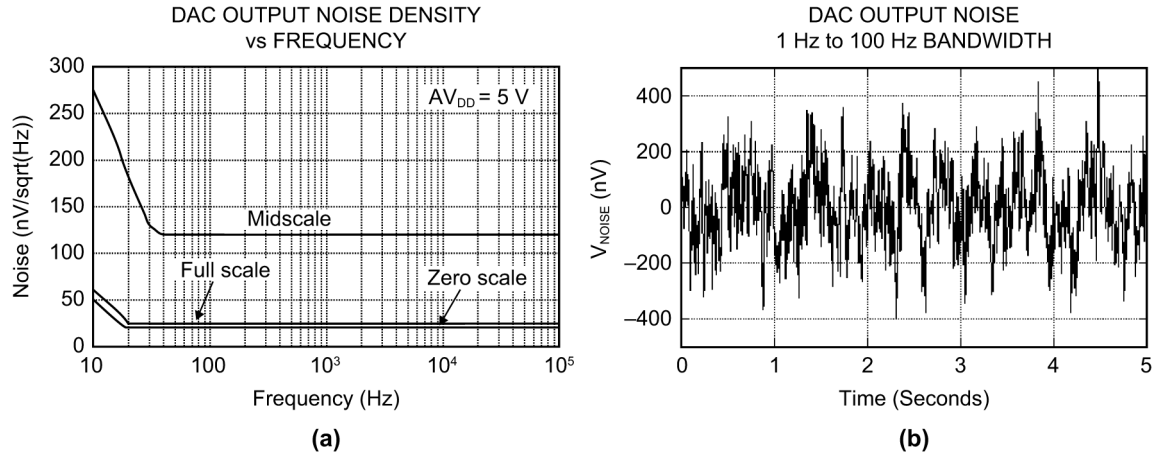


Figure 25—Low-frequency noise plots

11.3.3 Comments on $1/f$ noise

$1/f$ noise can complicate the measurement of INL, gain, and offset. There are many (often proprietary) tricks that are used to make the INL measurements immune to $1/f$ noise. It is beyond the scope of the standard to reveal the techniques.

11.4 Effective number of bits

For an input sine wave pattern of specified frequency and amplitude, the effective number of bits of a digital-to-analog converter, ENOB, is shown in Equation (41).

$$ENOB = N - \log_2 \left(\frac{\text{rms noise}}{\text{ideal rms quantization error}} \right) = \log_2 \left(\frac{FSR}{\text{rms noise} \cdot \sqrt{12}} \right) \quad (41)$$

where

FSR is the DAC full scale range

N is the number of bits in the DAC input word

The second equality in Equation (41) comes from Equation (42).

$$\text{ideal rms quantization error} = \frac{Q}{\sqrt{12}} \quad (42)$$

ENOB generally depends on the amplitude and frequency of the applied sine wave. The amplitude and frequency at which the measurement was made is to be specified.

The effective number of bits is a unit of measurement of the amount of rms noise and distortion produced by the DAC under test. If for a sine-wave pattern input of given frequency and amplitude, a real 10-bit DAC has an ENOB of 9.0, then it produces the same rms noise and distortion for that input as would an ideal 9-bit DAC. There is no claim that the DAC has quantization error, only that effective DAC shorthand for the performance of that DAC is that its performance is equivalent to that of an ideal 9-bit DAC. Note that by choosing the data rate, and the measure bandwidth, it is possible for ENOB to be greater than the resolution of the DAC under test.

11.4.1 Test method for effective number of bits

Find the actual rms noise and distortion by any of the methods described in this standard and apply Equation (41). The update rate and the pattern that is used to generate the DAC shall be specified. In order for the results to be meaningful, the following constraints shall be applied:

- The pattern shall excite all of the codes of the DAC ($2\pi \times 2^N$ appropriately computed codes should be applied to exercise every code in an N -bit DAC).
- The sine wave pattern used as input to the DAC shall contain an integer number of cycles of a sine wave, and the number of cycles shall be relatively-prime to the number of points in the pattern.
- Unless otherwise specified, the sine wave amplitude should encompass the full-scale range of the DAC.

11.4.2 Comment on ideal quantization error

There is some ambiguity concerning the meaning of ideal quantization error. On one hand, an ideal DAC delivers the exact signal that is specified by the DAC inputs. On the other hand, if one is trying to synthesize a sine wave, and one generates samples that are defined by 24-bit data then the 12-bit DAC that connects to this stream is introducing a quantization error. In some situations, it is accepted that the quantization error described by the second case is assigned to the process that generates the codes, and that no quantization error is assigned to the perfect DAC. However, for the definition of ENOB this quantization error is considered to be from the ideal DAC.

11.4.3 Comments on test conditions and results of the SNR and ENOB tests

The noise and distortion of a DAC depends on the data rate and the update rate of the DAC because of the rate dependent errors in the reconstruction process. The test conditions for a measurement shall be explicitly stated. Among the parameters that need to be specified are as follows:

- DAC data and update rates
- Pattern length of the input data
- DAC output filter characteristics
- The input filter effects on the instrument measuring SNR
- Test frequency and amplitude

11.4.4 Comment on the relationship of SINAD and ENOB

$SINAD_{dB}$ and effective number of bits are related by Equation (43) and Equation (44).

$$ENOB = \frac{SINAD}{20} \log_2 10 - \frac{\log_2 1.5}{2} - \log_2 \frac{2A}{FSR} \approx \frac{SINAD - 1.76}{6.02} - \log_2 \frac{2A}{FSR} \quad (43)$$

$$SINAD = 20 \log_{10} \left(\sqrt{1.5} \frac{2A}{FSR} 2^{ENOB} \right) \quad (44)$$

where

A is the amplitude of the fitted sine wave during the test
 FSR is the DAC Full Scale Range

Equation (44) takes into account the case when the amplitude of the test signal is below one half the FSR.

11.5 Noise power ratio

11.5.1 Adjacent channel leakage power ratio

Adjacent channel leakage power ratio (ACLR) is the ratio in dB between the measured total power centered on the assigned channel frequency to the measured total power centered on an adjacent channel frequency. For example, the channel bandwidth is 3.84 MHz with 5 MHz spacing between channels for WCDMA applications.

11.5.1.1 Test method for ACLR

Using the test setup described in 4.4, the stimuli signal is a WCDMA signal. Generate a stimulus signal according to ETSI standard 3GPP TS 25.141, Test model 1 [64 channels at 30 kSa/s distributed randomly across the code space, at random power levels and random timing offsets are defined so as to simulate a realistic traffic scenario that has a high peak-to-average ratio (PAR)]. The PAR value provides additional information on how the amplitude levels of the signal are distributed. Low PAR indicates a more uniform distribution. Set the carrier frequency so that assigned channel and adjacent channels are within the frequency band supported by the DAC. Measure the true rms voltage over the channel bandwidth (3.84 MHz) in the assigned channel and adjacent channel both side of channel frequency. Calculate the ACLR using Equation (45).

$$ACLR = 20 \log \left(\frac{\text{rms assigned channel}}{\text{rms adjacent channel}} \right) \quad (45)$$

12. Harmonic and spurious distortion

12.1 Total harmonic distortion

At a specified clock frequency and using an integer number of pure sine-wave cycles as input to the DAC, total harmonic distortion (THD) is defined as the total power of a specified set of harmonics divided by the power of fundamental. It is often expressed as a percent or in dBc.

The total harmonic power, THP, for the specified subset of harmonics, f_h , is defined as follows in Equation (46):

$$THP = \frac{1}{N_{dig}^2} \sum_h |X[f_h]|^2 \quad (46)$$

where

$X[f_h]$ is the complex value of the spectral component at frequency f_h , where f_h is the h^{th} harmonic frequency of the DFT of the waveform recorder output waveform computed using Equation (C.1) N_{dig} is the waveform epoch
 h is the set of harmonics over which the sum is taken

The summation is taken over all of the N_H harmonics used for the THD specification. THP is the square of the rms value of the signal consisting of the specified harmonics.

THP is an intermediate quantity used in computing THD and is not a critical DAC parameter in its own right. It is related to the square of THD by a scale factor and is a useful quantity in a number of calculations both in this subclause and in 11.2.

The members of the set of harmonics, f_h used in Equation (46) shall be specified. The choice of harmonic components included in the set is a compromise between the desire to include all harmonics with a significant portion of the harmonic distortion energy, but not to include DFT bins whose energy content is dominated by random noise. Unless otherwise specified, to estimate THD, the set is normally comprised of the lowest nine harmonics, i.e., second through tenth inclusive, of the input sine wave. The total harmonic distortion is given by the ratio shown in Equation (47).

$$THD = \frac{\sqrt{THP}}{A_{rms}} \quad (47)$$

where

A_{rms} is the rms value of the input sine wave

$$A_{rms} = \frac{|X_{avm}(f_{in})|}{N_{dig}} \quad (48)$$

where

f_{in} is the frequency of the input signal

The THD is also often expressed as a dB ratio with respect to rms amplitude of the fundamental component of the output or as a percent of the DAC full scale range.

The total harmonic distortion depends on both the amplitude and frequency of the applied signal. Thus, the amplitude and frequency of the input for which THD measurement(s) are made shall be specified.

12.1.1 Total harmonic distortion test method

Apply the generated sine-wave data set to the input of the DAC and use the test setup as described in 4.4.

Carefully choose a DAC data rate f_{DATd} and a test signal frequency f_{in} to fulfill any application requirements.

The test signal amplitude should be set to full scale value (0 dBFS). However, if the DAC final application targets significantly different typical amplitude, the typical expected value can be used instead.

The test signal frequency should be carefully chosen to fulfill any application requirements.

In the case of digitizer deployments compute the DFT of the measured waveform as described in Annex C. Identify the higher harmonic components (from the second to tenth) and use Equation (46) and Equation (47).

If a high-quality digitizer (as described in 4.4) is unavailable, a spectrum analyzer or lock-in amplifier can be used alternatively to acquire output harmonic components.

In all the above cases the first harmonic component may be attenuated by proper notch or high-pass filter to decrease dynamic range requirements for low-level harmonic component measurement as described in 4.4. In this case, the original amplitude value (before attenuation) shall be acquired to correctly compute Equation (47). In case the filter used affects even higher harmonic components, this effect shall be compensated properly before computing Equation (47).

The calculated THD value shall be always accompanied by listing test signal frequency, amplitude, and DAC clock signal frequency.

For very low distortion DAC, averaging could be used in order to reduce effect of random noise.

12.2 Intermodulation distortion

Intermodulation distortion (IMD) is the ratio of amplitude of the intermodulation product with the largest amplitude to the largest amplitude generated test tone.

12.2.1 Intermodulation distortion test method using two tones

Apply a quantized sum of two sine waves of specified frequencies and equal amplitudes (at least -6 dBFS each) to the input of DAC and use the test setup as described in 4.4. Identify in the spectral domain the second order intermodulation products $|(\pm f_1 \pm f_2)|$ and their images, and the third order intermodulation products $|(\pm 2f_1 \pm f_2)|$ and $|(\pm f_1 \pm 2f_2)|$ and their images.

For narrow band applications the possible list of IM products is sometimes restricted to $2f_1 - f_2$ and $2f_2 - f_1$. This restriction is made because these products fit to the frequency range of original f_1 and f_2 . In this case, the result is called the third order IMD. Similarly, higher order IMD can be tested and given if required by final DAC application.

The IMD result is given in dBc and shall be always accompanied by listing test signal frequencies, amplitudes, DAC clock signal frequency, and numbers of IMD orders examined. If the last one is not explicitly given, examination up to and including the third order is expected.

12.2.2 Comment on intermodulation distortion test method

IMD may be estimated using more than two tones. This is very common for DACs when used in base stations using many more than two tones. These are estimated with the same procedure described in 12.2.1, but will not be exhaustively described here.

12.3 Glitches

A glitch is the transient in the analog output outside of the expected ideal transition subsequent to a code change at the digital input.

12.3.1 General comments on DAC glitches

A DAC generally consists of many current or voltage sources that are switched on or off as defined by a digital input word. When a waveform is being synthesized, there is a transition between words at every tick of the f_s clock. Whenever one of the sources makes a transition from *on* to *off* or *off* to *on* there is the potential of a timing clash caused by differences in the turn-off and turn-on times of the switches. These transitions can introduce impulse like transitions in the DAC outputs. These impulses can affect both static and dynamic performance of a DAC. Several design strategies have been used to reduce the effect of the glitches. If the glitches are the same for all transitions, they will introduce an update-rate dependent offset. If they vary with the binary input word, they potentially add an update-rate dependent gain. If they are unpredictable, they introduce update-rate dependent and code-dependent noise.

Ideally, when a DAC output changes it should move from one value to a new one monotonically. In practice, the output is likely to overshoot, undershoot, or both (see Figure 26). This transient in the DAC output during a transition is known as a glitch (Kester [B19]).

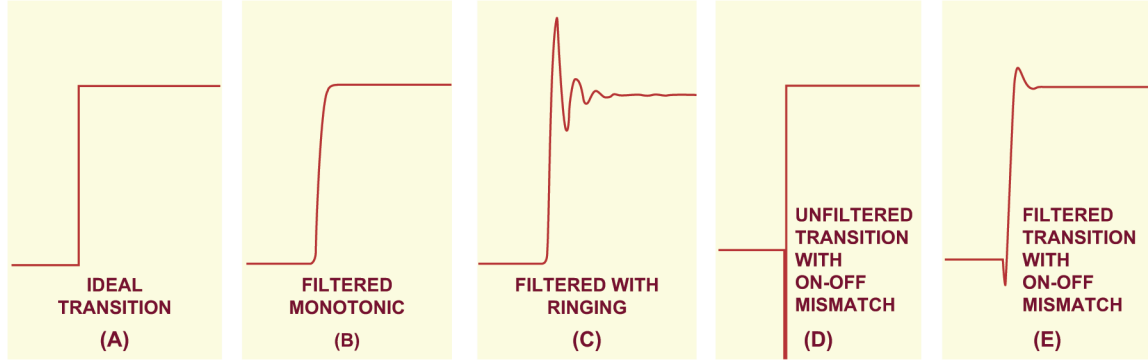


Figure 26—DAC transitions with (D) and (E) showing glitch

A DAC glitch can arise from several mechanisms: capacitive coupling of digital transitions to the analog output, transients in the DAC response, and the effects of some switches in the DAC producing temporary spurious outputs since they operate more quickly than others (Kester [B19]). Capacitive coupling frequently produces roughly equal positive and negative spikes that settle within a finite time interval. The glitch produced by switch timing differences is generally of greater concern (Kester [B19]).

Although the largest glitch is generally generated by a major carry transition around the MSB level (Stewart [B30]), it can be present at any code transition.

The transients observed in Figure 26 (D) and Figure 26 (E) is of particular concern. They are produced when some switches in the DAC are nominally turned off and others are turned on at the same time. These can introduce artifacts in THD and SINAD measurements.

The traditional measure of the glitch is the integral of the voltage output over the duration of the glitch in units of volt-seconds (Balestrieri [B3]). Typical glitch areas as described in data sheets are 0.5 nV s.

The glitch area (GA) is the time integral of the analog value of the glitch transient that falls outside a specified error band following a specified code change, as shown in Equation (49).

$$GA = \int_{T_G} (V_G - V_{ideal}) dt \quad (49)$$

where

T_G is the glitch duration, which starts at the time of the initiating clock edge and ends after the transient has settled to its final value

V_G is the analog value of the glitch transient

V_{ideal} is the ideal transition as modified by the analog response of the linearly modeled DAC response as illustrated in Figure 26 (B) and Figure 26 (C)

The glitch area is expressed in volt-seconds (Vs) or ampere-seconds (As) depending on the output units (Balestrieri [B2]).

Glitches are best observed with high bandwidth instruments often outside the capability of ATE. High-speed DACs are usually observed with a high bandwidth oscilloscope and rarely measured in production (Irons and Hummel [B14]).

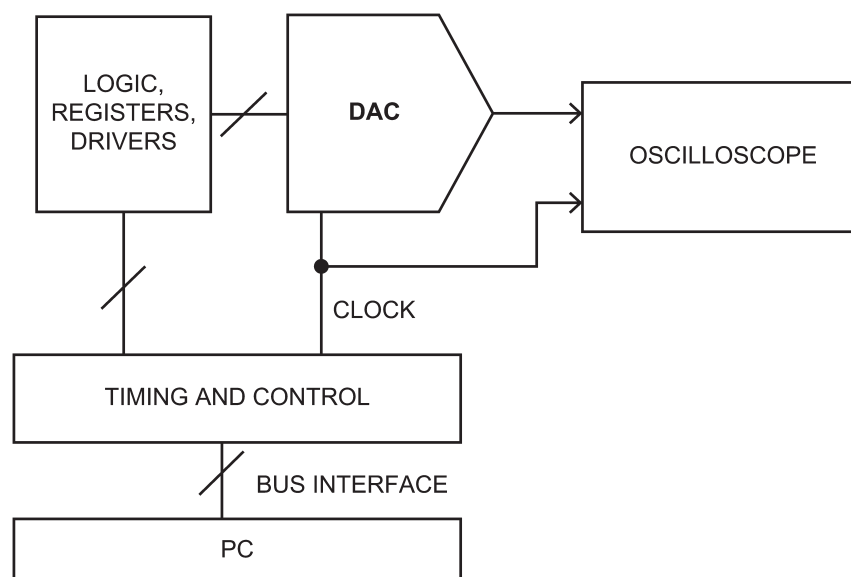


Figure 27—DAC glitch setup

Glitch parameters are estimated by using the test setup shown in [Figure 27](#), which is a specialized version of [Figure 17](#) for glitch testing. Due to the nature of the transients to be measured, a wideband fast-settling oscilloscope is needed to perform accurate measurements.

There are several considerations in selecting the proper scope. Preferably, the scope bandwidth should be at least three times the DAC output signal bandwidth in order to include the second and third harmonic components for a more accurate representation of the waveform. Overdrive should be checked by changing the scope sensitivity by a known factor and making sure that all portions of the waveform change proportionally. The sensitivity of the scope should be sufficient to measure the desired error band.

When making glitch measurements, it is generally better to make a direct connection between the DAC output and the 50-Ω scope input without the use of probes. If probes are required, compensated passive probes are preferable, but they should be used with care. In making the connection between the DAC and the scope, it is mandatory that a low impedance ground be maintained.

If the DAC output is designed to drive the virtual ground of an external current-to-voltage converter and does not have enough compliance to develop a measurable voltage across a load resistor, then an external op amp is required. Select an op amp that has a settling time that is at least three to five times faster than the DAC under test (Kester and Bryant [\[B20\]](#)).

12.3.2 Test method for DAC glitches

Set up the test configuration shown in [Figure 27](#). Program a transition between adjacent codes that are being tested. Generally, the measurement is most conveniently performed at major code transitions where the theoretical increment is 1LSB, and the deviations are not overwhelmed by the static DAC output changes. Collect the transient waveform in a waveform recorder. Observe the waveform. Calculate the glitch area from the net area of the glitch. An example of such a transition in a low-glitch DAC is shown in [Figure 28](#). Clock and data feedthrough may be present even if the transition is from one code to the same code. The glitches that are potentially troublesome occur when the codes are changing.

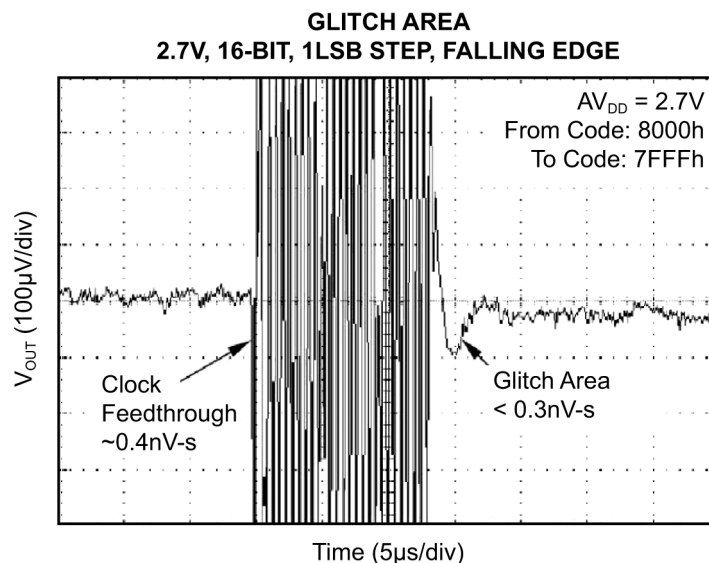


Figure 28—Glitch waveform

13. Step response parameters

13.1 General comments on step response parameters

A DAC's performance is a function of its dynamic characteristics. The DAC is assumed to have a configuration described by the diagram shown in Figure 5. Note that in the case no digital interpolation filtering is implemented (Figure 6 with $K = 1$), the DAC output is determined only by the reconstruction network response. If $K > 1$, the step response is affected by both the analog and digital (interpolation) filters.

13.2 Test method for acquiring an estimate of the step response

Use the test setup shown in Figure 17. Use a suitable waveform recording instrument as the measuring instrument. It should have step response aberrations no greater than one fourth of those expected from the DAC under test. The settling portion of the transient response will generally be measurable only with a specialized measurement setup. Using this 4:1 performance ratio as a guideline, a less than a 3% error in the transition duration estimate should be expected. For aberrations, the uncertainty in the measuring instrument transient parameters will add equal uncertainty in the corresponding estimated DAC quantities. If smaller uncertainties are required, then it may be necessary to construct specialized windowing circuits, which can reduce the effects of overdriving the waveform recorder input.

Before and after acquiring the actual transient response data, determine the (static) initial and final values of the step, which are respectively the baseline and topline or the topline and baseline. If the pre-acquisition measurements of the initial and final values differ significantly from the post-acquisition measurements of the initial and final values (due to system drift during the data acquisition), estimate the initial and final values of the step as the average of their respective pre- and post-acquisition measurements. Acquire a waveform or several waveforms of the DAC step response that are sufficiently long to include all features of interest, for example, precursors, and electrical and thermal settling. In order to produce high-resolution sampling of step response features and reduce errors due to aliasing, it may be necessary to use the method of equivalent time sampling, provided that the requirements on DAC output periodicity are met. To reduce the effects of random noise and aperture uncertainty in waveform recorder, it is recommended that multiple records of the transient response be ensemble averaged.

Significant error sources in acquiring the transient response estimate include aliasing, jitter, random noise, and measuring instrument imperfections.

The following clauses describe how to determine DAC parameters from measured step response data. The initial and final value of the applied input step signal should be specified in the test results.

13.2.1 Comment on test results

The transient response of some waveform recorders can be nonlinear when the slew rate of the DAC output signal exceeds the slew rate limit of the waveform recorder. The slew rate limit of the waveform recorder shall be higher than the slew rate of the DAC output step, in order to get accurate results. The statement of results should include the transition duration and amplitude of the applied step.

13.3 Slew rate limit

The slew rate is the limitation in the rate of change of output voltage, usually imposed by some basic circuit consideration such as that of a limited current charging a capacitor. The slewing speed of a DAC may be limited by the slew rate of the amplifier used as the output stage. Slew rate is measured in volts per second. A current output DAC will have slew rate limits given in units of amperes per second.

13.3.1 Slew rate limit test method

Record the step response for an output step having amplitude 10% of full-scale. Determine and store the maximum rate of change of the output transition. Repeat this process, increasing the amplitude of the input step each time. When the maximum rate of change ceases to increase with increasing step amplitude, slew limiting is taking place and the slew rate limit is the largest recorded value of the rate of change. If the measured slew rate continues to increase up to the maximum amplitude step, the slew rate limit can be reported as equal to or greater than the largest value measured.

13.4 Settling time parameters

One of the most common dynamic tests is settling time, defined as the amount of time it takes for a DAC to stabilize to its final static level within a specified error band after a DAC input code change has been applied. The straightforward approach to testing settling time is to digitize the DAC's output as the DAC input transitions from an initial code to a final code and then use the known time period between digitizer samples to calculate the settling time.

The final settled value is noted. The point at which the output last crosses the limits (e.g., $\pm 1/2$ LSB) is used to determine the settling time to that limit.

13.4.1 Test method for settling time and short-term settling time

Typically, settling time will be measured as the DAC transitions from minus full scale (–FS) to plus full scale (+FS) and vice versa, since these two tests represent the largest voltage swing. Settling time may be a difficult parameter to measure because DAC performance may be better than most available measuring instruments. A typical, modern digitizing oscilloscope does not have sufficient digitizing resolution or basic accuracy to measure DAC settling time directly. One commonly-used method uses an equivalent-time sampling oscilloscope with sufficient speed and sensitivity in conjunction with carefully devised limiting circuitry to help prevent overload of the vertical amplifier with its associated recovery time. An alternative method makes use of a tracker feedback loop as a fast and accurate sampling system.

Settling time is defined in reference to the 50% reference level instant of the step waveform. If the measurement is made using the special measurement setup described above, then there is no 50% instant with which to reference the measurement. In this case, the 50% reference level instant should be estimated.

Record the step response to an input code change using a waveform epoch sufficient to represent the step over the duration specified, or for at least one second when the duration is not specified.

Determine the time of occurrence of the first 50% point on the transition of the recorded waveform. Counting from that time, the settling time (or the short-term settling time) is the time at which the output waveform last enters the bound given by $V(t) \pm e$, where $V(t)$ is the value at the end of the specified duration, and e is the specified error. When the duration is not specified, $V(t)$ is the value one second after the beginning of the step.

To measure the long-term settling error, record the same step used to determine the short-term settling time, with a record that spans at least a one-second interval from the beginning of the step. The long-term settling error is the absolute difference between the value one second after the beginning of the step, and the value at the end of the specified duration following the step, expressed as a percentage of the step amplitude.

13.4.2 Comment on settling time

The term settling time refers to the time required to settle to the steady state, dc value, to within the given tolerance. The dc value is assumed to be the value after a constant input has been applied for at least one second. Changes that occur after one second are considered drift, and may be due to room temperature fluctuations, component aging, and similar effects.

The term *short-term settling time* refers to the time required to settle to a relative value (perhaps different from the steady-state value), defined as the value at the end of a specified duration, for waveform epochs less than one second. If static offset, gain, and linearity corrections are used to assign true values to short-term settling data, the results will have an uncertainty given by the long-term settling error. The uncertainty results because of longer-term settling phenomena, such as thermal imbalances that may occur after the short-term duration is complete, but that affect a steady-state measurement.

Note that only short-term settling time can be specified for ac-coupled DACs.

13.4.3 Filtering settling region of step for improved settling time measurement

The use of a suitable digital filter to reduce noise can improve the estimation of the settling time. The use of a filter is motivated by the fact that measuring settling time to within a small fraction of the applied step will be contaminated with the DAC output noise and the waveform recorder noise. Filtering is used to reduce noise commensurate with the uncertainty required of the measurement.

Since the combined noise standard deviation of the DAC and the waveform recorder is assumed to be larger than the waveform recorder's ideal quantization level, the noise may be described by a Gaussian noise process.

The effects of noise can be reduced by applying a time varying digital filter to the step response data. An easily implemented filter that is effective in many situations is an expanding window moving average filter. The window is chosen to have a duration of one sample (no averaging) before and during the region where the waveform is changing significantly, after which it expands linearly with data point index in the region where the waveform is changing slowly. The filtering operation may be written as shown in Equation (50).

$$y[n] = \frac{1}{2w[n] + 1} \sum_{j=-w[n]}^{w[n]} x[n+j] \quad (50)$$

where

$x[n]$ is the digitized step response
 $y[n]$ is the filtered digitized step response
 $w[n]$ is the filter window function

where

$w[n]$ is 0 for $1 \leq n \leq n_0$

$w[n]$ is $R \cdot \text{fix}(n/L)$ for $n > n_0$

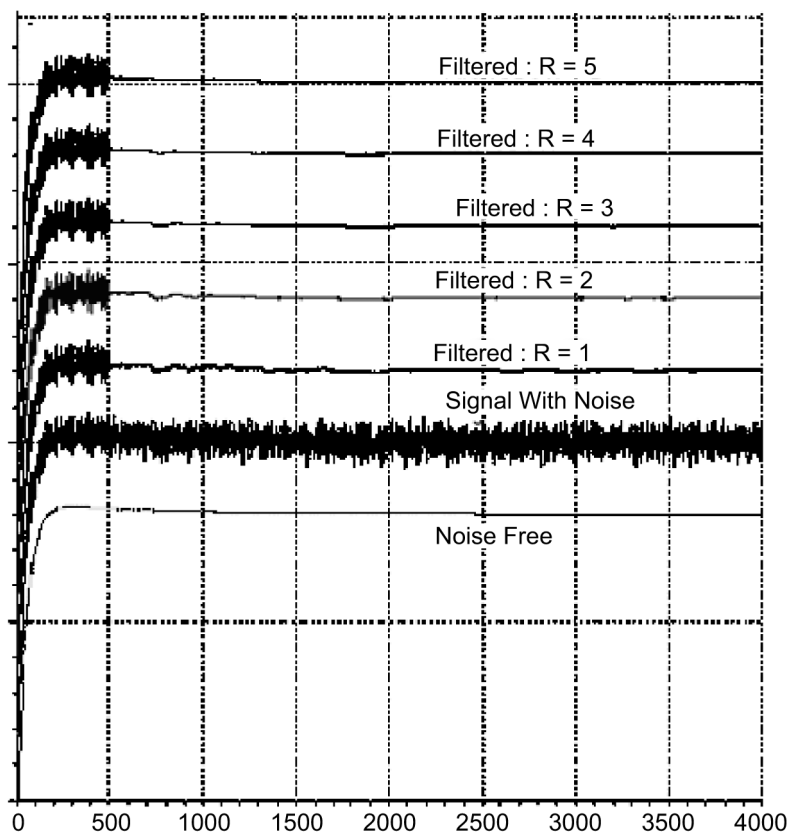
R is a filter parameter. The filter length increases by R samples every L points after the first n_0 samples

L is a filter parameter. The filter length increases by R samples every L points after the first n_0 samples

$\text{fix}(\alpha)$ returns the integer part of the argument α . When such a filter is used in the measurement of frequency response, the filter parameters should be reported along with the measurement results.

Figure 29 shows the effect of applying an expanded window moving average filter on a step-like waveform with additive noise. In this case, n_0 was 500, and L was 32. The effect of reducing noise in the part of the waveform beyond the transition region is the reduction of noise at all points in the frequency response.

Note that some caution in applying time variant filters is prudent. For example, if the window expansion is begun too early in the record, useful spectral information may be lost. A good rule of thumb is to begin the moving average at approximately the place in the waveform where the rate of change has decreased to the point that local variations in the waveform are comparable in size to the noise.



NOTE—The waveform contains two time constants separated by a factor of 10.

Figure 29—Example of noise reduction using an expanding-window moving average on step data for different values of R

13.5 Transition duration

Transition duration is the difference between the two reference level instants of the same transition. Unless otherwise specified, the two reference levels are the 10% and 90% reference levels.

13.5.1 Test method

Record the step response and determine the 10% reference level and the 90% reference level of the output transition. Linear interpolation should be used to determine the two reference levels when insufficient data points are available on the transition. The transition duration of the step response is the difference between the first 10% reference level instant and the last 90% reference level instant on the transition.

13.5.2 Comment on pathological test results

On some recorders, typically those that do not employ sharp cutoff anti-aliasing filters, the step response can be nonlinear. The degree of nonlinearity increases with the steepness of the applied step. Because of these nonlinear effects, the step response recorded on such a recorder can be misleading. To eliminate the gross nonlinearities, the transition duration of the applied step must be limited to above a minimum value. The value of this limiting transition duration is dependent upon the recorder under test. The statement of results should include the transition duration of the applied step.

13.5.3 Effects of a non-ideal step on transition duration and overshoot

The transition duration of the digitizer should be equal to or less than one-fourth of the transition duration of the DAC under test. If not, the observed transition duration will appear longer than the actual duration and the calculated frequency response will roll off faster than the actual response. Use of the one fourth rule should keep errors in transition duration and frequency response (out to the 3 dB frequency) to less than 3%.

13.5.4 Effect of jitter on transition duration

Jitter contributes both a systematic and a random component of error to a transition duration measurement. The random component can be reduced with averaging, but a bias will remain.

In the frequency domain, the bias can be represented by a low pass filter. In the time domain, averaging a signal, $x(t)$ in the presence of jitter results in the signal $x_j(t)$, which can be approximated by [Equation \(51\)](#).

$$x_j(t) = x(t) + \frac{\sigma_t^2}{2} \frac{d^2 x(t)}{dt^2} \quad (51)$$

where

σ_t is the standard deviation of the jitter

For a transition duration measurement, this relation provides a good approximation to the bias in the time domain when the jitter distribution spans a time window that is small compared to the true transition duration of the signal.

The transition duration of a signal averaged in the presence of jitter is increased according to the relation shown in [Equation \(52\)](#).

$$t_{d,jitter} \approx \sqrt{t_d^2 + (2.5 \sigma_t)^2} \quad (52)$$

This relation assumes that both the jitter distribution and the applied step signal are Gaussian or nearly so. The factor of $2.5\sigma_t$ is the 10% to 90% transition duration of the integral of the Gaussian jitter distribution.

Conversely, the true transition duration of a signal can be approximated from a jittered measurement, $t_{d,jitter}$, using the relation shown in Equation (53).

$$t_d \approx \sqrt{t_{d,jitter}^2 - (2.5 \sigma_t^2)} \quad (53)$$

This expression should be applied with some caution. The closer σ_t is to $t_{d,jitter}$, the more sensitive this expression becomes to one's knowledge of σ and to variance in the estimate of $t_{d,jitter}$. As a general rule, the expression is most useful for cases when σ is less than one fourth of $t_{d,jitter}$.

These relations assume that both the jitter distribution and the applied step signal are Gaussian or nearly so.

13.6 Overshoot and precursors

Pre-transition and post-transition overshoot and undershoot can be calculated from the samples collected during the DAC settling time test. These are defined in IEEE Std 181.

13.6.1 Test method

Record the step response. Calculate the parameters of interest following the method in Clause 5 of IEEE Std 181. Overshoot and undershoot can be strongly influenced by the data rate, update rate, and the filters used in the DAC step response measurement. These quantities shall be specified along with the overshoot and undershoot.

14. Interference-related DAC parameters

This clause discusses a number of parameters that represent some type of interference: interference between channels, interference between frequency bands, and interference between digital and analog signals. It also includes errors due to channel mismatch.

14.1 Multitone power ratio

Multitone power ratio (MTPR) is defined as a figure of merit for applications such as Asymmetric Digital Subscriber Line (ADSL), where groups of frequency domain impulses (tones) are uniformly spaced over the bandwidth of interest, with the characteristic that periodically a tone is “missing.” The output waveforms are analyzed to determine how much power has bled into the “missing” tone. The ratio of the power in the spectral notches to the power of the frequency components is the MTPR. Figure 30 illustrates the system input and system output of a MTPR test.

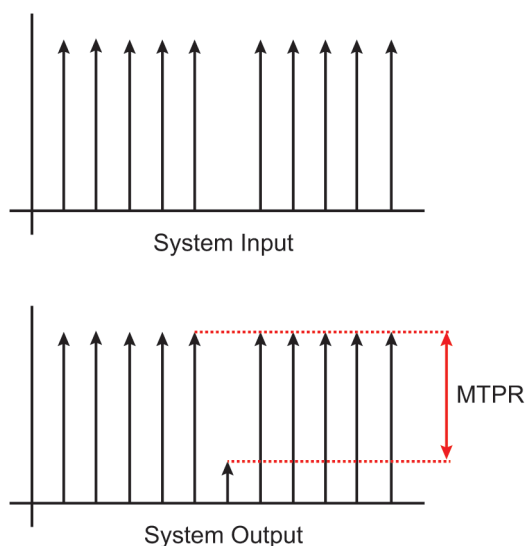


Figure 30—MTPR

14.1.1 MTPR test method

A MTPR test is commonly performed to determine the non-linearities of a discrete multitone (DMT) system, also known as orthogonal frequency division multiplexing (OFDM). A DMT waveform is a signal consisting of multiple discrete frequency components. To execute a MTPR test, a DMT waveform is created with frequency components that span the region to be tested. Additionally, this waveform contains missing frequency components, or spectral notches. The frequency range, number of frequency component, and missing tones are specified by the corresponding standard.

Once the DMT waveform has been created, it is loaded onto the DAC under test, using the test setup described in 4.2. Measure the true rms voltage at one of the carriers and at the frequency of the missing tone. The MTPR is given by Equation (54).

$$MTPR = 20\log\left(\frac{rms\ carrier}{rms\ missing\ tone}\right) \quad (54)$$

14.1.2 Comments

The drawback to DMT in general is the bothersome peak-to-average (PAR) ratio associated with the channel waveform due to possible subcarrier instantaneous summation. Each carrier is given a starting phase to constrain the PAR. Specifically, each tone's starting phase is adjusted to establish a desired PAR.

14.2 Crosstalk (channel separation, channel isolation)

Crosstalk occurs in devices that contain multiple DACs or multiple-channel DACs, when the channels are imperfectly isolated. Typical coupling mechanisms that cause crosstalk include the following:

- Power supply coupling
- Capacitive or inductive coupling
- Internal ground coupling

Digital input currents can couple into the output signal via the power supplies or ground currents.

Capacitive coupling between adjacent channels or device pins can couple unwanted high-frequency components into the outputs.

Ground currents internal to the device can cause the internal ground to be different from the device zero reference causing spurious frequency components to be injected into the output. This may be a function of the rate at which the digital inputs are switched.

14.2.1 dc crosstalk measurement

DC crosstalk is the dc change in the output level of one DAC channel in response to a change in the output of another DAC channel. It is measured with a full-scale output change on one DAC channel while monitoring another channel. It is expressed in V/V or in dB for a voltage output DAC.

A DAC output channel can experience dc shifts when another DAC channel is subjected to loading. The output load on one channel is varied while the reference DAC is monitored. The output voltage level is monitored at the output of another DAC channel relative to a full-scale output change of the reference DAC. It is expressed in V/mA. This coupling is called load crosstalk.

This method is then repeated for all possible permutations based upon the number of DAC channels per device or system setup. See Figure 31 for the block diagram of a test configuration.

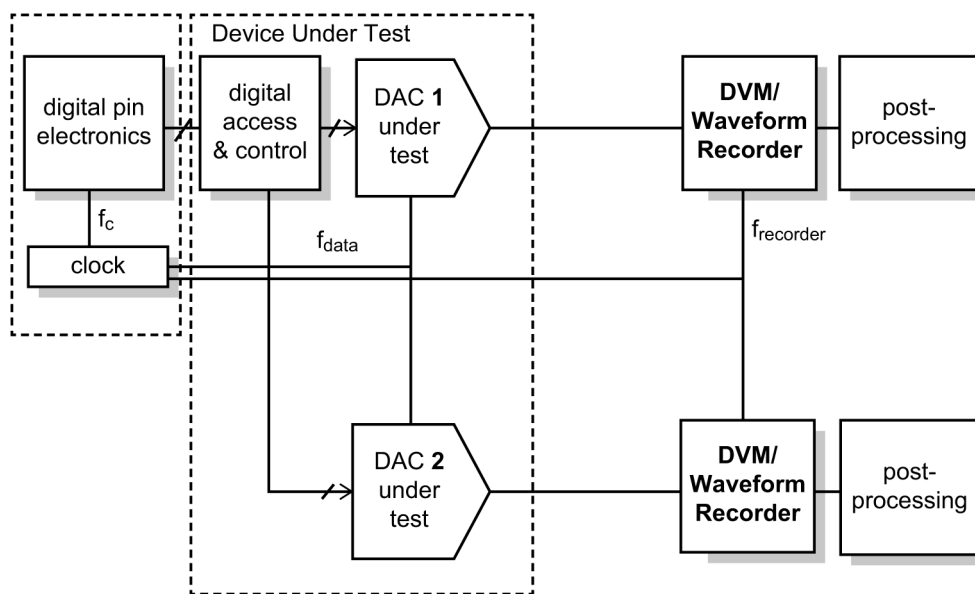


Figure 31—Block diagram of multi-channel DAC test showing two channels

The following series of steps effectively tests dc crosstalk:

- 1) Program DAC 1 to +FS
- 2) Program DAC2 to +FS
- 3) Measure the value at the DAC1 output into variable V1
- 4) Program DAC2 to −FS

- 5) Measure the value at the DAC1 output into variable V2
- 6) Compute the dc crosstalk as shown in Equation 55

$$\text{Voltage Crosstalk} = \frac{V2 - V1}{FSR} \quad (55)$$

Repeat these steps for all other DAC channels.

Load crosstalk is measured by traversing the following steps:

- 1) Program DAC 1 to +FS
- 2) Program DAC2 to +FS
- 3) Measure the value at the DAC1 output into variable V1
- 4) Apply a load to DAC2 of current I1
- 5) Measure the value at the DAC1 output into variable V2
- 6) Compute the dc crosstalk as shown in Equation 56

$$\text{Load Crosstalk} = \frac{V2 - V1}{I1} \quad (56)$$

14.2.2 ac channel-to-channel isolation test method

The ratio, in dB, of the amplitude of the signal at the output of one DAC channel to a sine wave on the output of another DAC channel is a measure of the ac channel-to-channel isolation.

The DAC channel being monitored is set to a dc static level (typically a nominally 0 output) while another DAC is sourcing a full-scale sine wave. The level of isolation is typically frequency- and amplitude-dependent and directly proportional to the sine-wave frequency and amplitude.

Sources of crosstalk include packaging, the DAC reference voltage, supply lines, layout, etc.

Collect a waveform from the waveform recorder that is monitoring the output of the DAC with a nominal constant value. Take a DFT of the waveform and estimate the rms value of the component at the frequency that is being generated by the other DAC. Record this value as V1. Collect a waveform of the output of the DAC that is generating the sine wave. Take a DFT of the waveform and estimate the rms value of the component at the frequency that is being generated as V2. The ac channel-to-channel isolation in dB is given by the following expression:

$$\text{ac channel-to-channel isolation} = 20 \log \frac{V_2}{V_1} \quad (57)$$

14.2.3 Digital crosstalk test method

Digital crosstalk refers to the glitch impulse transferred to the output of one DAC channel in response to a full-scale code change in the input register of another DAC channel. It is measured in standalone mode. The area of the glitch is calculated using a time integral of the glitch impulse and is expressed in units of versus

Unless otherwise specified, the glitch is measured by monitoring the output of the channel under test while it is programmed to a nominally zero output. Other channels in the device are exercised with code transitions. The output of the DAC is digitized by a waveform recorder and the output in the time domain is examined for

glitches. The area of the glitch is calculated using a time integral of the glitch impulse and is expressed in units of versus

14.2.4 Comment on crosstalk parameters measurement

All crosstalk tests should be performed by using a high bandwidth waveform recorder capable of a sampling rate of several times the f_s of the DAC and sufficient voltage sensitivity.

14.3 Channel matching

Channel matching is a measure of the relative amplitude difference between one channel and another channel both statically and as a function of frequency. It is typically expressed in percent of full-scale-range. The method used is to measure the full scale amplitude of each DAC output while the DAC is generating a sine-wave.

14.3.1 Channel matching test method

The test is implemented by performing the following sequence of steps:

- 1) Measure the amplitude of the frequency response of the first channel. Record this as $F[1, f_1]$, $F[1, f_2]$, $\dots$, $F[1, f_{M-1}]$, $F[1, f_M]$, where $f_1, f_2, \dots, f_{M-1}, f_M$ is the set of M frequencies that have been selected for measurement.
- 2) Measure the frequency responses of all other channels, 2 to N , as $F[2, f_{1:M}]$, $\dots$, $F[N, f_{1:M}]$ where $F[i, f_j]$ is the response for channel i at frequency f_j .
- 3) Compute $\bar{F}[i]$, where:

$$\bar{F}[i] = \frac{1}{M} \sum_{j=1}^M F[i, f_j] \quad \text{for } i = 1 \text{ to } N \quad (58)$$

Channel matching, $MATCH$, for each channel and frequency is given by the following expression:

$$MATCH[i, f_j] = \frac{F[i, f_j] - \bar{F}[j]}{\bar{F}[j]} \quad (59)$$

14.3.2 Comment on channel matching

For the purpose of simplicity, this definition and test method ignores the matching of phase, which can be important.

14.4 Channel skew

Some types of DACs are designed for use in matched groups. In a circuit with multiple DACs, the channel skew corresponds to the time mismatch between the transition times of two DACs. This parameter is used when many DACs work synchronously (Maloberti [B21]).

For example, a color palette RAM DAC is a device that is used to produce colors on video monitors. A RAM DAC uses a random access memory (RAM) lookup table to turn a single color value into a set of three DAC output values, representing the red, green, and blue intensity of each pixel. These DAC outputs need to change almost simultaneously to produce a clean change from one pixel color to the next (Roberts and Burns [B27]). A possible skew between the time responses gives rise to blur pictures or ghost effect (Maloberti [B21]). The channel skew, representing the degree of mismatch between the three DAC outputs, is measured by

digitizing each DAC output and comparing the timing of the 50% point of each output to the 50% point of the other outputs (Roberts and Burns [B27]). There are three skew values (R-G, G-B, and B-R) as illustrated in Figure 32.

DAC channel skew is an important parameter for I/Q DAC channel pairs, too. It is also a difficult parameter to measure considering the large potential skew in instruments and transmission paths. If limits are to be tested in production, the best approach is to use focused calibration techniques to software deskew the sample clock and DAC output instrument and transmission paths for the DACs. Normally, skew is fixed by design and need only be observed with a high bandwidth scope, being careful to place scope probes equidistant from the DUT (Jasper [B15]). Measurement is typically specified as an absolute time.

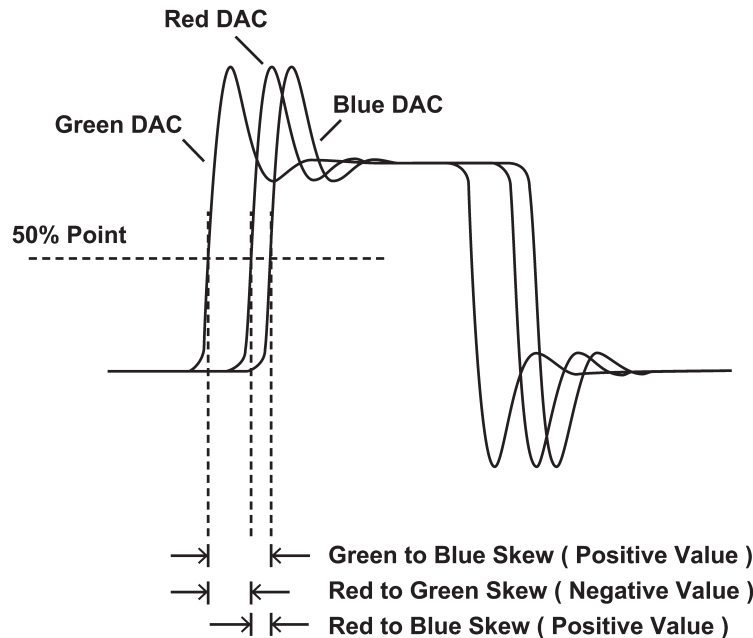


Figure 32—Three RAM DAC channel skew values (see Roberts and Burns [B27])

15. Frequency response parameters

15.1 General comments on frequency response parameters

The frequency response parameters that are of interest for a DAC depend on the intended application of the DAC and on its architecture. DAC frequency response describes its output signal characteristic versus its input, which could be a digital input or, for a multiplying DAC, the analog (reference) input.

The DAC response with respect to its digital input is typically specified differently depending on the architecture and application. For instance, resistor string DACs, R-2R ladder DACs or non-interpolated current source switching DACs are usually characterized by their settling time. For a zero-order-hold DAC, the digital-to-analog frequency response typically will not be specified, because the response is dominated by the sinc response of the zero-order hold. On the other hand, a DAC employing any kind of interpolation, delta-sigma or pulse width modulation, as well as DAC having built-in sinc correction, frequency-response flatness is usually specified.

Some DACs are intended to produce useful output at the image frequencies of the digital input. For these it is common to specify bandwidth-related parameters for each Nyquist band in which the DAC is intended to function.

15.2 Reference input bandwidth

Generally, analog bandwidth is only applied to reference input for a multiplying DAC. Like its analog circuit counterpart, bandwidth, measured in Hertz, is defined as the difference of upper and lower cutoff frequencies, for which, the signal gain is -3 dB or sometimes -1 dB versus the maximum gain. dc is the lower cutoff frequency for most of DACs. This parameter is similar to its equivalent for analog linear circuit, like amplifier or attenuator.

15.2.1 Reference input bandwidth test method

In order to measure the reference input bandwidth with best accuracy, a constant digital code corresponding to highest analog output shall be used to drive the digital-to-analog converter input. This code is usually all 1s at all bits. The reference input of the converter is connected to a signal generator, while its output shall be connected to a digitizer or a spectrum analyzer. The signal received by the converter reference input shall be within normal converter operating range. Commonly used input signal is sine wave. Output signal amplitude is measured while input sine-wave frequency changes. For each different frequency used, the ratio of the output amplitude to that of input is calculated. The cutoff frequencies are usually defined as amplitude gain drop to $\sqrt{2}/2$ times the device maximum gain. This value is often approximated to -3 dB. A constant gain demanding system requires the cutoff frequency be defined at -1 dB.

For bandwidth characterization, cutoff frequency could be determined by successive approximation method. First send a sine wave at a frequency close to one of the device spec cutoff frequency. If the gain is greater than the target cutoff frequency gain, then choose a second frequency greater than the first frequency for upper cutoff frequency or lower than the first frequency for lower cutoff frequency. Otherwise, choose a second frequency lower than the first frequency for upper cutoff frequency or later than the first frequency for lower cutoff frequency. If the gain is greater than the target cutoff frequency gain, then choose a third frequency greater than the second frequency for upper cutoff frequency or lower than the second frequency for lower cutoff frequency. Otherwise, choose a third frequency between the first frequency and the second. Repeat this process until the gain reaches the target value.

The previous iterative method is time consuming for production test. Device test time also becomes unpredictable. To accelerate test time, interpolation or extrapolation method could be used.

15.2.2 Reference input bandwidth test method based on step response

Record the step response of the DAC using a sampling and digitizing instrument having a bandwidth at least four times greater than the expected bandwidth of the DAC. If necessary, use equivalent-time sampling. Select the sampling rate high enough to give negligible aliasing errors based on the bandwidth of the recorder.

The signal used at the reference input is a step. The signal generator used to generate the step should have a bandwidth at least four times that which is to be measured.

Take a record of data with an epoch long enough to ensure that the step has settled to within the desired accuracy. Take the first difference of this record, that is, take the discrete time derivative, to find the impulse response. Obtain a Fourier transform of the impulse response using a non-weighted (rectangular) window. The result is the frequency response of the DAC. DAC bandwidth is the frequency at which the frequency response is attenuated by $1/\sqrt{2}$ from the dc value of the frequency response.

15.3 Digital-to-analog-conversion frequency response

The digital-to-analog-conversion frequency response is the response for the entire DAC, including the interpolation filter (if any) and the analog filter. Just as for an analog circuit, such as an amplifier, the DAC frequency response is the ratio of the magnitude of an output sine wave to that of the corresponding input sine wave. The frequency response can also be defined as the Fourier transform of the impulse response, i.e., the response to an input that is a discrete-time impulse. There are some differences between DAC frequency responses and analog-circuit frequency responses. The most significant difference is that a DAC produces multiple sinusoidal outputs, one in each Nyquist band, for a single sinusoidal input.

A plot of the impulse response of an ideal zero-order-hold DAC is shown in Figure 33.

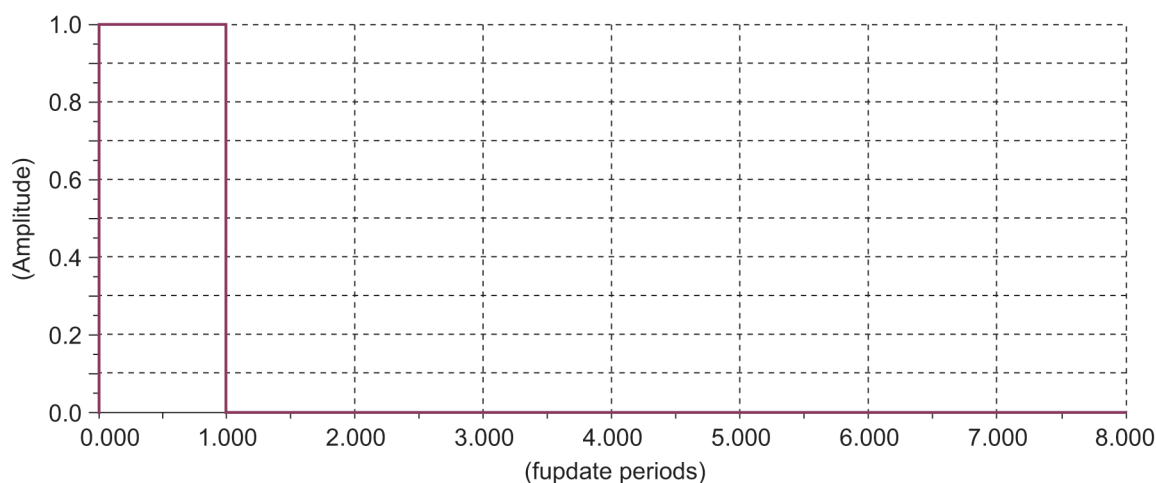


Figure 33—Ideal internal DAC impulse response

The digital interpolation filter (Figure 6) produces an output from its input via the transfer function shown in the following equation:

$$\text{filter out}[n] = \sum_{k=0}^{N-1} h[k] \cdot \text{filter in}[n - k] \quad (60)$$

where

n	is the index into a continuous stream of data
N	is the number of defined points in the filter response. Generally, N is finite and the filter is categorized as a Finite Impulse Response (FIR) filter
$h[k]$	is the filter response at tap point k
$\text{filter in}[n - k]$	is the filter input at update point $n - k$

A typical digital interpolation filter with finite impulse response exhibits the impulse response shown in Figure 34. Note that the major part of the response is delayed significantly from the input signal.

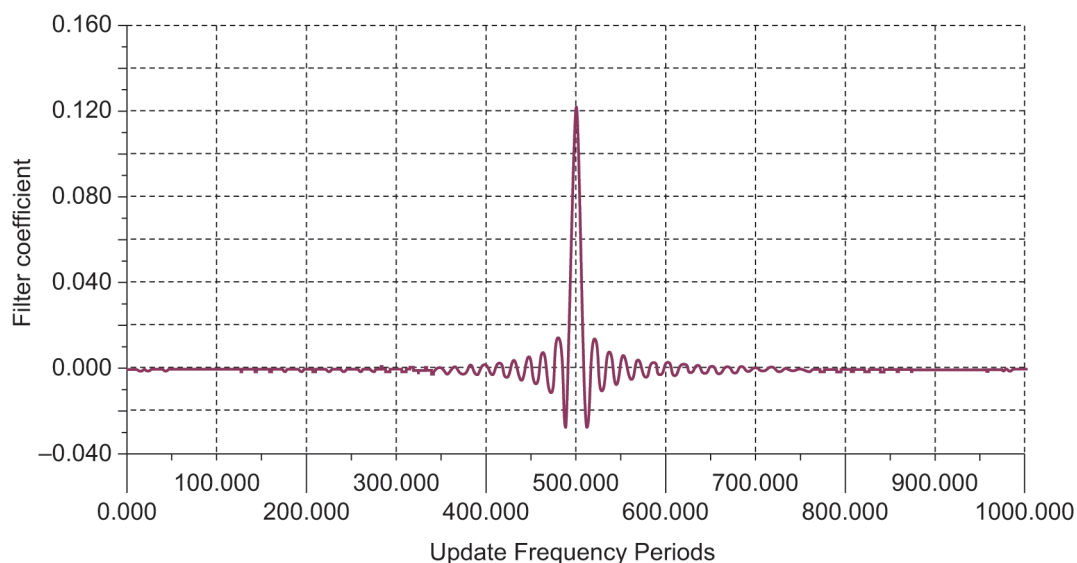


Figure 34—Impulse response of 1000 tap FIR digital interpolation filter

A measured impulse response can be translated into the frequency response by taking the DFT of the impulse response. The result for the example is shown in [Figure 35](#). The response is mirrored around multiples of the update frequency when applied to a perfect DAC.

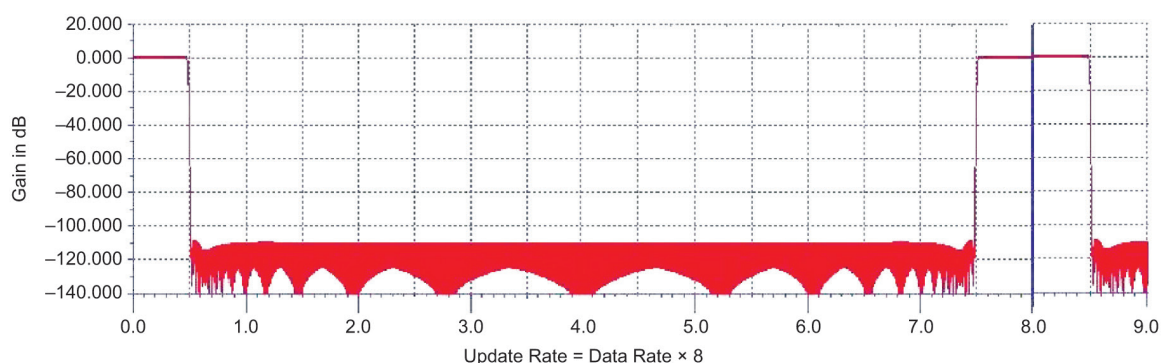


Figure 35—Frequency response corresponding to impulse response of [Figure 34](#)

The frequency response can generally be divided into two regions—the pass-band and the reject-band. Notice that the reject-band in this example, starts from slightly above $0.5 f_{DATA}$ and continues up to $(K \times f_{DATA} - 0.5 \times f_{DATA})$. The rejection in the reject-band is about 110 dB. In the example shown, $K = 8$.

The pass-band response is shown in detail in [Figure 36](#). Note that the ripple in the pass-band was one of the design parameters for the digital interpolation filter (0.1 dB pk-pk).

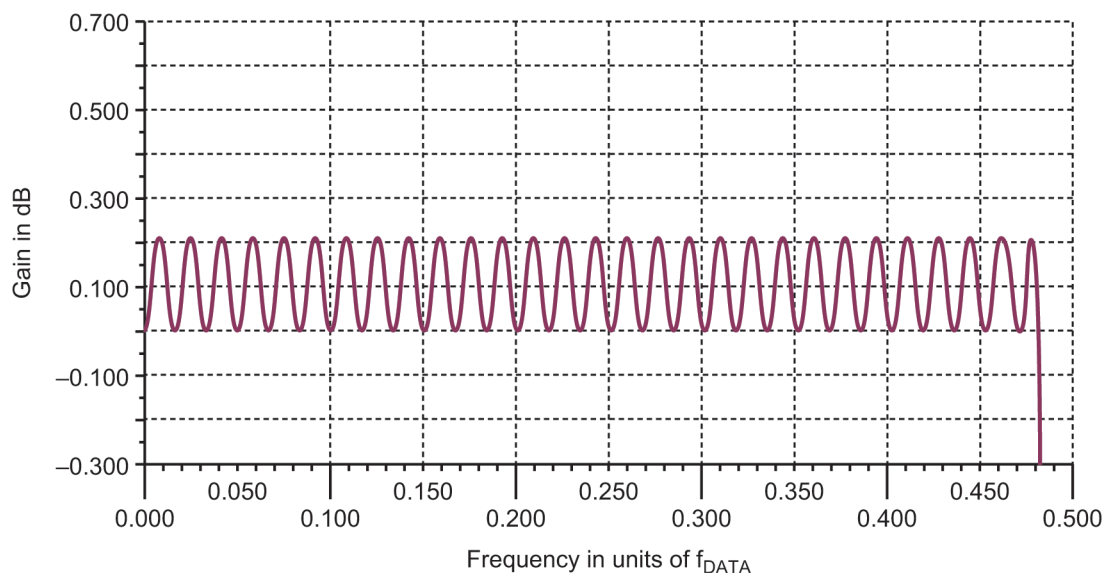


Figure 36—Plot of Pass-band response of filter shown in Figure 34

The spectrum shown in Figure 35 continues indefinitely for multiples of the update rate, but the digital interpolation filter output is modified by the analog filter. The resultant frequency response will generally attenuate the image frequencies heavily. The frequency response of a five-pole analog filter is shown in Figure 37.

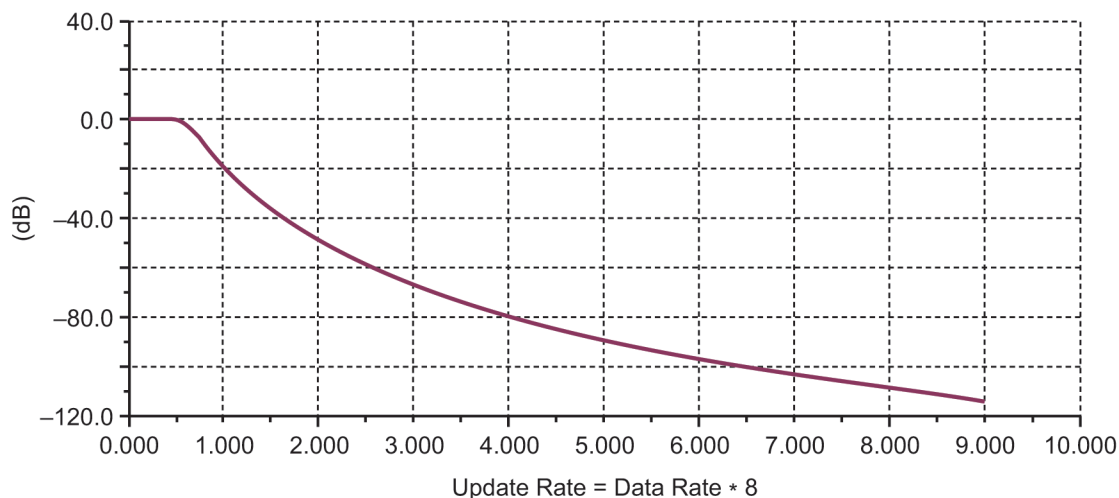


Figure 37—Plot of 5 pole Butterworth analog low pass filter frequency response

The composite response of the filters is shown in Figure 38.

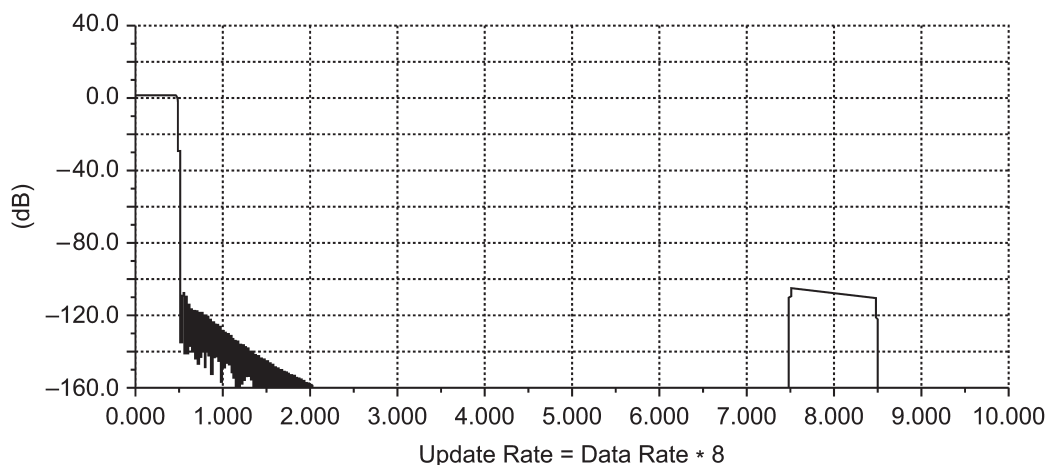


Figure 38—Plot of composite interpolation filter and analog filter response

The impulse response of the analog filter and the digital interpolation filter combination is illustrated in Figure 39.

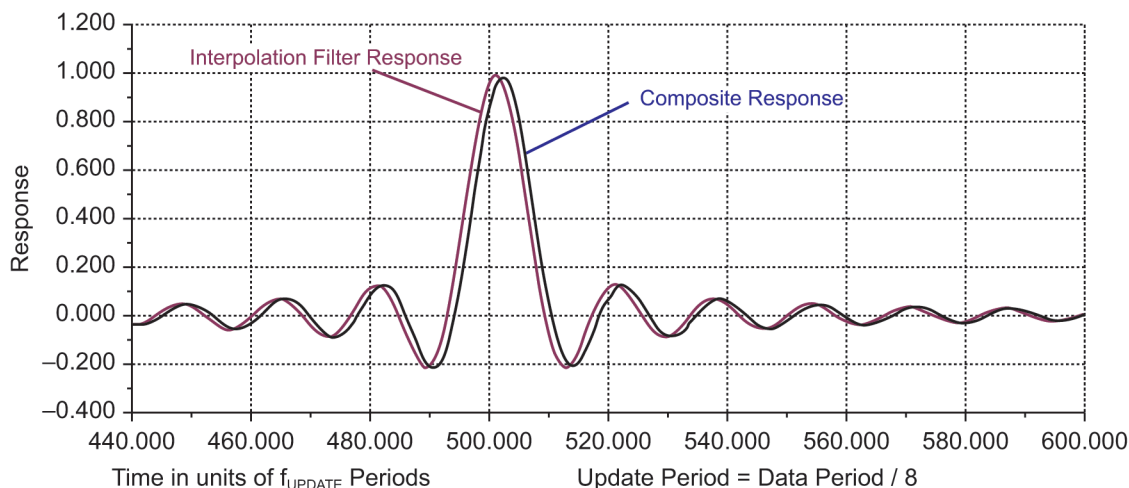


Figure 39—Plot of composite digital interpolation filter and analog filter impulse response shown together with digital interpolation filter response

The connection between the impulse response and the frequency response suggests a test method for computing the frequency response from the impulse response. One should be able to measure the impulse response, and compute from it the frequency response by taking a DFT of the impulse response.

The computation of the out-of-band attenuation from the impulse response can be very sensitive to quantization error and noise, and it may be impractical to use the impulse response for this purpose except to get a coarse lower bound on the attenuation. A perfect 18-bit resolution measurement introduces enough quantization error to raise the floor of the computed reject band to -90 dB, instead of the -110 dB shown in Figure 35. This effect is illustrated in Figure 40.

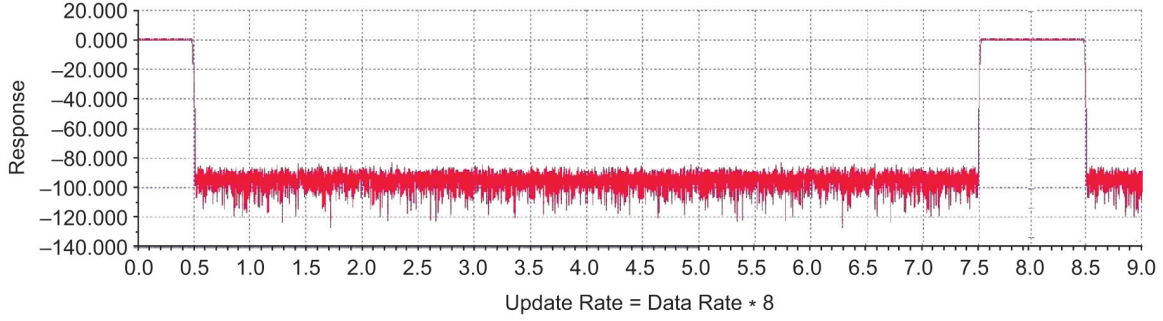


Figure 40—Plot of frequency response computed from impulse response using a perfect 18-bit resolution waveform recorder

In many instances, it may be preferable to measure the frequency response with a sine wave source.

15.3.1 Digital-to-analog conversion frequency-response test using sine waves

This test requires a digital code generator that will produce a sinusoidal output waveform from the DAC under test. The code generator should be capable of producing valid data codes at a rate at least as fast as the maximum update rate supported by the DAC. It is important that waveform samples be generated so that a maximum number of levels are included in the test code pattern. For a test pattern consisting of N_{test} samples and J waveform periods, waveform samples may be computed with Equation (61).

$$y_n = \text{round}\left(\frac{2^N - 1}{2} \cos\left(\frac{2\pi n J}{N_{test}}\right) - 0.5\right) \quad n = 0, 1, \dots, N_{test} - 1 \quad (61)$$

The round function helps to guarantee that a sine pattern will be applied that will include all codes from 100000...₂ (Negative Full Scale) to 011111...₂ (Positive Full Scale). If the DAC is unipolar or offset binary, the sign bit should be complemented. The test pattern will contain the maximum number, 2^N , of distinct levels if N_{test} is a sufficiently large power of 2 and J is odd, but it is only necessary that N_{test} and J be mutually prime and for N_{test} to be sufficiently large. The Nyquist frequency limits the value of J to be less than $N_{test}/2$. It is often considered useful to make N_{test} large enough so that all DAC codes are excited. To excite every code in the sine wave, $\pi/2$ times as many samples are needed as there are codes. Typically, N_{test} is chosen to be four times the number of codes.

The frequency response at any image frequency of the input signal is the ratio of the magnitude of the output signal at that frequency to the magnitude of the input signal converted to output units.

15.3.2 Test method for frequency response from impulse or step response

This method produces the same result as the sine-wave test method, as described in 15.3.1, to the extent that the DAC is linear and the measurement uncertainty is low. The immediately following instructions are for the use of a step response. The minor corrections for using an impulse response follow.

- 1) Record the step response of the DAC as described in 13.2. Use averaging, if necessary, to reduce the effects of noise. The digitizer making the recording should have adequate bandwidth and sampling rate, which may require operating it in equivalent time sampling mode.
- 2) Take a waveform with an epoch long enough to ensure that the impulse response is negligible outside of the epoch, i.e., the step response differs negligibly from a constant.
- 3) Calculate the impulse response from the step response using Equation (62).

- 4) Obtain a Fourier transform of the impulse response using an unweighted (rectangular) window. The result is the frequency response of the DAC.

The linear (delay) term in the phase portion of the transform is arbitrary, since the equivalent delay of the recorded signal with respect to the input waveform is indeterminate. The nonlinear phase components can be made more apparent by rotating the phase plot (subtracting a linear fit to the plot) so that the phase shift is zero at a selected frequency. The result is a plot of the nonlinear phase contribution.

If an impulse response is used Step 3) is eliminated, but the resulting frequency response will have to be appropriately normalized. If a step response is used, the impulse response is calculated by using [Equation \(62\)](#).

$$h(t) = \frac{s(t + t_{\text{DATA}}) - s(t)}{t_{\text{DATA}}} \quad (62)$$

where

$h(t)$ is the impulse response
 $s(t)$ is the measured step response
 t_{DATA} is the reciprocal of f_{DATA}

Note that if the sampling rate of the waveform recorder is not an integer multiple of the data rate of the converter under test, interpolation will have to be used to obtain $s(t + t_{\text{DATA}})$.

16. Power supply parameters

16.1 Power supply currents

Measuring the power supply current of each of the DAC supplies is a fast way to help ensure that the device has no gross defects. Power supply currents should also be measured to verify that the device draws the power specified in the DAC data sheet.

16.1.1 Power supply current test method

This test is used for measuring power supply currents of a DAC. Equipment capable of applying prescribed voltages to the test circuit power supply terminals and measuring the currents flowing in these terminals shall be provided. If environmental testing is to be performed, the test chamber has to be capable of maintaining the device under test at a specified temperature.

After stabilizing the device at the specified test temperature, DAC inputs should be conditioned to put the device into its worst-case power dissipation. The supply current flowing into the DAC should settle to a stable value before it can be measured. The tester and the device interface board circuits should also settle to a stable value. The supply current will generally vary with the load supplied to the DAC under test. The load conditions should be specified when performing the measurements.

The current flowing in the supply terminals shall be measured with the supply at their specified operating levels.

16.2 Power supply voltage effects

Changes in the power supply voltage can affect the DAC output. Different effects may be observed if the power supply voltages are changed statically or if an ac waveform is superimposed on the power supply voltages. The effect of power supply changes is usually a function of the digital input code.

Although information about performance of the DAC with different power supply voltages can be achieved by performing gain, offset, and linearity measurements at the minimum, nominal and maximum supply voltages given in the DAC specifications, most of the information about the effects of power supply changes can be measured by simply observing the effect on the output level when the power supply voltages are changed.

AC signals superimposed on the power supply voltages can modify the output of the DAC. Most effects of high-frequency injection are mitigated and complicated by properly bypassing and filtering the supplies; however, it is fairly straightforward to measure the effects of low-frequency ripple superimposed on the supplies, and these effects are usually described by evaluating the power supply rejection ratio (PSRR_{AC}).

16.2.1 Power supply voltage effects test method

This test requires adjustable power supplies powering the DAC under test, and a suitable output measuring device (such as a voltmeter). Static PSRR is measured by implementing the following steps:

- 1) Set the power supply voltages to their nominal values. The power supply voltage is defined as V_{PS}
 - 2) Apply a full scale code to the DAC input
 - 3) Measure the corresponding DAC output as $M1$
 - 4) Change one of the power supply voltages to a new value (typically the specified tolerance on the power supply voltage). The change in the power supply voltage is defined as ΔV_{PS}
 - 5) Measure the new DAC output as $M2$
 - 6) Calculate the PSRR using Equation (63):
- $$PSRR = \frac{(M2 - M1)/FSR}{\Delta V_{PS}/V_{PS}} \quad (63)$$
- 7) Restore the power supply to the nominal voltage
 - 8) Repeat step 3) through step 6) for all the other power supplies. If desired, repeat at other input codes
 - 9) The PSRR should be less than the limits set by the data sheet

Note that PSRR is a dimensionless ratio. It can be expressed in units of %/% or in dB.

If it is desired to measure the effects of a ripple voltage superimposed on the supply voltages, then PSRR_{AC} can be measured. PSRR_{AC} at low frequencies is usually tested by adding a low-frequency (typically 1 kHz) ripple directly through the regulator of the DAC power supply. A suitable ac output measuring device (such as a spectrum analyzer) is connected to the device output. The component of the ripple at the output is noted. The PSRR_{AC} is then calculated using Equation (64). High-frequency ripple can be injected either with transformers or with resistive ladders. Because high-frequency ripple injection often conflicts with bypass capacitor requirements, there is no general way to describe how to superimpose high-frequency noise on the power supply.

DAC ac PSRR is defined as shown in Equation (64):

$$PSRR_{AC} = \left(\frac{\Delta V_{out}/FSR}{V_{ripple}/V_{PS}} \right) \quad (64)$$

where

V_{ripple} is the rms ripple voltage added to power supply
 ΔV_{out} is the corresponding change in the rms output level as measured by a spectrum analyzer
 FSR is the full scale range of the DAC under test

V_{PS} is the nominal value of the power supply

The units of $PSRR_{AC}$ can be simply noted as “%FSR / %Power Supply Change” or it can be noted in dB by the relationship shown in Equation (65):

$$PSRR_{dB} = 20\log(PSRR) \quad (65)$$

This test requires an adjustable power supply that can accept a low-frequency ac ripple source as an input.

16.2.1.1 Comments on power supply rejection ratio

PSS (power supply sensitivity) and PSR (power supply rejection) are terms that are often used to describe the effect of power supply changes on the output. This standard considers only the PSRR parameter, defined according to the previous clauses, to describe the effect of power supply changes on the DAC output.

16.3 Ground currents

DAC ground represents the reference node for all DAC measurements, and it should be as stable and noise free as possible. There are different philosophies that claim to be the best way to ground a device; any ground system that provides a low noise floor is a good one. In connecting and grounding a DAC, determining where the currents flow is essential. Most precision DAC have separate digital and analog return pins. Grounding problems can be reduced if proper attention is paid to where the currents flow and where voltage drops are being generated. A block diagram of the returns in a typical DAC application is shown in Figure 41.

In Figure 41, seven nodes are identified that are nominally at “ground” potential. These nodes are described as follows:

- a) The return of one of the power supplies that are powering the digital portion of the DAC.
- b) The Digital Return of the DAC. Note that the digital input currents are nominally returned through this node.
- c) The Analog Return of the DAC. This node is usually considered the reference node. The internal workings of the DAC try to maintain the specified level of the DAC output with reference to this node. DAC data sheets usually demand that the Digital Return and the Analog Return be connected somewhere in the system to chassis ground.
- d) The Load Return. The load that is applied to the output returns current to this node.
- e) Instrument Low. The instrument that is measuring the DAC output has its return at this node.
- f) A second Power Supply Return. Current drawn by the DAC returns through this line. This power supply powers the analog portion of the DAC.
- g) Chassis Ground. The system return that is eventually connected to the ac power line return.
- h) Return of the Digital Stimulus Generator.
- i) Return of the reference source.

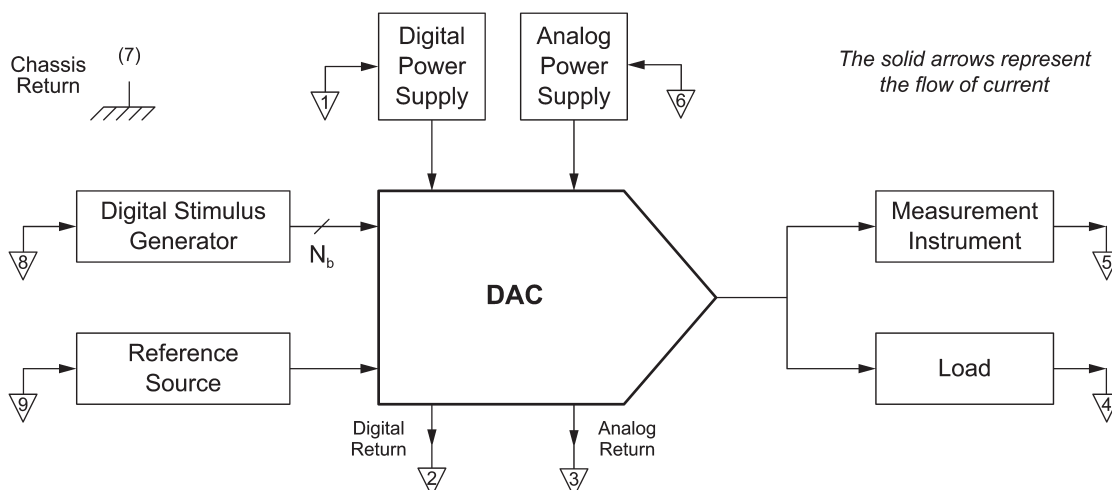


Figure 41—Block diagram of DAC “ground” nodes

Some general rules are vital for good measurement quality. These include the following:

- 1) Connect (5)—the Measure Instrument LO to the Analog Return.
- 2) Connect the (7)—Chassis Return to (2)—Digital Return and to (1)—Digital power supply return.
- 3) The Reference input to the DAC should have its return referenced to (3)—Analog Return.
- 4) The analog Power Supply, PS2, should have its return (6) connected through an isolated line to (3)—the Analog Return of the DAC.
- 5) The (2)—Digital Return and the (3)—Analog Return should be strongly connected through a ground plane. Note that many DACs have the analog and digital grounds internally connected or merged into a single pin.
- 6) The (4)—Load Return should be connected with an isolated connection to (3)—The Analog return.

Many of these rules can be judiciously violated, but the engineering of the ground interconnect should always take into consideration the finite impedance of the ground interconnects, and the paths of the current flows—both static and dynamic.

Annex A

(informative)

DAC architectures

Over the years, several architectures have been developed to implement the digital-to-analog converter (DAC) function. This annex describes some of the architectures. [Table A.1](#) summarizes the architectures.

Table A.1—DAC architectures

Subclause	Description	Features	Variations	Advantages	Disadvantages
A.1	Binary Weighted	N Resistors N switches	Current sources or capacitors can replace resistors	Simple, Minimum components. Minimal Digital encoding	Difficult to manufacture Not necessarily monotonic
A.2	Resistor String	2N–1 Resistors, 2N Switches	Parallel current sources can replace resistors	Simple, all resistors are identical. Inherently monotonic	2N Resistors
A.3	Segmented DAC	Tapped potentiometer fed into tapped potentiometer	Current or voltage output	Inherently monotonic. Fewer resistors than resistor string	Resistor matching errors may degrade INL
A.4	R-2R (Ladder)	2N resistors, 2N switches	Equal current sources can drive R-2R ladder	Easy to fabricate, all resistors are similar in value. Constant impedance	Not inherently monotonic. Impedance lower than other DAC architectures
A.5	Delta-Sigma	Input data stream feeds a digital interpolation filter, which drives a 1-bit DAC followed by a filter	DAC can be more than 1 bit	High linearity. Easily integrated into standard digital ICs	Low bandwidth Can be noisy May have artifacts
A.6	Pulse-Width Modulation	Pulse width proportional to input data	Current or voltage pulses	Very low cost Low internal power dissipation	Low bandwidth Nonlinear frequency response

A.1 Binary weighted

The binary-weighted (or binary-encoded or binary-scaled) DAC utilizes a number of elements (current sources, resistors, or capacitors) that are binary weighted (Kester [B17]). The voltage-mode binary-weighted resistor DAC is shown in [Figure A.1](#). This DAC architecture is not inherently monotonic and is actually quite hard to manufacture successfully at high resolutions. In addition, the output impedance of the voltage-mode binary DAC changes with the input code (Kester [B17]).

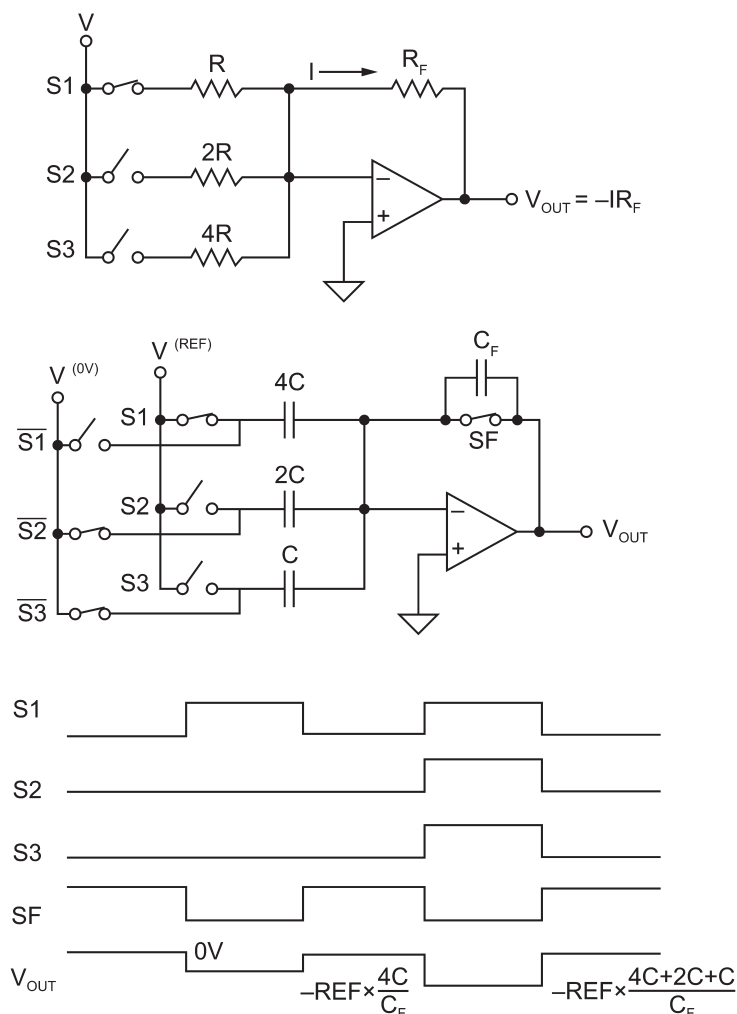


Figure A.1—A binary resistor (a)/capacitor (b) DAC (see Morrison [B22])

The general architecture of a binary weighted current-steering DAC is shown in Figure A.2. The switches are controlled by the input bits, b_i , where $i = 0, 1, \dots, N-1$ and N is the number of bits. b_0 is the LSB and the corresponding current source has the dc value I_{LSB} .

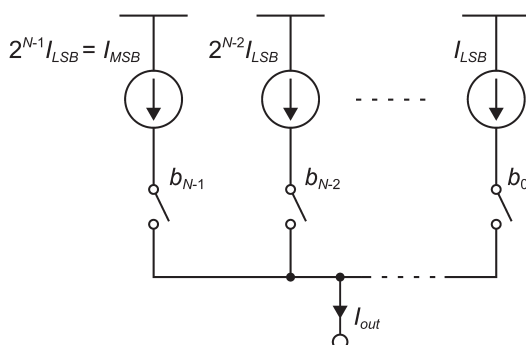


Figure A.2—An N-bit current-steering binary weighted DAC (see Nuttall [B23])

A.2 Segmented

Segmented architectures can be used for both current- and voltage-output DACs. The voltage across the decoded resistor in a Kelvin divider circuit can be further subdivided to build a voltage-segmented DAC (Figure A.3). This subdivision can be achieved through a second Kelvin divider circuit (in which case the architecture is called a Kelvin-Varley divider). The output of the overall DAC remains monotonic as long as the individual segments are monotonic. Monotonicity is achieved when the offset of the buffer amplifiers is sufficiently small.

Segmentation has the added benefit of reducing the number of resistors required to achieve a given resolution, allowing smaller die sizes. Thus, it is common for high-resolution DACs to be segmented. Overall linearity is still determined by the resistor matching.

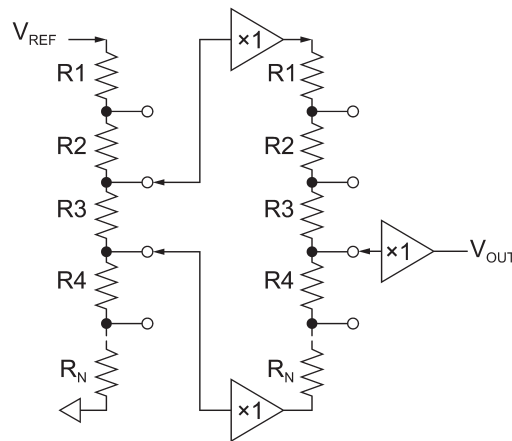


Figure A.3—Voltage segment DAC (see Morrison [B22])

A.3 R-2R

One of the most common DAC building-block structures is the R-2R resistor ladder network shown in Figure A.6. It uses resistors of only two different values, and their ratio is 2:1. An N -bit DAC requires $2N$ resistors, and they are quite easily trimmed. There are also relatively few resistors to trim. There are two ways in which the R-2R ladder network may be used as a DAC—known respectively as the voltage mode and the current mode. Each mode has its advantages and disadvantages. In the voltage mode, R-2R ladder DAC shown in Figure A.4, the “rungs” or arms of the ladder are switched between V_{REF} and ground, and the output is taken from the end of the ladder. The output may be taken as a voltage, but the output impedance is independent of code, so it may equally well be taken as a current into a virtual ground (Maloberti [B21]).

The voltage output is an advantage of this mode, as is the constant output impedance, which eases the stabilization of any amplifier connected to the output node. Additionally, during switching the arms of the ladder between a low impedance V_{REF} connection and low impedance ground, the capacitive glitch currents tend not to flow in the load. On the other hand, the switches should operate over a wide voltage range (V_{REF} to ground). This is difficult from a design and manufacturing viewpoint, and the reference input impedance varies widely with code, so that the reference input should be driven from a very low impedance. In addition, the gain of the DAC cannot be adjusted by means of a resistor in series with the V_{REF} terminal (Maloberti [B21]).

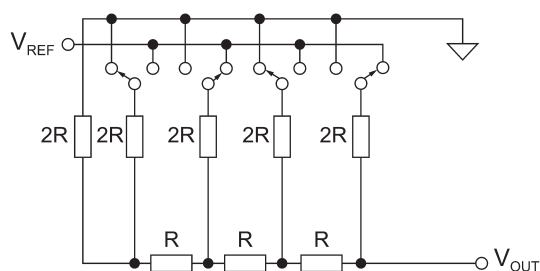


Figure A.4—R-2R DAC in voltage mode (see Morrison [B22])

In the current-mode R-2R ladder DAC, shown in Figure A.5, the gain of the DAC may be adjusted with a series resistor at the VREF terminal, since in the current mode, the end of the ladder, with its code-independent impedance, is used as the VREF terminal; and the ends of the arms are switched between ground (or, sometimes, an “inverted output” at ground potential) and an output line, which shall be held at ground potential. The normal connection of a current-mode ladder network output is to an op amp configured as current-to-voltage (I/V) converter, but stabilization of this op amp is complicated by the DAC output impedance variation with digital code. This architecture is sometimes referred to as an “inverted R-2R” DAC. Current-mode operation has a larger switching glitch than voltage mode since the switches connect directly to the output line(s) (Maloberti [B21]).

However, since the switches of a current-mode ladder network are always at ground potential, their design is less demanding and, in particular, their voltage rating does not affect the reference voltage rating. If switches capable of carrying current in either direction (such as CMOS devices) are used, the reference voltage may have either polarity, or may even be ac. Such a structure is one of the most common types used as a multiplying DAC (MDAC) (Maloberti [B21]).

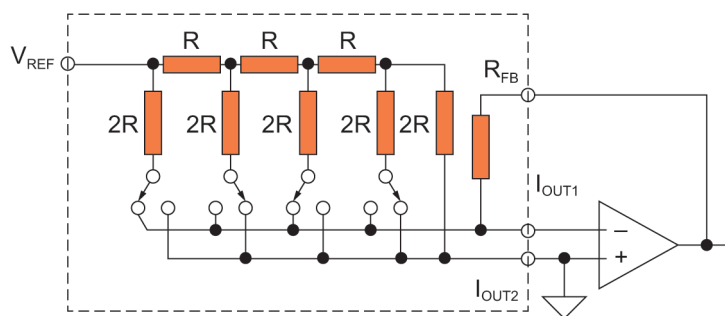


Figure A.5—R-2R DAC in current mode (see Morrison [B22])

A.4 Delta-Sigma

Delta-Sigma architectures can be used to trade bandwidth for resolution. The architecture consists of a digital Delta-Sigma modulator and a DAC of lower resolution. (See Figure A.6.) In the extreme case, (e.g., in audio DACs) the DAC could have a single bit.

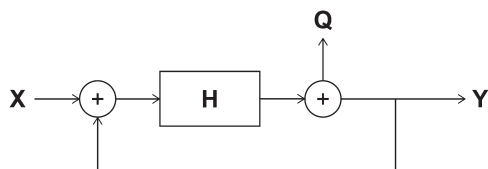


Figure A.6—DAC Delta-Sigma architectures

In this diagram, X denotes the input, Y denotes the output, H denotes the loop filter, and Q denotes the error or noise introduced during quantization. Thus, we have the following equation:

$$Y = \frac{H}{1+H}X + \frac{1}{1+H}Q \quad (\text{A.1})$$

When H tends to zero, the output tends to Q while H tends to be larger than 1, the output tends to the input. H can be used to redistribute the quantization noise outside the band of interest. For instance, if H is a low pass filter with a high gain, the output will tend to the input with very little quantization noise at low-frequency band.

A simple implementation of a Delta-Sigma architecture, the duty-cycle DAC, operates as follows.

Consider a series of 8-bit numbers that correspond to a 1 kHz sine wave sampled at 64 kHz. Proceed through the following steps:

- a) Load the first value into a counter
- b) Count down the counter value to zero at a rate of 100 MHz. While the counter is greater than 0, turn on a 1-bit DAC. When the counter reaches 0, turn off the 1-bit DAC until the 100 MHz clock has completed 256 counts
- c) Repeat this process for all the numbers after the 256 counts have been completed
- d) Low-pass-filter the waveform so as to generate the desired output

The unwanted spectral content of the unfiltered waveform is large so the filter design may be difficult. Many of the properties of modern Delta-Sigma DAC implementations can be understood by noting the spectra produced by a duty-cycle DAC.

A.5 PWM DAC

The generation of an analog voltage using a digital Pulse-Width-Modulated signal is known as a PWM DAC. PWM DACs are widely used in very low-cost applications, where accuracy is not a primary concern. Standalone DACs, however, are used for applications requiring higher accuracy.

Figure A.7 shows a basic configuration of the PWM DAC. The digital microcontroller (MCU) outputs a PWM signal to an RC low pass-filter. The PWM pulse train's digital value becomes an analog voltage, when it passes through the RC filter. At a given period of time, the analog output is proportional to the PWM pulse's high durations.

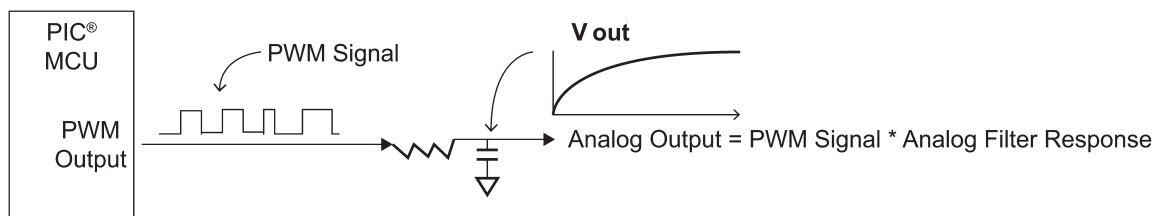


Figure A.7—Block diagram of a PWM DAC (see Oppenheim and Schafer [B24])

A PWM signal is defined as a digital signal with a fixed frequency, but a varying duty cycle. Figure A.8 illustrates a PWM signal. The PWM period (T) is the time interval required to complete one full PWM cycle. The duty cycle is the ratio of the high duration (t) to the total period (T).

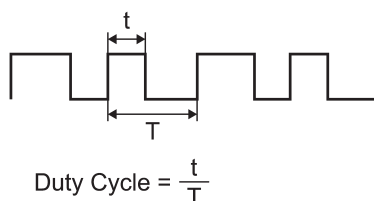


Figure A.8—PWM signal (see Oppenheim and Schafer [B24])

The PWM signal and RC-filter circuit parameters affect the analog output's resolution, amplitude, settling time, and ripple. The PWM DAC's limitations are clearly demonstrated by analyzing the interaction of the PWM parameters and the RC filter. A better understanding of the relationship between these parameters enables designers to optimize the PWM to best suit their application's requirements, while minimizing design time.

A.6 Multiple DAC architecture

Different types of DAC architectures have their different advantages and disadvantages. An intuitive conclusion from this is to use a combination of different architectures to improve the overall performance. This is illustrated in Figure A.9, where the different sub-converters can be of completely different types. Each sub-DAC converts M_i bits and in the simplest configuration, the total resolution is equal to the sum of the resolutions of the sub-converters. The sub-converters have different reference levels (Nuttall [B23]).

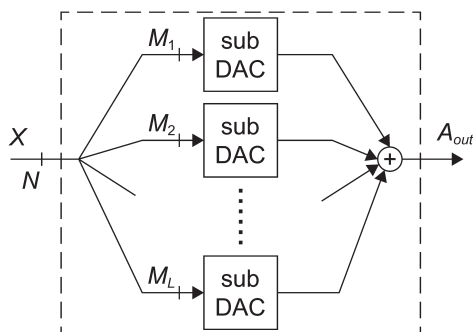


Figure A.9—A multiple DAC architecture using a combination of a number of different types of DACs (see Nuttall [B23])

Annex B

(informative)

Sine-wave fitting algorithms

B.1 An algorithm for three-parameter (known frequency) least-squares fit to sine wave data using matrix operations

This algorithm provides a least squares method for fitting digitized waveform data to a sine wave in the case where the frequency of the sine wave is known. The algorithm is presented using matrix notation.

The equation for the sine wave is given in [Equation \(B.7\)](#).

Assuming the data record contains the sequence of M samples $x[1], x[2], \dots, x[M]$ taken at times $t_1, t_2, \dots, t_M$, this algorithm finds the values of A_0 , B_0 , and C_0 that minimize the following sum of squared differences [see [Equation \(B.1\)](#)].

$$\sum_{n=1}^M [x[n] - A_0 \cos(2\pi f_0 t_n) - B_0 \sin(2\pi f_0 t_n) - C_0]^2 \quad (\text{B.1})$$

where

f_0 is the known frequency of the sine wave applied to the ADC input

Since the equation is linear, a closed form (noniterative) solution can be computed. To find the values for A_0 , B_0 , and C_0 , first create Matrix (B.2), Matrix (B.3), and Matrix (B.4).

$$D_0 = \begin{bmatrix} \cos(2\pi f_0 t_1) & \sin(2\pi f_0 t_1) & 1 \\ \cos(2\pi f_0 t_2) & \sin(2\pi f_0 t_2) & 1 \\ \vdots & \vdots & \vdots \\ \cos(2\pi f_0 t_M) & \sin(2\pi f_0 t_M) & 1 \end{bmatrix} \quad (\text{B.2})$$

$$\mathbf{x} = \begin{bmatrix} x[1] \\ x[2] \\ \vdots \\ x[M] \end{bmatrix} \quad (\text{B.3})$$

$$\mathbf{s}_0 = \begin{bmatrix} A_0 \\ B_0 \\ C_0 \end{bmatrix} \quad (\text{B.4})$$

In matrix notation, the sum of squared differences in [Equation \(B.1\)](#) is given by that shown in [Equation \(B.5\)](#).

$$(\mathbf{x} - D_0 \mathbf{s}_0)^T (\mathbf{x} - D_0 \mathbf{s}_0) \quad (\text{B.5})$$

where

$(\cdot)^T$ is the transpose of the vector or matrix $(\cdot)$

The least-squares solution, $\hat{s}_0$, that minimizes Equation (B.5) is given by Equation (B.6).

$$\hat{s}_0 = (D_0^T D_0)^{-1} (D_0^T x) \quad (\text{B.6})$$

Details regarding the implementation of the three-parameter and four-parameter sine wave fitting methods can be found in by Renczes et. al. in [B26].

The fitted sine wave comprises the following samples:

$$y[n] = A_0 \cos(2\pi f_0 t_n) + B_0 \sin(2\pi f_0 t_n) + C_0 \quad (\text{B.7})$$

The residuals, $r[n]$, of the fit are given by:

$$r[n] = x[n] - A_0 \cos(2\pi f_0 t_n) - B_0 \sin(2\pi f_0 t_n) - C_0 \quad (\text{B.8})$$

The root-mean-square (rms) error is given by:

$$e_{rms} = \sqrt{\frac{1}{M} \sum_{n=1}^M r[n]^2} \quad (\text{B.9})$$

B.2 An algorithm for four-parameter (general use) least-squares fit to sine-wave data using matrix operations

This algorithm provides a least-squares method for fitting digitized waveform data to a sine wave in the case where the frequency of the sine wave is not known. The algorithm is given in matrix notation. The equation for the sine wave is the same as that given in Equation (B.7) except that the input sine-wave frequency, f_0 , is not known. Since the equation is no longer linear, a closed form solution cannot be computed, and iterative techniques should be used.

Assuming the waveform contains the sequence of M samples, $x[1], x[2], \dots, x[M]$, taken at times $t_1, t_2, \dots, t_M$, this algorithm uses an iterative process to estimate the parameters A_0 , B_0 , C_0 , and f_0 that minimize the following sum of squared differences:

$$\sum_{n=1}^M [x[n] - A[i] \cos(2\pi f[i] t_n) - B[i] \sin(2\pi f[i] t_n) - C[i]]^2 \quad (\text{B.10})$$

where

- i is the iteration index
- $A[i]$ is the i -th estimate of A_0
- $B[i]$ is the i -th estimate of B_0
- $C[i]$ is the i -th estimate of C_0
- $f[i]$ is the i -th estimate of f_0

The algorithm is given below.

- 1) Set iteration index $i = 0$.
- 2) Make an initial estimate $f[0]$ of the frequency f_0 of the waveform. The frequency may be estimated by using a DFT (either on the full record or a portion of it), or by taking the inverse of the average time

between zero crossings, or simply by using the best measurement of the applied input frequency. A very effective general method is to use the interpolated Fast Fourier Transform described in Schoukens, et al. [B28] and Bilau, et al. [B6]. This uses an interpolation formula on the DFT coefficients of the waveform and is used in the software available at Virosztek, et al. [B31].

- 3) Perform a pre-fit using the three-parameter matrix algorithm to estimate $A[i]$, $B[i]$, and $C[i]$.
- 4) Set $i = i+1$ for the next iteration.
- 5) Create the following matrices:

$$\mathbf{x} = \begin{bmatrix} x[1] \\ x[2] \\ \vdots \\ x[M] \end{bmatrix} \quad (\text{B.11})$$

$$\mathbf{D}[i] = \begin{bmatrix} \cos(2\pi f[i-1]t_1) & \sin(2\pi f[i-1]t_1) & 1 & -A[i-1]t_1 \sin(2\pi f[i-1]) \\ \cos(2\pi f[i-1]t_2) & \sin(2\pi f[i-1]t_2) & 1 & -A[i-1]t_2 \sin(2\pi f[i-1]) \\ \vdots & \vdots & \vdots & \vdots \\ \cos(2\pi f[i-1]t_M) & \sin(2\pi f[i-1]t_M) & 1 & -A[i-1]t_M \sin(2\pi f[i-1]) \end{bmatrix} \quad (\text{B.12})$$

$$\mathbf{s}[i] = \begin{bmatrix} A[i] \\ B[i] \\ C[i] \\ \Delta f[i] \end{bmatrix} \quad (\text{B.13})$$

- 6) Compute the least-squares solution, $\mathbf{s}[i]$.

$$\mathbf{s}[i] = (\mathbf{D}[i]^T \mathbf{D}[i])^{-1} (\mathbf{D}[i]^T \mathbf{x}) \quad (\text{B.14})$$

Note that although the value of the least squares fit is given by Equation (B.14), the introduction of the frequency update may result in an ill-conditioned matrix, $\mathbf{D}[i]$. The issue can be handled by the appropriate scaling of the corresponding matrix column (see Renczes [B26]). Additionally, the required pseudo-inverse matrix may be computed by a more numerically stable method, such as the Q-R decomposition (see Stewart [B30]).

- 7) Update the frequency estimate using:

$$f[i] = f[i-1] + \Delta f[i] \quad (\text{B.15})$$

where

$\Delta f[i]$ is obtained from Equation (B.13)

- 8) Repeat Steps d) through g), re-computing the model based on the new values of $A[i]$, $B[i]$, $C[i]$, and $f[i]$, calculated from the previous iteration. It appears to be best to iterate a fixed number of times. Based on experience, six iterations have proven more than adequate. This method doubles the number of significant digits in $f[i]$ at each iteration and converges very rapidly.

The residuals, $r[n]$, of the fit are given by:

$$r[n] = x[n] - A[i]\cos(2\pi f[i]t_n) - B[i]\sin(2\pi f[i]t_n) - C[i] \quad (\text{B.16})$$

The rms error is given by:

$$e_{rms} = \sqrt{\frac{1}{M} \sum_{n=1}^M r[n]^2} \quad (\text{B.17})$$

Annex C

(informative)

Discrete Fourier transforms and windowing

C.1 Discrete Fourier transforms and windowing

A discrete Fourier transform (DFT) is a mathematical operation that converts sampled data from the time domain to the frequency domain (Stearns and Hush [B29], Oppenheim and Willsky [B25]). (See IDFT below for the inverse operation.) In this standard, the DFT is defined for a waveform $x[n]$ made of N samples as:

$$X[k] = \sum_{n=0}^{N-1} x[n] \times \exp(-j2\pi kn/N) \quad \text{for } 0 \leq k \leq N-1 \quad (\text{C.1})$$

where

N is the number of sequential samples in the waveform

The inverse discrete Fourier transform (IDFT) is the mathematical operation that converts sampled data from the frequency domain to the time domain. The formula is as follows in Equation (C.2):

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \times \exp(+j2\pi kn/N) \quad \text{for } 0 \leq k \leq N-1 \quad (\text{C.2})$$

In this standard, the IDFT includes the requisite $(1/N)$ normalization.

The frequency corresponding to $X[k]$, f_k , is

$$f_k = \frac{k}{N} f_s \quad \text{for } 0 \leq k \leq N-1 \quad (\text{C.3})$$

where

f_k is the frequency corresponding to the k th DFT output complex value
 f_s is the sampling frequency

Note that the frequency components above the $\frac{f_s}{2}$ are the equivalent representation of the negative spectrum of the sampled signal, and for real signals their transform values are the complex conjugates of the values for the corresponding positive frequencies. The frequencies f_k are separated by an interval $\Delta f = f_s/N$. The interval of frequencies from $f_k - \Delta f/2$ to $f_k + \Delta f/2$ is called the k^{th} frequency bin. Associated with any frequency, f , is the bin number, $f/\Delta f$. If f is at the center of the k^{th} frequency bin, the bin number will be the integer, k . Non-integer values of the bin number correspond to frequencies that are not at the center of a bin.

$X[f_k]$ may be written in place of $X[k]$ to make the frequency dependence more obvious. For more information on properties of the DFT, see Stearns and Hush [B29] or Oppenheim and Willsky [B25].

An example of DFT was used to generate the amplitude spectrum shown in Figure C.1. It shows the amplitude spectrum of a signal sampled at 100 samples per second. The Nyquist frequency is 50 samples per second. The signal is a 21 Hz sine wave with amplitude 1 with second harmonic distortion with amplitude 0.5 and third

harmonic distortion with amplitude 0.25. The third harmonic, which is at 63 Hz is aliased down to 37 Hz. Note that one of the third harmonic terms is at the correct frequency of 63 Hz.

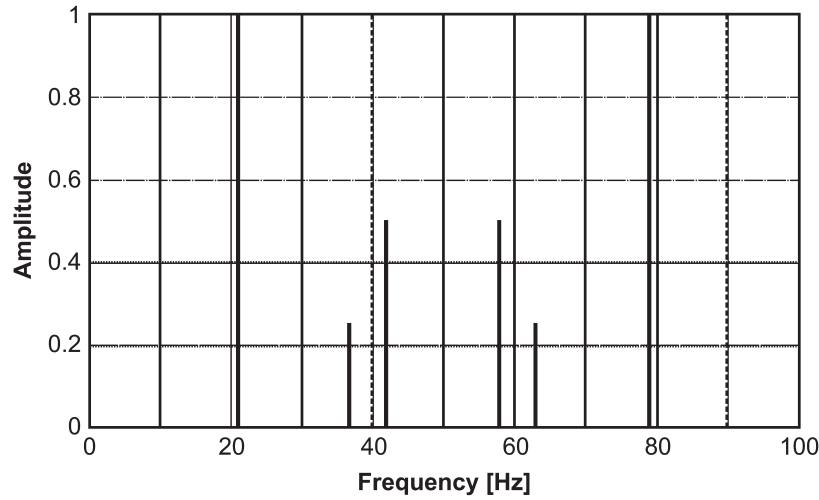


Figure C.1—Example DFT showing the negative frequencies mapped up to the frequencies between the Nyquist frequency and twice the Nyquist frequency

The “energies” in the time and frequency representations of the data are related by the Parseval relation shown in Equation (C.4).

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2 \quad (\text{C.4})$$

C.2 Windowed DFT

A windowed DFT is defined in Equation (C.5). The window elements, $w[n]$, are usually maximized around the center of the data interval and approach zero at the ends of the interval.

$$X_w[k] = \sum_{n=0}^{N-1} w[n]x[n]\exp(-j2\pi nk/N) \quad (\text{C.5})$$

Windowing is multiplication in the time domain and corresponds to convolution in the frequency domain. Some authors refer to the function w as the temporal weighting function and its Fourier transform, W , as the window function; whereas some refer to w as the window function. When all the window factors are equal to one, i.e., no windowing, the window is called the rectangular window. Window functions other than the rectangular are used to reduce the effects of spectral leakage, which is described next in C.4.

C.3 Spectral averaging

It is sometimes desirable to average the spectral components from a number of waveforms. To do this:

- Acquire K waveforms of N points each at sample frequency f_s .

- Let $x_l[n]$ represent the l -th waveform of sine-wave data for $l = 1, 2, \dots, L$.
- For each $x_l[n]$ waveform, compute the DFT, $X_l[k]$, where k is an integer from 0 to $N-1$. The L sets of data are used to compute an averaged magnitude spectrum of the DFT at each basis frequency f_k by Equation (C.6):

$$X_{avg}[k] = \frac{1}{L} \sum_{l=1}^L |X_l[k]| \quad k = 0, 1, 2, \dots, N-1 \quad (\text{C.6})$$

The averaged spectral magnitude, X_{avg} , is used because it has a smaller variance than the non-averaged spectral magnitude, X . The standard deviation of the random errors in X_{avg} is less than $|X|$ by a factor approximately equal to the square root of L (Jenkins and Watts [B16]).

C.4 Spectral leakage

In case of coherent sampling, the rectangular-windowed DFT of a sine-wave produces a single spectral line. If the frequency of the sine-wave is not at the center of a bin, the DFT will produce lines in all frequency bins (Stearns and Hush [B29], Oppenheim and Willsky [B25]). The leakage can be observed in the Fourier transform, W , of the window function. Two examples are shown in Figure C.2. The solid line is the Hann window (defined in C.6) and the dashed line is the rectangular window. The horizontal axis is the DFT bin number, and the vertical axis is W expressed in dB. Also since $W[-f] = W[f]^*$, where $(\cdot)^*$ is the complex conjugate, the data are not shown for negative frequency values. If the signal being analyzed is a sine-wave with amplitude A and frequency with a bin number of x (not necessarily an integer), the k^{th} spectral line will have an amplitude of $A \times W[k-x]$.

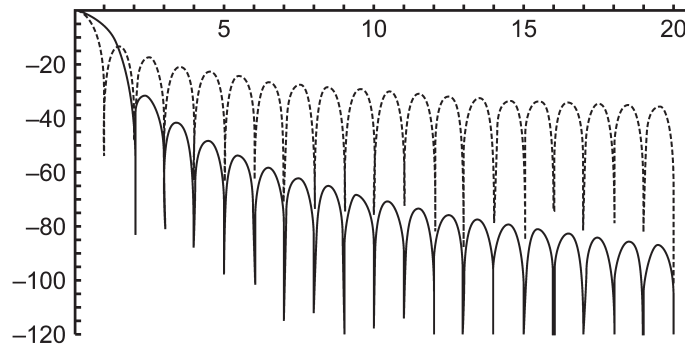


Figure C.2—Fourier transform of the Hann and the rectangular windows

The negative spikes in the plot are where the function is zero. For the rectangular window, the function is zero at all integers except zero, while the function for the Hann window is zero at all integers except zero and ± 1 . Thus, for a coherently sampled sine-wave (one with an integer bin number), the rectangular window will produce a single line, while the Hann window will produce three lines. However, for incoherent sampling, both windows produce lines at all frequencies. The amplitudes of the lines fall off much more slowly for the rectangular window than for the Hann window.

C.5 Coherent sampling, sine fitting, and other means of dealing with spectral leakage

There are several methods of dealing with spectral leakage other than the judicious choice of windows. The basic problem is that the energy of some large sinusoidal component of the signal leaks into spectral bins that

have smaller components that need to be identified. This is often solved simply by choosing a sufficiently long waveform epoch. As seen from the figure in C.4, the simplest window (Hann) is down by more than 87 dB at an offset of 20 frequency bins. As the epoch becomes longer, the frequency deviation corresponding to 20 frequency bins becomes smaller. For example, with a waveform epoch of 4096, 20 frequency bins are less than 1% of the Nyquist frequency.

Coherent sampling, in which the bin number of the applied signal is an exact integer, is usually the best approach for dealing with spectral leakage. It eliminates the leakage problem completely if exact coherence is obtained. Also, the recommended frequencies given elsewhere in this standard for best achieving other ends all result in coherent sampling. However, sometimes the accuracy and/or precision of the frequency setting of available oscillators does not allow for coherent sampling.

C.6 Useful windows and their characteristics

The only window parameter used in this standard is the normalized noise power gain, *NNPG*. The normalized noise power gain is given by Equation (C.7).

$$NNPG = \frac{1}{N} \sum_{n=0}^{N-1} w[n]^2 \quad (C.7)$$

The *NNPG* is a conversion factor that gives a Parseval-like relationship for the windowed DFT:

$$\frac{1}{N} \sum_{n=0}^{N-1} x[n]^2 = \frac{1}{N \times NNPG} \sum_{k=0}^{N-1} X_w[k]^2 \quad (C.8)$$

where

$X_w[k]$ is the DFT of a windowed input sequence $w[n]x[n]$

The value of *NNPG* is equal to one for the rectangular window and is less than one for all other windows.

Useful windows and their characteristics are described in Harris [B9] and Nuttall [B23]. These windows are of the form shown in Equation (C.9).

$$w[n] = \sum_{l=0}^L a_l (-1)^l \cos(2\pi n l / N) \quad (C.9)$$

where

a_l is the l th coefficient used to define the window function. The sum of the coefficients equal one
 L is the order or number of coefficients in the window

These windows are commonly used and are discussed at length in Nuttall [B23]. Any window of this form is referred to as a cosine window of order L .

For these windows the coefficients of the DFT of the window function are proportional to the coefficients multiplying the cosine terms. There is one nonzero coefficient at zero frequency, L nonzero terms at positive frequency and L at negative frequency. Therefore, a coherently sampled sine wave produces L lines on each side of the central bin.

From Equation (C.7), the Parseval relation and the fact the a_l are the Fourier coefficients of the window function, Equation (C.10) is obtained.

$$NNPG \cong a_0^2 + \frac{1}{2} \sum_{l=1}^L a_l^2 \quad (\text{C.10})$$

The windows shown earlier are described in Table C.1.

Table C.1—Properties of various data windows

Name	L	$NNPG$	a_0	a_1	a_2
Rectangular	0	1	1	0	0
Hann	1	0.625	0.5	0.5	0
Blackman	2	0.305	0.42	0.5	0.08

A.1 Choosing a window

The spectral leakage from a large sinusoidal component can interfere with some other feature one wishes to measure. The magnitude of the problem is determined by inspection of the DFT of the window sequence, as described earlier, and depends on the size of the interfering component(s) and their distance from the desired smaller components. If the chosen window causes an interference that is too large, one should either choose a window that is sufficiently small at the location that is important for the particular problem or use one of the other methods for reducing spectral leakage.

Annex D

(informative)

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